

Time Series Analysis and Forecasting

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1 Foundations

1.1 Learning outcomes

By the end of this chapter you should be able to:

1. Explain what a time series is and how it differs from cross-sectional data, and say why temporal dependence calls for its own methods.
2. Treat an observed series as one realisation of a stochastic process.
3. Explain what forecasting is, judge when a process is predictable, and keep forecasts, goals and plans separate.
4. Match a forecast horizon to the decision it supports, and read a forecast as a random variable rather than a single number.
5. Identify trend, seasonality, cycle and irregular variation in a plot, and describe the shapes each of them takes in practice.
6. Produce the four benchmark forecasts, and evaluate any competing model against them on data it has not seen.
7. Use the lag and lead operators, and the algebra built on them, to write models compactly.
8. Define weak stationarity, say which of its conditions a given series violates, and choose a transformation that restores it.
9. Define white noise, MA, AR, ARMA, ARIMA and SARIMA processes, derive their basic properties, and identify them from an ACF and a PACF.
10. Fit, diagnose and forecast these models in R, following the workflow of Box and Jenkins, and test residuals against the white noise standard.
11. Explain volatility clustering and fit ARCH and GARCH models to it.

1.2 Notation and conventions

One set of symbols is used throughout. Where a competing convention is common in the wider literature it is noted, but the version in Table 1 is the one used here.

Symbol	Meaning
y_t	The observed value of the series at time t
n	Number of observations in the series
T	Forecast origin, the time of the last observation used
h	Forecast horizon, the number of steps ahead
m	Number of observations in one full cycle
$\mathcal{T}$	Index set of a stochastic process
$\mathcal{J}_T$	Everything known up to and including time T
$\hat{y}_{T+h T}$	Point forecast for time $T+h$ given $\mathcal{J}_T$
e_{T+h}	Forecast error $y_{T+h} - \hat{y}_{T+h T}$
ε_t	Innovation, the unpredictable part of the process
w_t	A differenced series, $w_t = \nabla y_t = y_t - y_{t-1}$
x_t	An explanatory variable, never the series itself
L	Lag operator, $Ly_t = y_{t-1}$
F	Lead operator, $Fy_t = y_{t+1}$, equal to L^{-1}
∇	Difference operator, $\nabla = 1 - L$
$\gamma(h), \rho(h)$	Autocovariance and autocorrelation at lag h
ϕ_{hh}	Partial autocorrelation at lag h
ϕ, θ	Autoregressive and moving average coefficients
c	Constant or drift term
σ_t^2	Conditional variance at time t
Q_{LB}	Ljung and Box portmanteau statistic

Table 1: Symbols used throughout the chapter

A few conventions are used consistently throughout this chapter.

The observed series is always written as y_t . The symbol x_t is reserved for explanatory variables.

The lag operator is written as L . Some textbooks use B (for backshift); the two notations are equivalent.

The cycle length is denoted by m . This leaves P available for the seasonal autoregressive order in SARIMA models.

Three symbols are easy to confuse. The symbol T denotes the forecast origin, T_t denotes the trend component in a decomposition, and $\mathcal{T}$ denotes the index set of a stochastic process. Unless stated otherwise, forecasts are made from the end of the observed series, so $T = n$.

In R, the `frequency` argument in `ts()` specifies the number of observations per cycle, namely m . It is not the mathematical frequency $f = 1/m$. For monthly data, `frequency = 12`, whereas the corresponding frequency is $f = 1/12$.

1.3 The nature of time series data

1.3.1 Why the ordering matters

Modern institutions generate enormous quantities of observations collected sequentially through time: daily exchange rates, hourly electricity demand, monthly unemployment, quarterly GDP, heart rate recordings sampled by the minute, satellite vegetation indices, share prices. What these have in common is not their subject matter but their structure.

Unlike the data met in introductory statistics, observations collected through time are rarely independent. Today's value usually depends on yesterday's. That dependence changes which tools are available: ordinary regression typically assumes independent errors (or weaker conditions), not necessarily independent observations, whereas time series analysis models the dependence explicitly rather than assuming it away.

A **time series** is an ordered collection of observations recorded sequentially through time,

$$\{y_t : t = 1, 2, \dots, n\},$$

where y_t is the observation at time t . Throughout this course we assume that observations are equally spaced, and n is the number of observations.

1.3.2 Where these methods are used

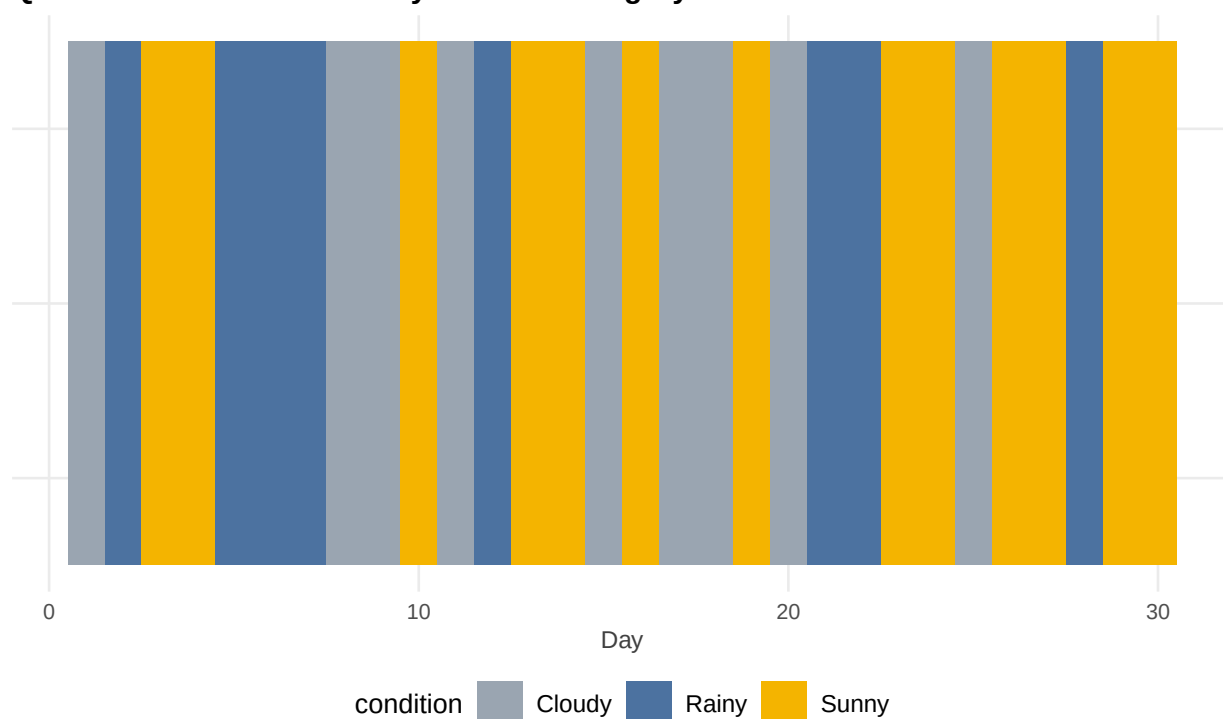
Discipline	Typical series
Economics	Gross domestic product, inflation
Finance	Share prices, returns, volatility
Medicine	ECG traces, patient admissions
Epidemiology	Case counts under surveillance
Environmental science	Rainfall, temperature, river flow
Engineering	Sensor and process monitoring
Ecology	Animal population counts

Table 2: Fields in which time series methods are standard

The subject matter differs completely across these fields. The statistical problem does not.

1.3.3 Time Series Types: Qualitative

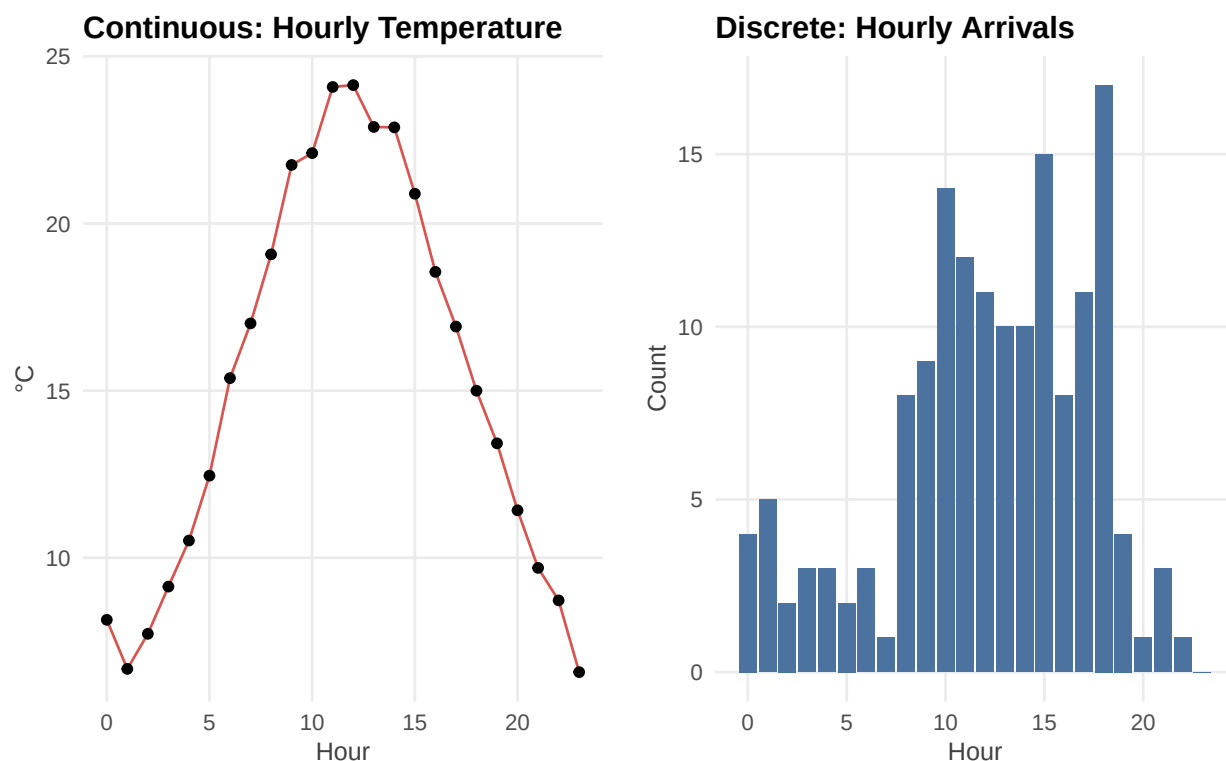
Not every time series is a number. Weather condition, machine status, credit rating, all of these get recorded at each point in time as a category, not a measurement.

Qualitative Time Series: Daily Weather Category

You cannot hand “Sunny” to the ACF formula and expect a sensible number back; there is no natural arithmetic on weather categories for a covariance to act on. The classical ACF and the models built on it in this chapter apply to *quantitative* series. A categorical series is not beyond analysis, but it calls for a different toolkit, typically a Markov chain, where the object of interest is a transition probability between states rather than a correlation. Recoding categories as indicator variables is the usual bridge between the two worlds, and it is more than enough reason to check, before opening any textbook chapter, which kind of series is actually in front of you.

1.3.4 Quantitative: Continuous vs. Discrete

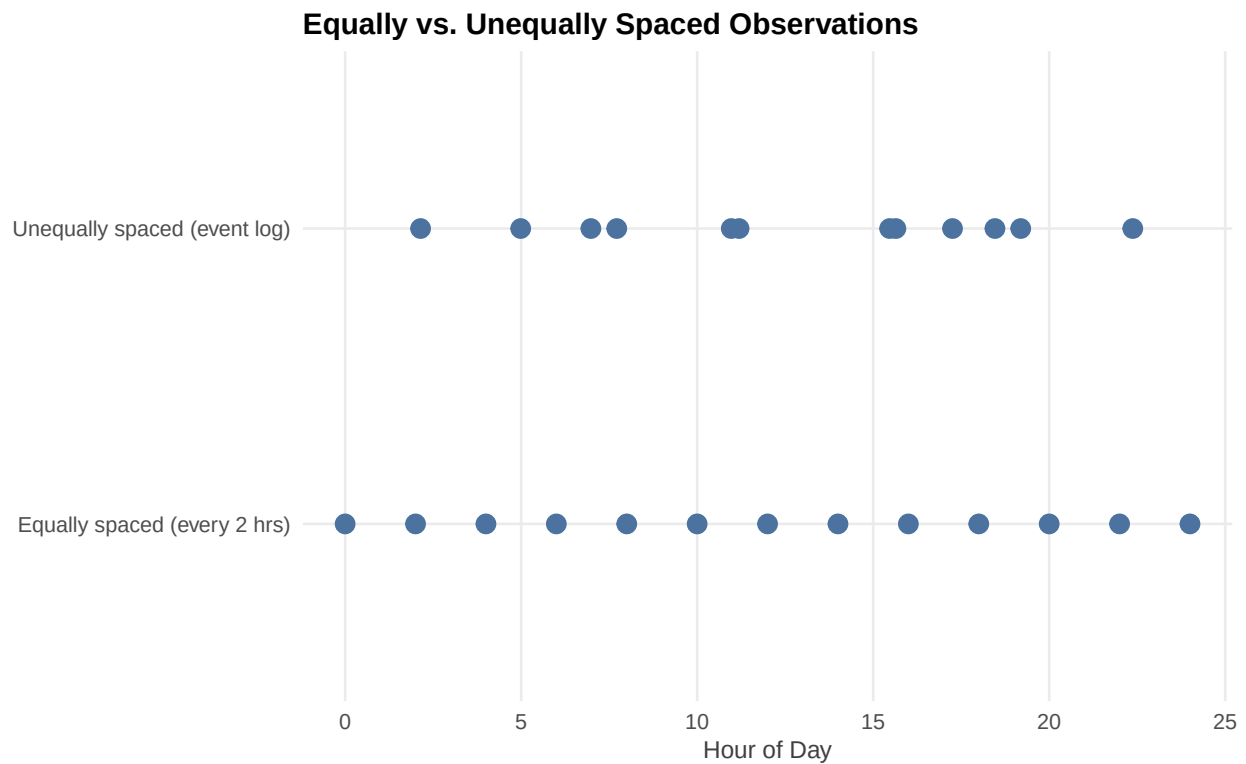
Within numeric time series, there’s still a meaningful split. Temperature can, in principle, take any value. A count of customer arrivals cannot, it’s a non-negative integer, and it looks different on a plot because of it.



Notice the plot choice itself tracks the distinction, a smooth line for something that varies continuously, bars for something that only ever comes in whole numbers. Models built for one (say, a Gaussian ARIMA) will usually need adjusting before they're appropriate for the other (Poisson or count-based approaches).

1.3.5 Equally vs. Unequally Spaced

Everything so far has quietly assumed observations arrive on a regular clock, every hour, every day. Plenty of real data doesn't: a transaction log, a hospital admission record, a server error log, all fire whenever the underlying event happens, not on any fixed schedule.

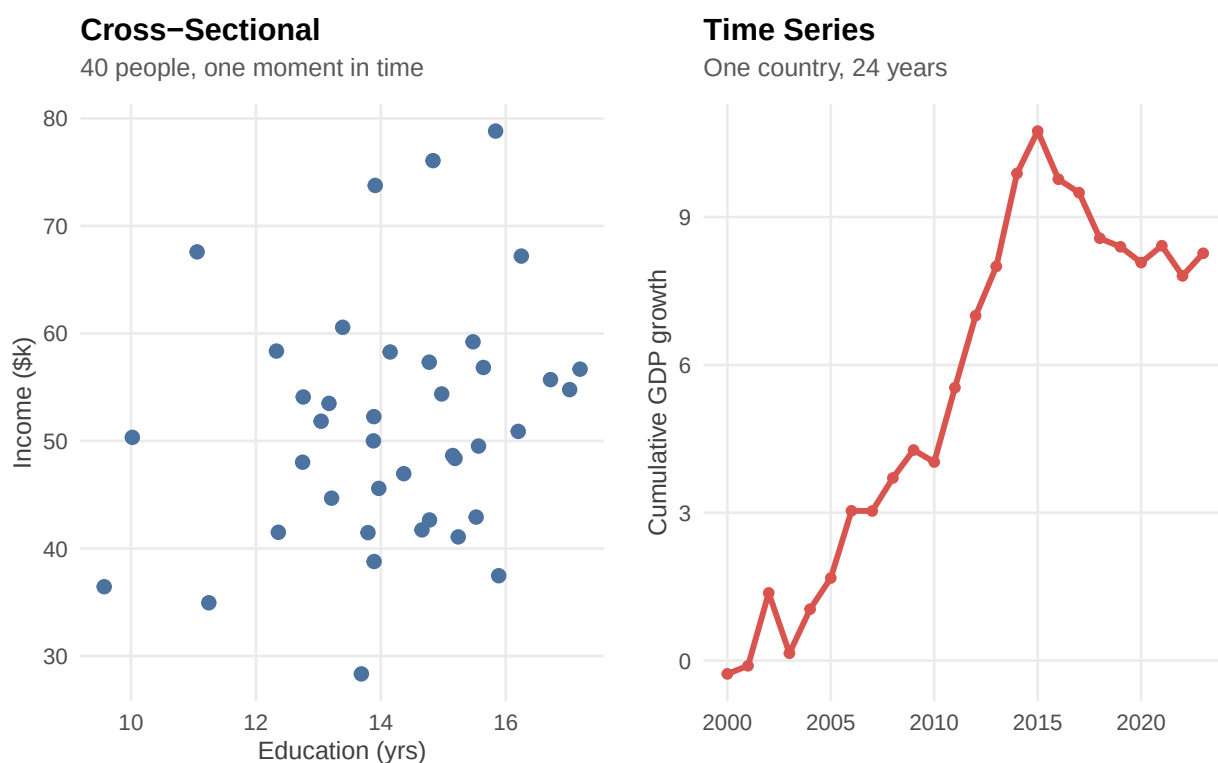


Most of the classical machinery, `ts()`, ACF, ARIMA, quietly assumes the top row. The bottom row needs different tools entirely (point process models, for instance), which is why the very first thing worth checking about any new dataset is simply: are these gaps between observations actually constant?

1.3.6 Cross-Sectional vs. Time Series Data

Cross-sectional data record many units at a single point in time, for example the grades of forty students in one semester or the GDP of one hundred countries in a particular year. The order in which the observations are listed carries no statistical information.

A time series records repeated observations on the same process over time, for example the annual GDP of one country or monthly rainfall at a weather station. Here the ordering is fundamental. Reordering the observations destroys the temporal structure, such as trends and seasonality, and can lead to incorrect statistical conclusions because time itself is an important explanatory variable.

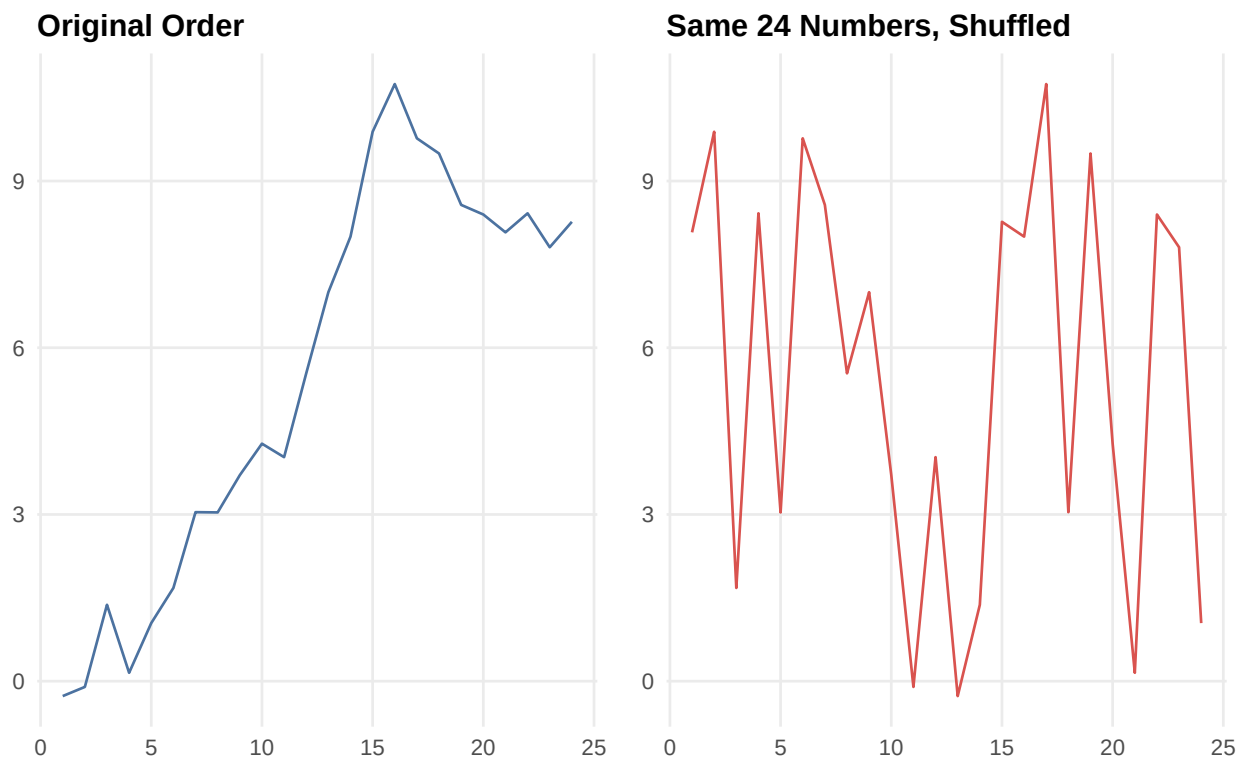


Same number of dots, roughly. Completely different questions being asked of them. The scatter on the left is asking “does education relate to income, across people?” The line on the right is asking “what happened to this one economy, over time?”

1.3.7 Key differences

	Cross-Sectional	Time Series
Unit of observation	Many, at one time	One, across many times
Does row order matter?	No	Yes, fundamentally
Typical assumption	Independence across units	Autocorrelation across time
A “neighbor” observation	Arbitrary	Yesterday, last week, last year

That second row is worth proving, not just asserting.



Exact same 24 values in both plots. Reshuffle a cross-sectional dataset's rows and nothing about your regression changes. Reshuffle a time series and you've destroyed the one thing that made it a time series in the first place, the trend is simply gone. Order isn't a formatting detail here, it's where the information actually lives.

1.3.8 Dependence made visible

The clearest way to see what dependence does is to simulate a series with none and a series with a great deal of it.

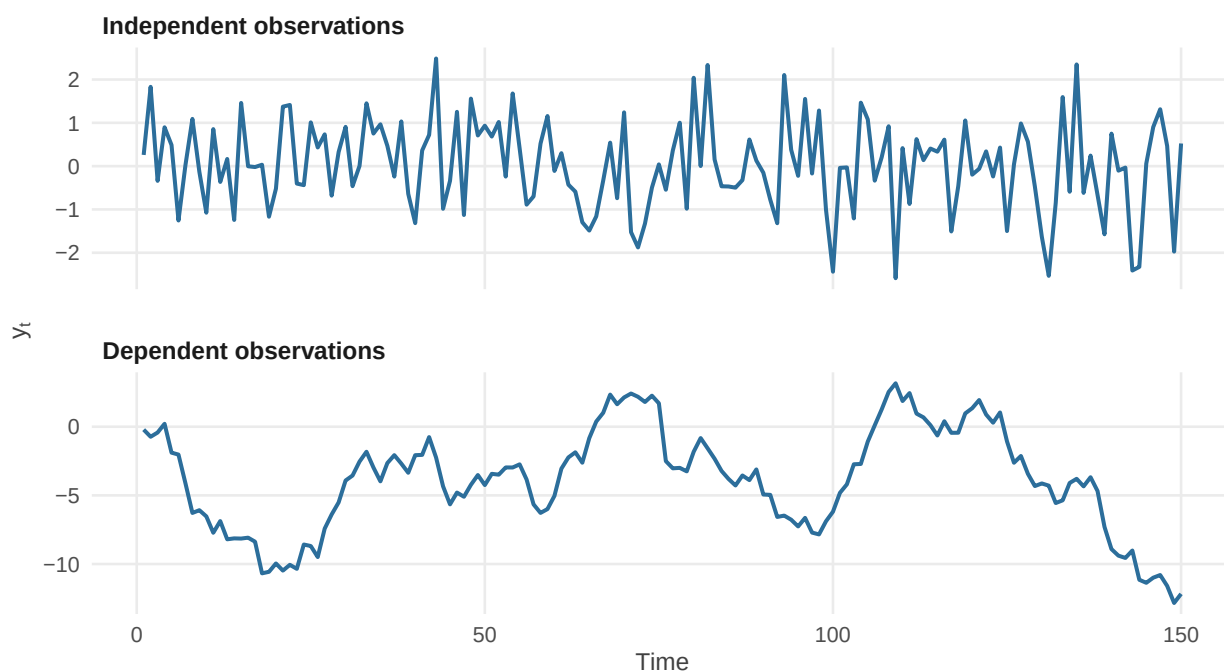
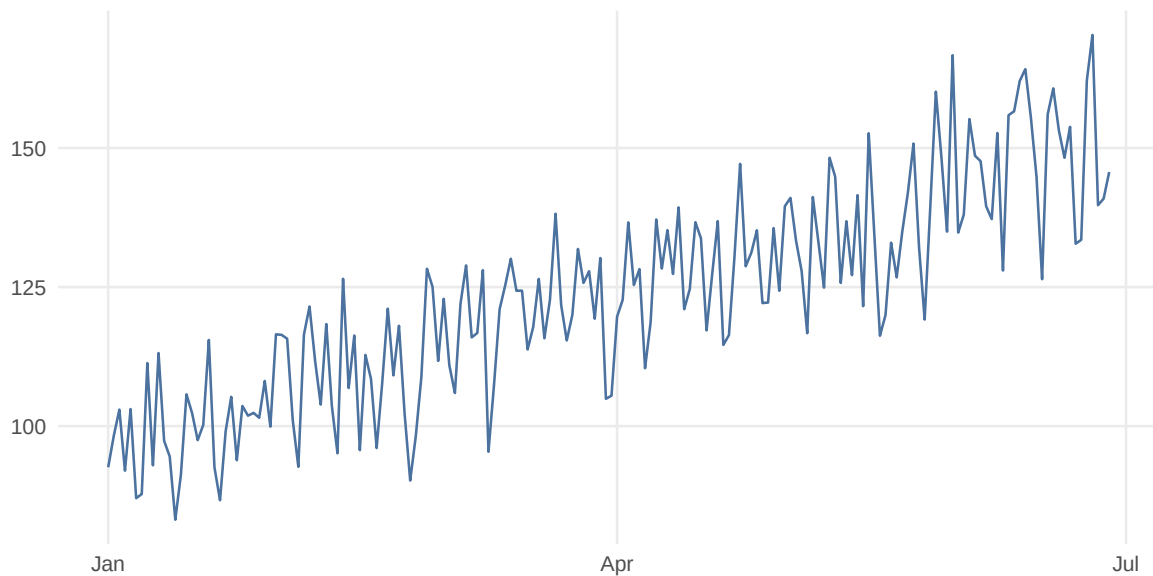
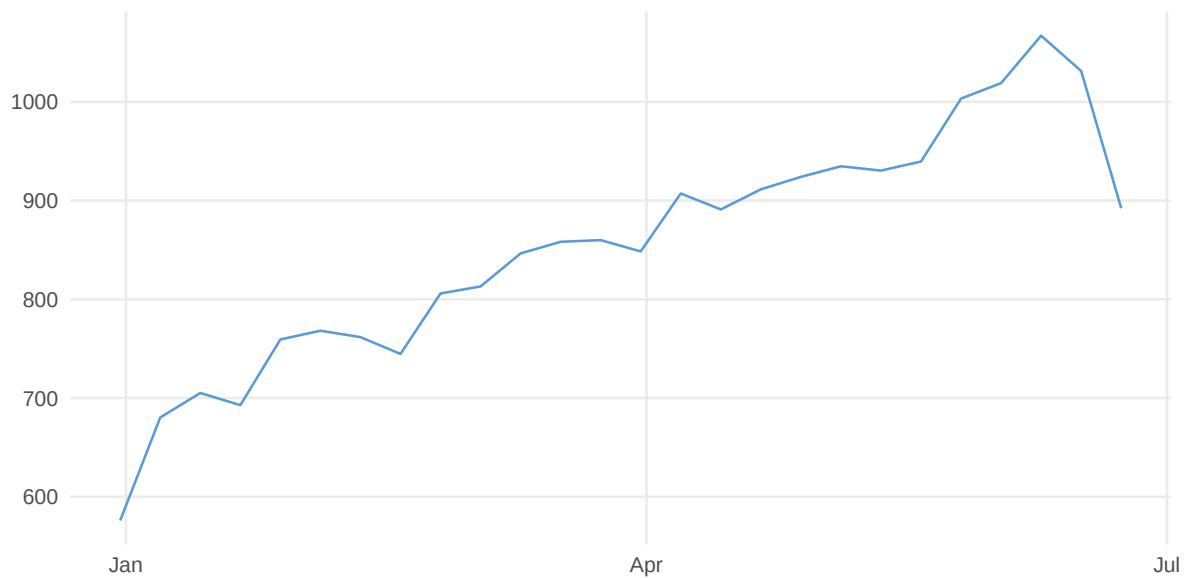
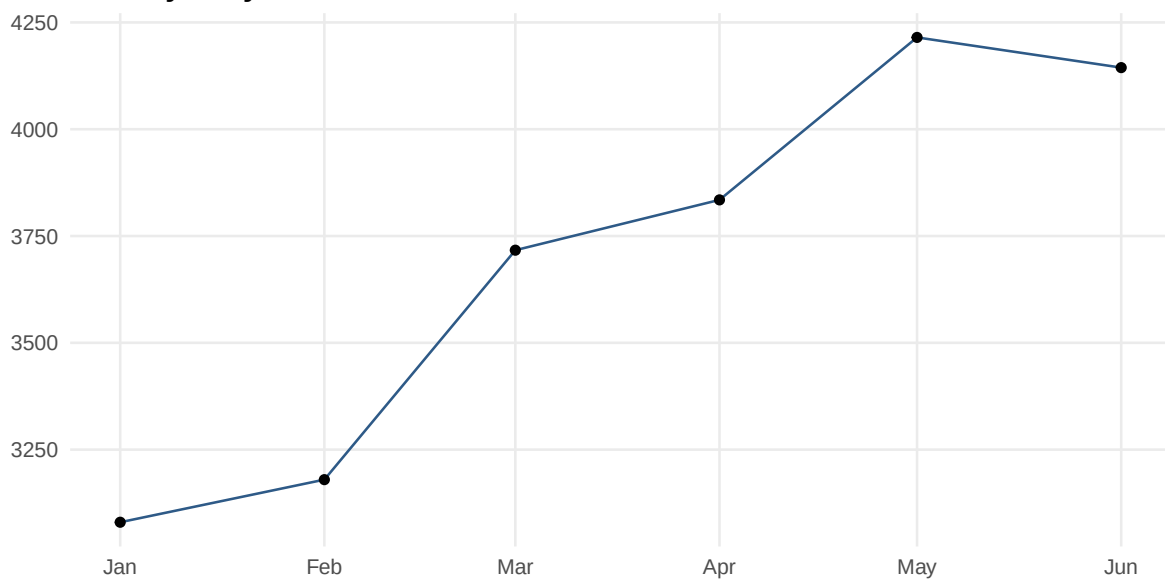


Figure 1: Independent observations scatter about a fixed level. Dependent observations wander.

The independent series shows no persistence: knowing one value tells you nothing about the next. The dependent series shows long runs of increase and decrease, which is the behaviour most natural processes display and the behaviour every model in this chapter is built to describe.

1.3.9 Temporal Granularity and the Aggregation Effect

The same underlying process looks like a different series depending on how finely you sample it. Watch what happens to a daily sales series with a weekday dip as we zoom out to weekly, then monthly totals.

Daily: the weekday dip is visible**Weekly: the dip is averaged away****Monthly: only the trend survives**

Nothing about the underlying process changed across those three panels, only the resolution we're viewing it at. Aggregating up smooths out high frequency structure (the weekday effect vanishes) while making low frequency structure (the trend) easier to see. This cuts both ways: whichever granularity you choose to work with has already decided, before you've run a single model, which patterns you're even capable of finding.

1.3.10 One realisation of a stochastic process

Suppose annual rainfall at a weather station has been recorded for a hundred years. Only one sequence $\{y_1, \dots, y_{100}\}$ is observed, yet nature could in principle have produced a great many other sequences under the same physical mechanism. The data in hand are one outcome drawn from a collection of possible outcomes, and it is that collection, not the single sequence, that a model describes.

A **stochastic process** is a collection of random variables indexed by time, $\{y_t : t \in \mathcal{T}\}$. Before observation each y_t is a random variable. After observation it is a realised value.

Stochastic process	Observed time series
A collection of random variables	A collection of recorded values
A theoretical object	The data in front of you
Infinitely many possible paths	One path that happened

Table 3: The process and its realisation

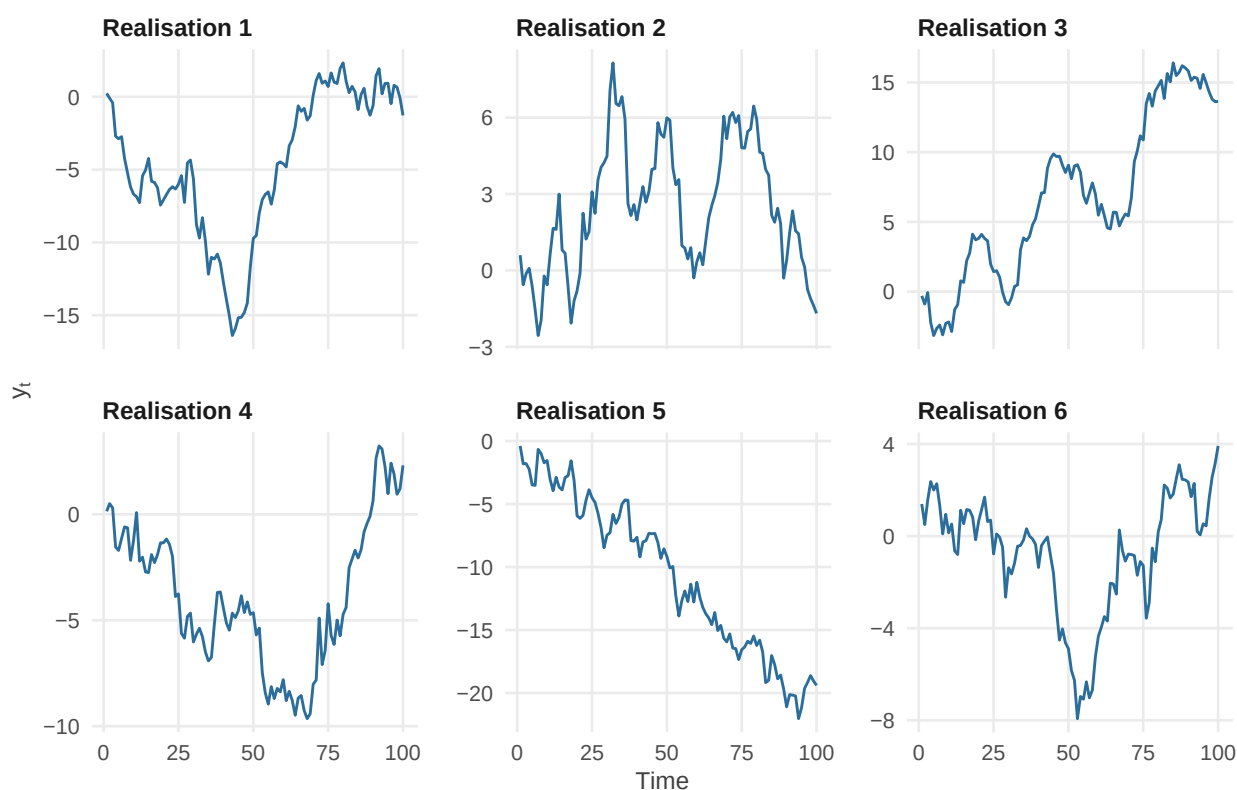


Figure 2: Six realisations of one model. The paths differ; their statistical character does not.

Every panel came from exactly the same model, and no two are alike. This is why inference targets the process rather than the particular dataset, and why a model that reproduces one realisation exactly is almost certainly overfitted.

It is worth fixing, once and for all, the chain of ideas that the rest of these notes relies on:

$$\text{Model} \longrightarrow \text{Stochastic process} \longrightarrow \text{Realisation} \longrightarrow \text{Observed series}.$$

A **model** is a mathematical specification, such as the AR(1) equation met later in this chapter, together with values or distributions for its parameters. It defines a **stochastic process**, the entire family of random variables $\{y_t : t \in \mathcal{T}\}$ consistent with that specification. A **realisation** is one complete sample path drawn from that process; Figure 2 above shows six. The **observed series** is the finite stretch of one realisation that has actually been recorded and is available for analysis. Keeping these four ideas distinct is what makes it possible to ask whether a fitted model describes the process well, rather than merely the one dataset in hand.

Deterministic and stochastic behaviour

A deterministic process such as $y_t = 5 + 0.2t$ is fully known once t is known. A stochastic process such as $y_t = 5 + 0.2t + \varepsilon_t$, with $\varepsilon_t \sim N(0, 1)$, cannot be pinned down exactly even when t is known, because of the random term.

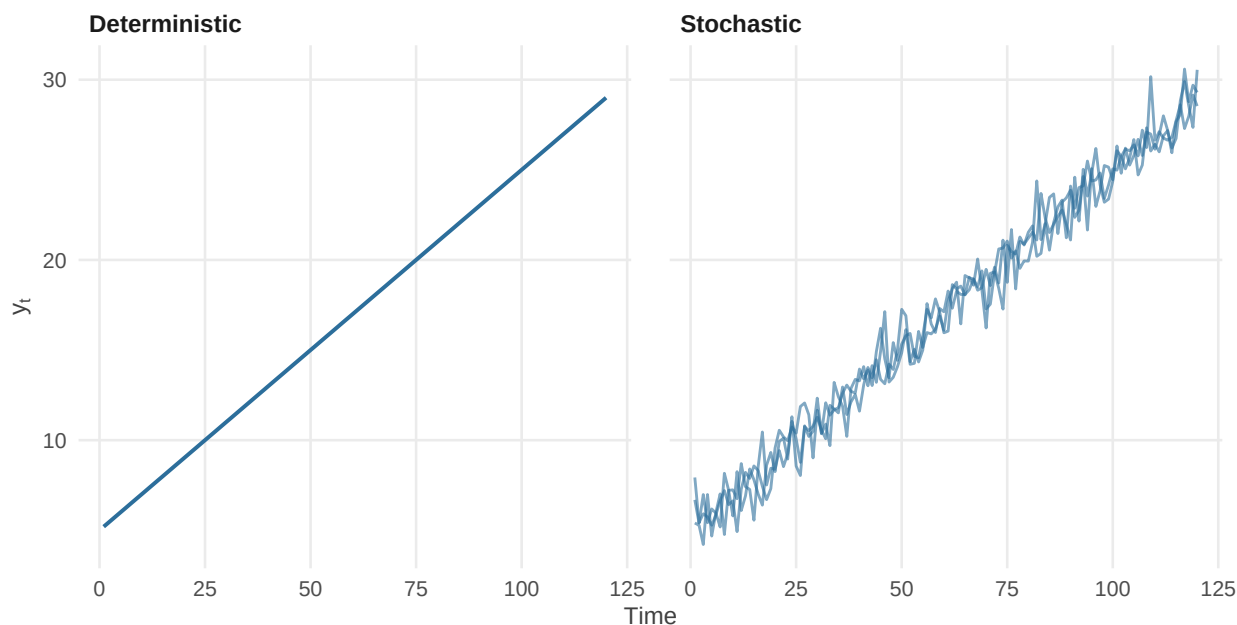


Figure 3: The same straight line: deterministic on the left, and three independent realisations of its stochastic counterpart on the right. Re-running the left panel reproduces it exactly; re-running the right panel never does.

Real data resemble the right hand panel. This motivates writing a series as signal plus noise,

$$y_t = s_t + \varepsilon_t,$$

where s_t is the systematic part and ε_t the random part.

This decomposition is intended as a simple conceptual starting point. Many useful models can be understood as describing a systematic pattern together with random variation. As the chapter develops, this idea will be extended. In models such as AR, MA and ARMA, the systematic component is generated dynamically through dependence on previous observations and innovations rather than by a fixed deterministic function. The underlying principle, however, remains the same: modelling seeks to explain as much of the observed behaviour as possible while leaving only unpredictable random variation in the residuals.

Monthly electricity demand is described by $y_t = 250 + 12 \sin(2\pi t/12) + \varepsilon_t$.

The constant 250 is average demand. The sine term is the seasonal signal, with a peak twelve months apart and an amplitude of 12 units. The term ε_t absorbs weather, consumer behaviour and everything else not written down.

2 The Forecasting Problem

2.1 Forecasting

2.1.1 What forecasting is

People have always wanted to know what comes next. Long before statistics developed as a scientific discipline, people attempted to predict the future through prophecy, astrology and other traditional practices. Modern forecasting has replaced these approaches with systematic analysis of data, probability and statistical models. Although the methods have changed dramatically, the underlying problem has remained the same: important decisions must often be made before the outcome is known.

History shows that even well informed forecasts can be wrong. Economies experience unexpected recessions, new technologies transform industries in unforeseen ways, and natural events occur that no model could fully anticipate. These examples are reminders that forecasting is not about eliminating uncertainty. It is about making the best possible use of the information available at the time a decision is made.

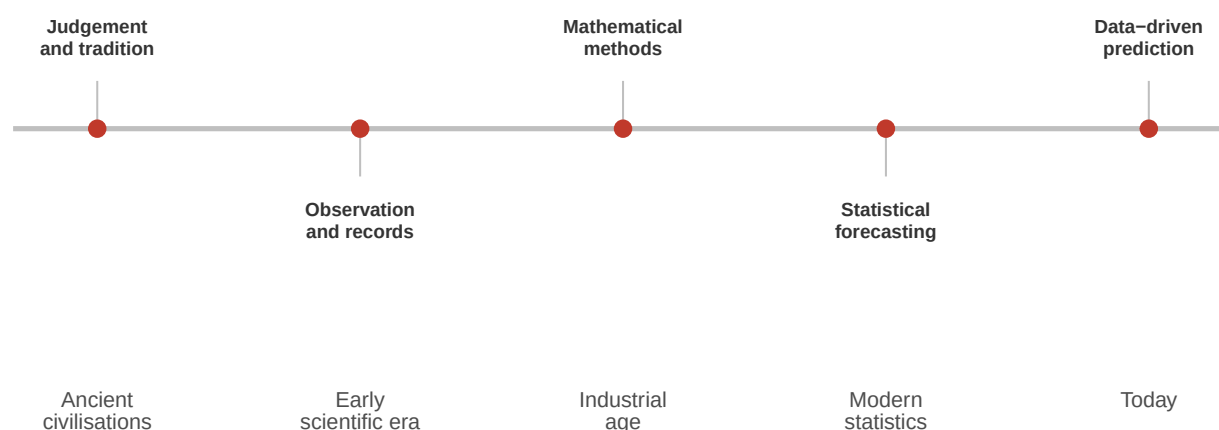


Figure 4: Forecasting has evolved from judgement and tradition to statistical modelling, but uncertainty remains an unavoidable part of predicting the future.

The purpose of the figure is not to present a detailed history of forecasting. Instead, it illustrates the gradual shift from judgement-based prediction towards evidence-based statistical methods. Despite advances in modelling and computing, uncertainty has never disappeared. Every forecast remains conditional on the information available when it is made.

Forecasting is the use of available information to estimate a quantity that has not yet been observed.

The objective of forecasting is not to predict the future with certainty. Rather, it is to provide the best estimate that can be made from the available evidence, together with a realistic assessment of the uncertainty surrounding that estimate. Good forecasts support better decisions because they describe both what is expected to happen and how confident we can be in that expectation.

Sector	Quantity forecast	Decision supported
Retail	Sales	Stock levels and promotions
Health	Patient admissions	Staffing and bed allocation
Agriculture	Crop yield	Planting, harvesting, storage
Finance	Exchange rates	Investment and risk management
Transport	Passenger demand	Timetables and fleet allocation
Energy	Electricity demand	Generation and distribution

Table 4: Forecasts and the decisions that depend on them

Every method in this chapter, from the mean forecast to a seasonal ARIMA model, exists for one reason: to improve decisions taken under uncertainty.

2.1.2 What can actually be forecast

Not everything can. Tomorrow's sunrise can be predicted almost perfectly because planetary motion is stable and well understood. Tomorrow's lottery numbers cannot be predicted at all, because the mechanism is built to be random.

Most practical problems sit between those extremes, and where a problem sits depends on four questions.

1. Do we understand the factors that drive the process?
2. Do we have enough relevant historical data?
3. Will the important relationships hold in the future?
4. Will publishing the forecast change the outcome?

Understanding the process. Forecasts improve when the drivers are known. Electricity demand responds to temperature, time of day and the calendar, all measurable. Asset prices respond to a very large number of interacting influences, most of which are not.

Relevant data. Models learn patterns from observations, but more data is not automatically better data. If the process itself has changed, observations from a decade ago may say nothing useful about next month.

Stability. Forecasting does not assume the future repeats the past exactly. It assumes the main relationships remain reasonably stable. Daily electricity use varies, yet the shape of the daytime peak and the overnight trough persists.

Independence from the forecast. If publishing a prediction changes behaviour, the target moves. A widely reported forecast of a housing shortage can push buyers to act at once, lifting prices today and undermining the forecast that caused it.

Predictability also depends on the horizon and on the sampling interval, which is why the labels in Figure 5 name both. Daily rainfall one day ahead is moderately forecastable; daily rainfall three weeks ahead is close to unforecastable, even though monthly rainfall totals remain seasonal and reasonably predictable. Always state the interval and the horizon before calling a series predictable.

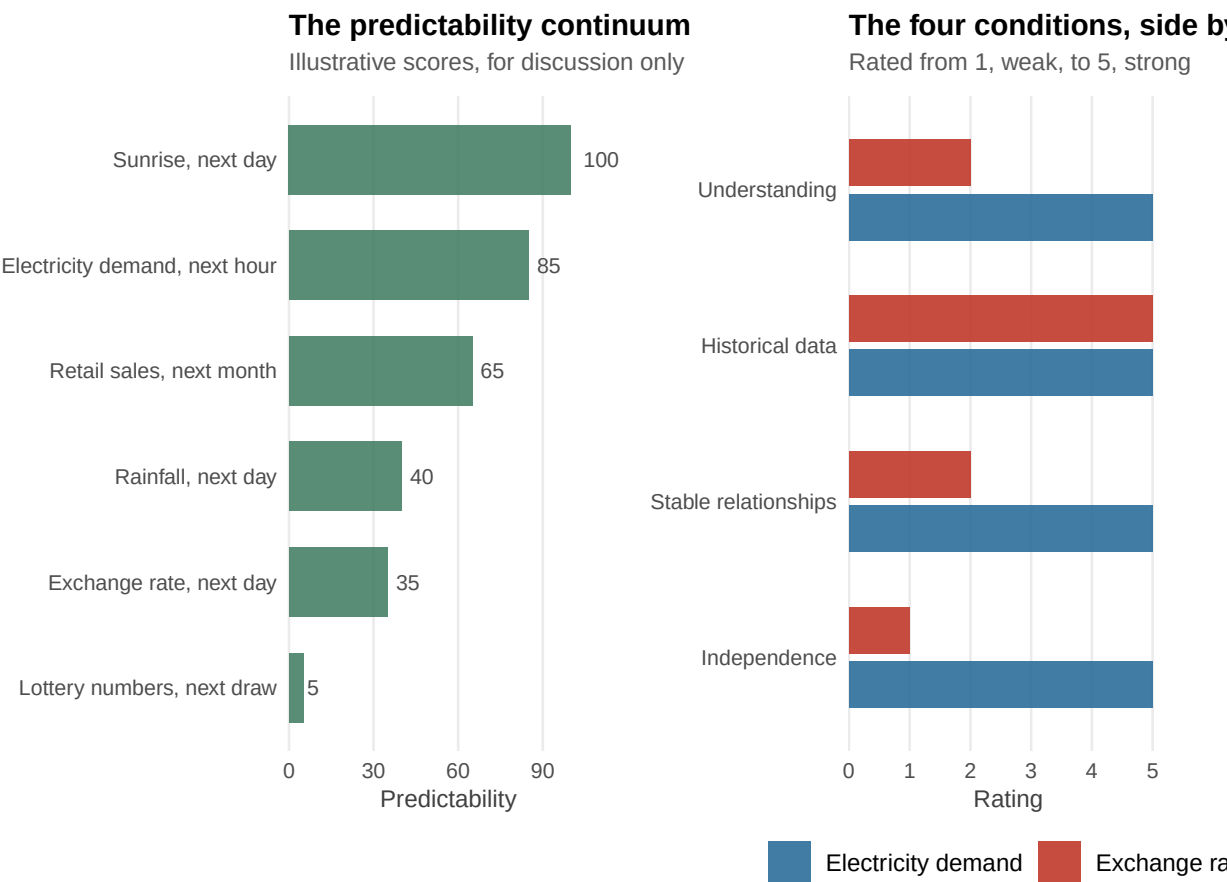


Figure 5: Processes differ in how well they meet the four conditions, and so in how predictable they are. Scores are illustrative and assume a short horizon at the stated sampling interval.

Two series of identical length can behave completely differently, which is worth seeing directly.

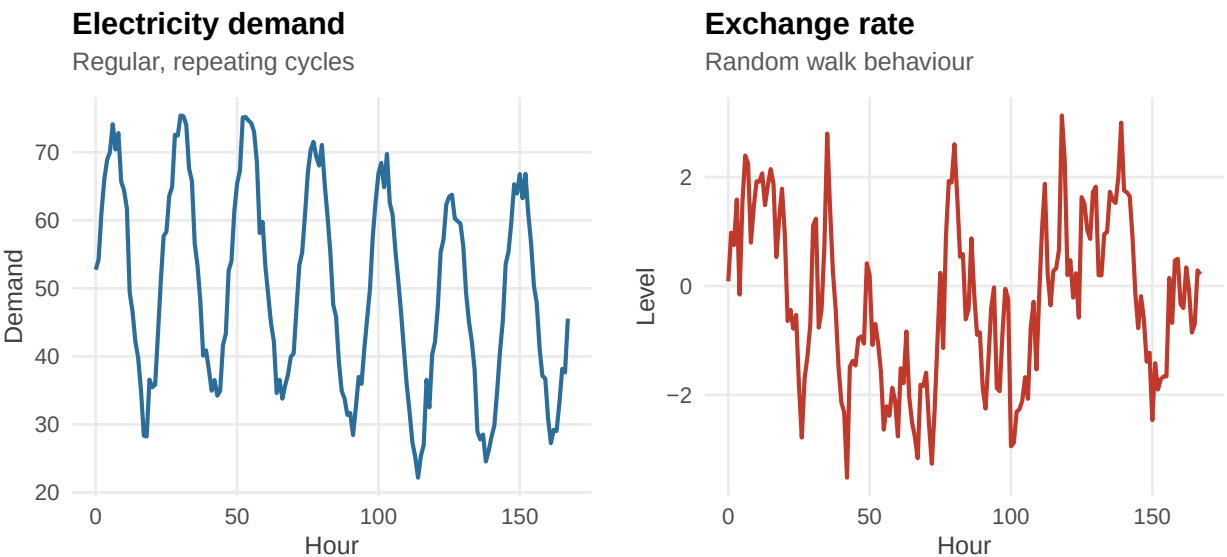


Figure 6: Both series contain 168 hourly observations. Only one of them contains a usable pattern.

The electricity series has structure a model can exploit. The exchange rate wanders. What matters is whether the data contain stable, informative patterns, not how many rows the spreadsheet

holds.

2.1.3 Forecasting, goals and planning

Treating these three as the same activity is a common and expensive mistake in practice.

Activity	Question answered	Character
Forecasting	What is likely to happen?	Evidence based estimation
Goal setting	What do we want to happen?	Ambition and strategy
Planning	What should we do about it?	Action and intervention

Table 5: Three activities that are often confused

A forecast describes the expected future, a goal describes the preferred future, and planning is the bridge between them.

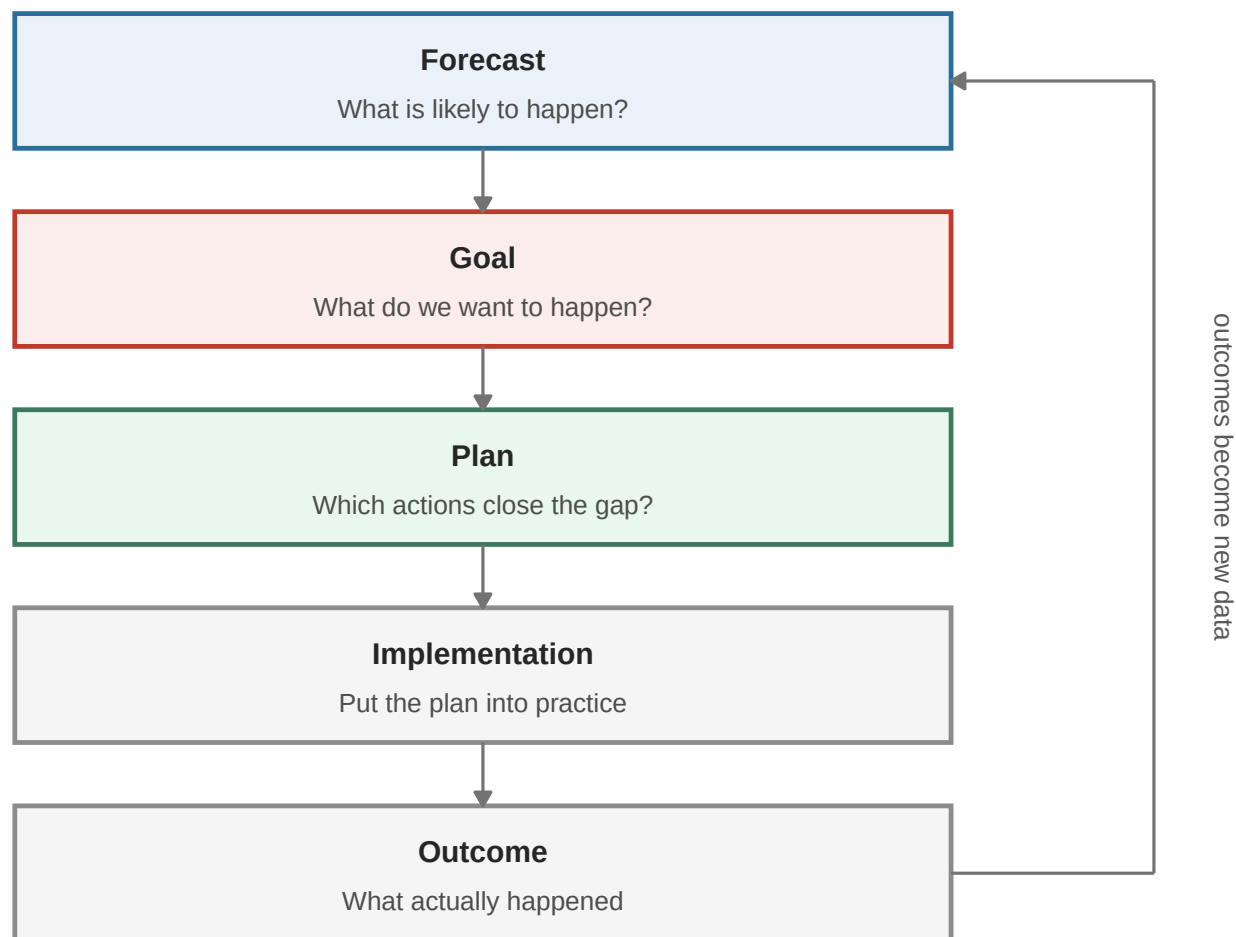


Figure 7: The three activities form a loop. Outcomes eventually become the data for the next forecast.

Suppose an objective forecast points to declining sales. Management may dislike the answer, but editing the forecast upward to match a growth target of ten per cent does not change what will

happen. It only breaks the instrument. The useful response is to accept the forecast and then plan, through advertising, new products or pricing, so that the outcome differs from the baseline.

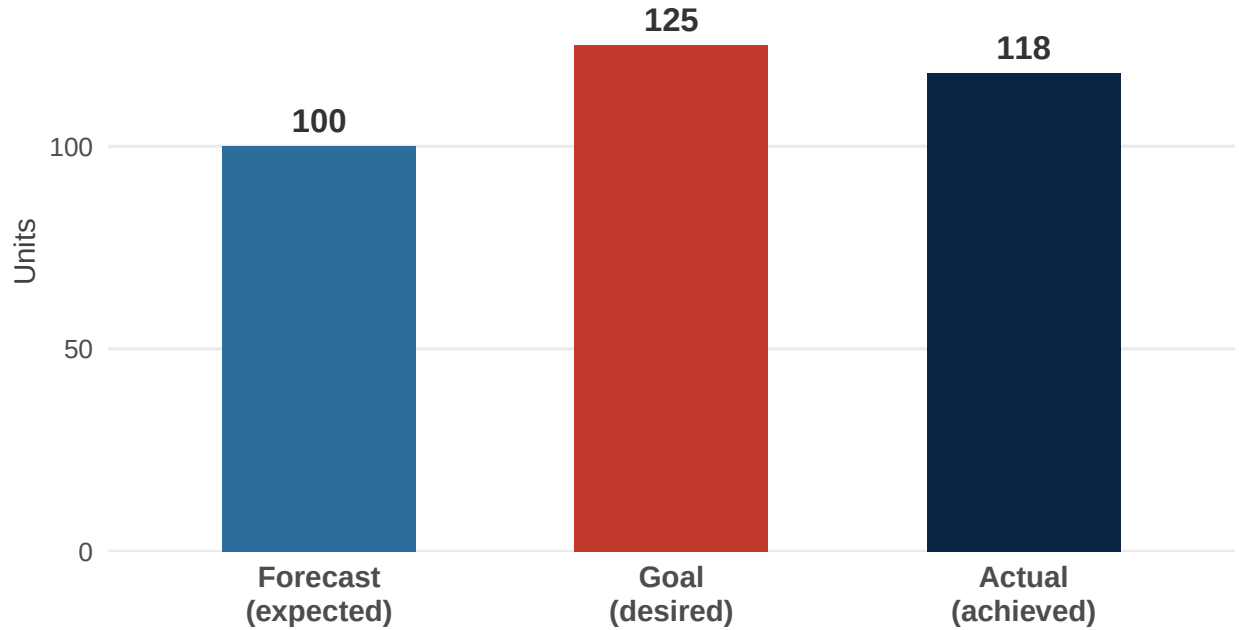


Figure 8: Planning is the effort that closes the distance between the baseline forecast and the goal.

2.1.4 Forecast horizons

The forecast horizon h is the distance between the forecast origin T and the point being predicted. The decision sets the horizon, and the horizon narrows the choice of model.

An electricity utility needs demand for tomorrow so that generation can be scheduled, and demand twenty years out so that capacity can be built. The variable is identical in both cases. The horizons, and therefore the models, are not.

Horizon	Typical scale	Decisions supported
Short term	Minutes to weeks	Scheduling, inventory, dispatch
Medium term	Months to two years	Budgets, hiring, equipment
Long term	Years to decades	Infrastructure, policy, capacity

Table 6: Horizons and the decisions they serve

How fast uncertainty accumulates depends on the model, and the two cases behave quite differently. For a series with a unit root, such as a random walk with innovation standard deviation σ , the h step forecast variance is $h\sigma^2$, so the interval width grows without limit, in proportion to $\sqrt{h}$. For a stationary model the forecast variance instead converges to a finite limit.

AR(1) forecast error variance. For a stationary AR(1) process with $|\phi| < 1$ and innovation variance σ^2 , the h step forecast error variance is

$$\text{Var}(e_{T+h}) = \sigma^2 \frac{1 - \phi^{2h}}{1 - \phi^2} \rightarrow \frac{\sigma^2}{1 - \phi^2} \quad \text{as } h \rightarrow \infty,$$

the limit being the unconditional variance of the process.

A long horizon forecast from a stationary model is simply the unconditional mean, reported with the unconditional standard deviation. Intervals therefore widen with the horizon, but they do not widen indefinitely unless the series is non-stationary.

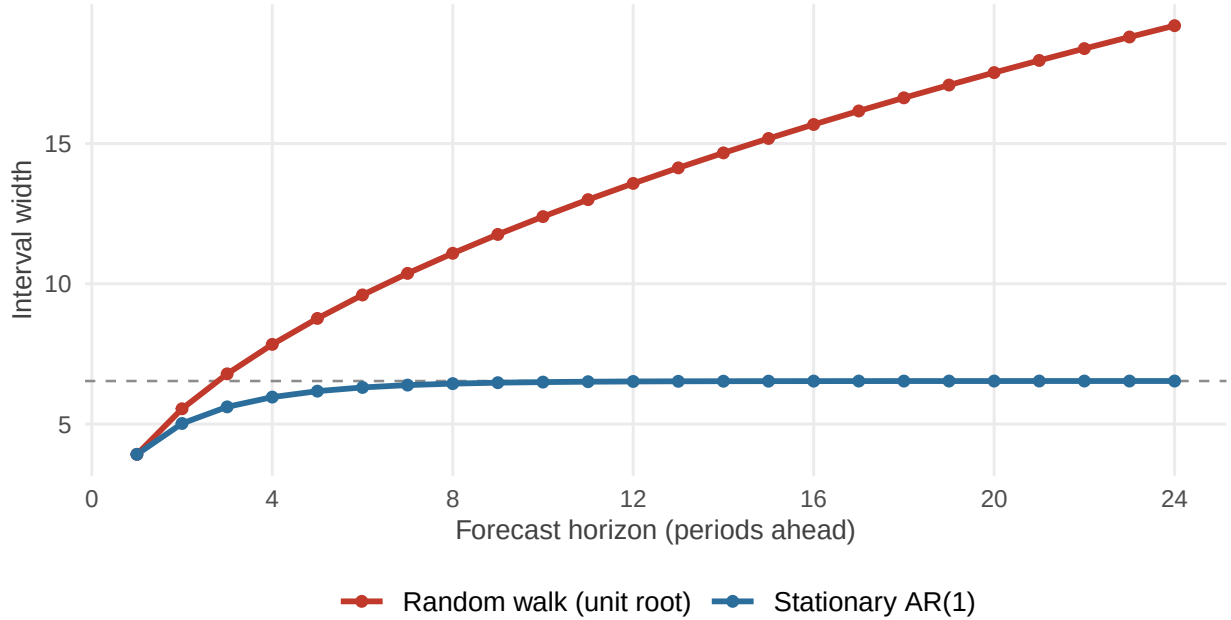


Figure 9: Ninety-five per cent interval width against horizon for a random walk and for a stationary AR(1) with $\phi = 0.8$, both with unit innovation variance. The dashed line is the AR(1) limit.

Long horizon work with a non-stationary series is therefore less about a precise number and more about a set of plausible scenarios. One further caution applies to both cases: the intervals produced by standard software condition on the estimated parameters and on the model being correct, so in practice they understate true uncertainty, sometimes substantially.

2.1.5 Where forecasts come from

Method families divide first on a single question: are there usable historical data?

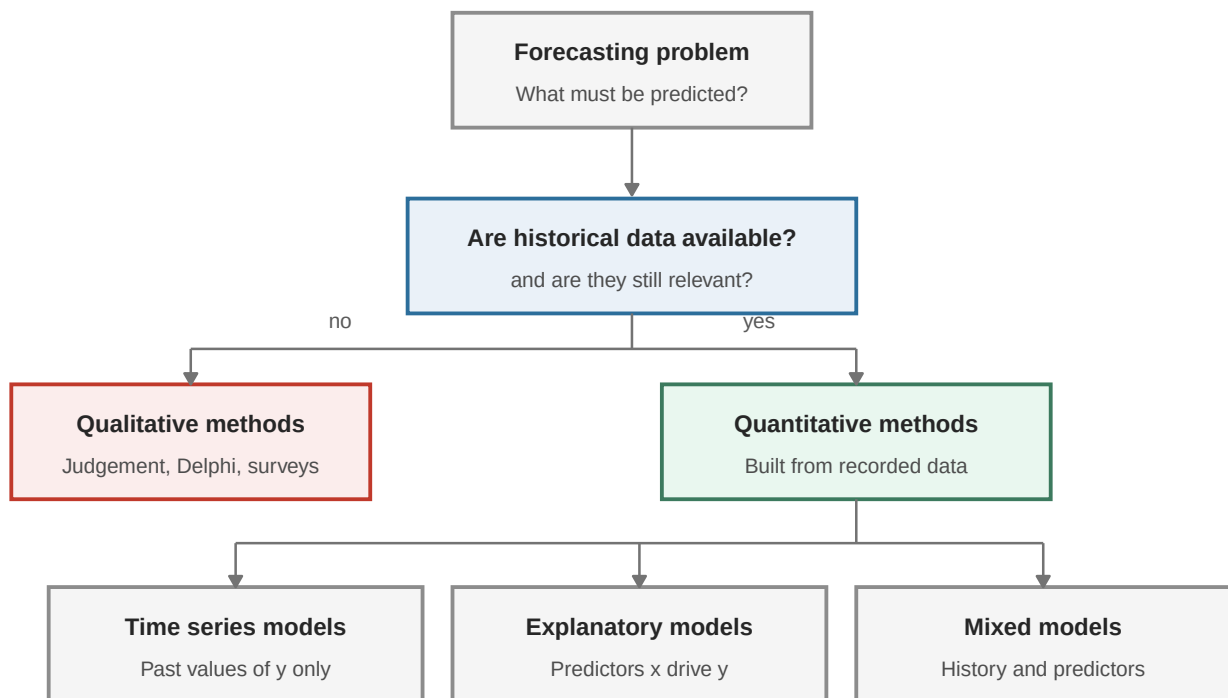


Figure 10: Method families, organised by the data available to the forecaster.

Time series models assume the history of y_t contains what is needed to project it forward. Exponential smoothing, ARIMA and SARIMA all belong here, and they occupy most of this chapter.

Explanatory models assume outside variables x_t drive the outcome. Regression is the standard example.

Mixed models use both, letting external drivers enter while still respecting the temporal structure of the series.

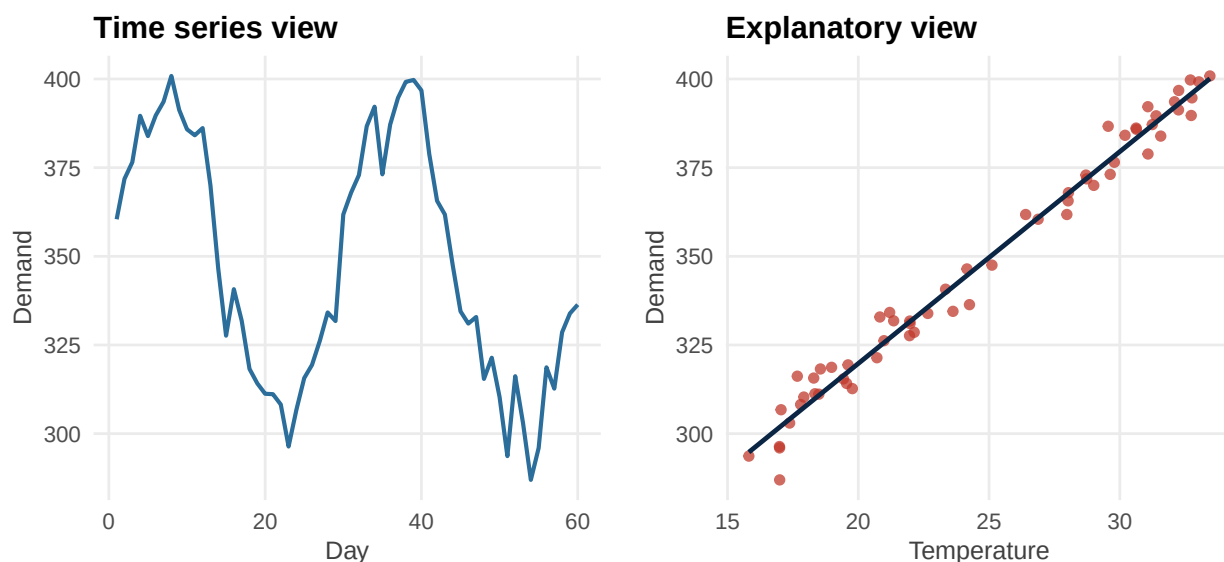


Figure 11: The same data seen two ways: as a sequence on the left, as a relationship on the right.

One caution applies to every explanatory model: the predictors must themselves be forecast. If demand depends on temperature, a demand forecast is only as good as the weather forecast behind it.

2.1.6 The five stages of a forecasting project

Fitting a model is one step inside a longer process, and it is rarely the step that decides whether the exercise succeeds.

1. **Define the problem.** What is being predicted, for whom, and how far ahead?
2. **Gather information.** Historical data, plus the context needed to read them. Knowing that a sharp dip in 2020 was a lockdown rather than a change in the market changes how the series is treated.
3. **Explore the data.** Plot everything. Look for trend, seasonality, missing values and structural breaks.
4. **Build and compare models.** Test candidates against the benchmarks given later in this chapter, on data the models have not seen. A complex model that cannot beat a naive one is not worth keeping.
5. **Evaluate and iterate.** Compare forecasts with what happened, and feed the errors back into the next round.

2.1.7 Forecasts are random variables

A forecast is not a single fixed number. Because the future is uncertain, the quantity being forecast is a random variable, and a forecast is a statement about its distribution.

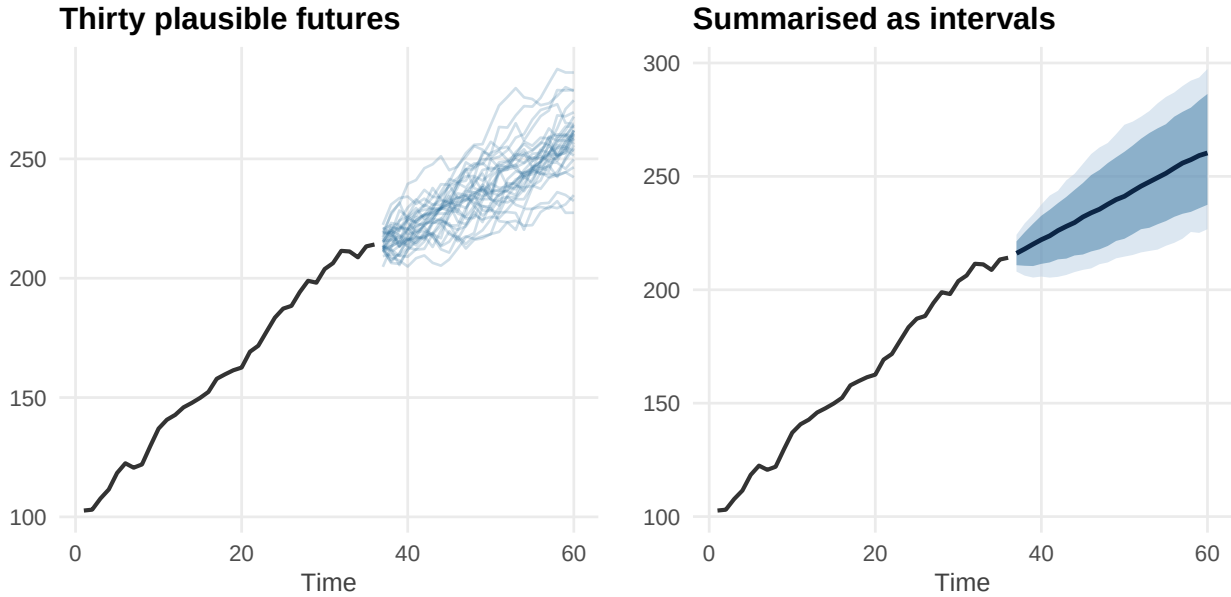


Figure 12: Simulated futures on the left, the same information summarised as prediction intervals on the right. Thirty paths are drawn; the quantiles are computed from a thousand.

A **point forecast** $\hat{y}_{T+h|T}$ is the single estimate drawn as the dark line. A **prediction interval** is the shaded region expected to contain the true value with a stated probability, usually 80 or 95 per cent. Decisions need both. A forecast of 250 units plus or minus 5 supports very different planning from a forecast of 250 units plus or minus 100.

Optimal point forecast under squared-error loss. Let $\mathcal{J}_T$ denote everything known at the forecast origin T , and judge a forecast by its expected squared error, $E[(y_{T+h} - \hat{y}_{T+h|T})^2 | \mathcal{J}_T]$. This loss is minimised by the conditional expectation,

$$\hat{y}_{T+h|T} = E(y_{T+h} | \mathcal{J}_T).$$

Every point forecast used in this chapter, the benchmark methods, ARIMA forecasts and volatility forecasts alike, is this quantity or a working approximation to it; only the information set $\mathcal{J}_T$ and the model generating the conditional distribution of y_{T+h} change from one method to the next. The vertical bar in the notation is doing real work: change what is known at T and the forecast changes with it.

Two consequences follow directly, and both recur throughout the chapter. First, because $\hat{y}_{T+h|T}$ is itself the conditional mean, the forecast error $e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$ satisfies $E(e_{T+h} | \mathcal{J}_T) = 0$; it can nonetheless only be computed once time $T+h$ has arrived, which is precisely why forecast evaluation is harder than model fitting. Second, deriving the point forecast requires only that y_{T+h} have a finite conditional mean and variance; turning it into a prediction interval requires more, typically an assumption that the conditional distribution of y_{T+h} is approximately Gaussian, which the residual diagnostics later in this chapter are used to check.

3 Exploring a Series

3.1 Two ways of looking at a series

Every time series analysis, at heart, answers one of two questions.

Perspective	Primary question
Time domain	How do earlier observations influence later ones?
Frequency domain	Which cycles explain the observed behaviour?

Table 7: The two perspectives

Neither is superior. They examine different features of the same stochastic process, and serious work usually needs both.

3.1.1 The time domain

The time domain approach explains y_t using $y_{t-1}, y_{t-2}, \dots$, and its central concern is serial dependence. This is the route that leads to AR, MA, ARMA and ARIMA models, which are the workhorses of forecasting and the bulk of what follows.

A **lag** is the separation between two observations measured in time units. If two observations are h periods apart, the lag is h .

Plotting a series against its own first lag is the simplest possible diagnostic for dependence.

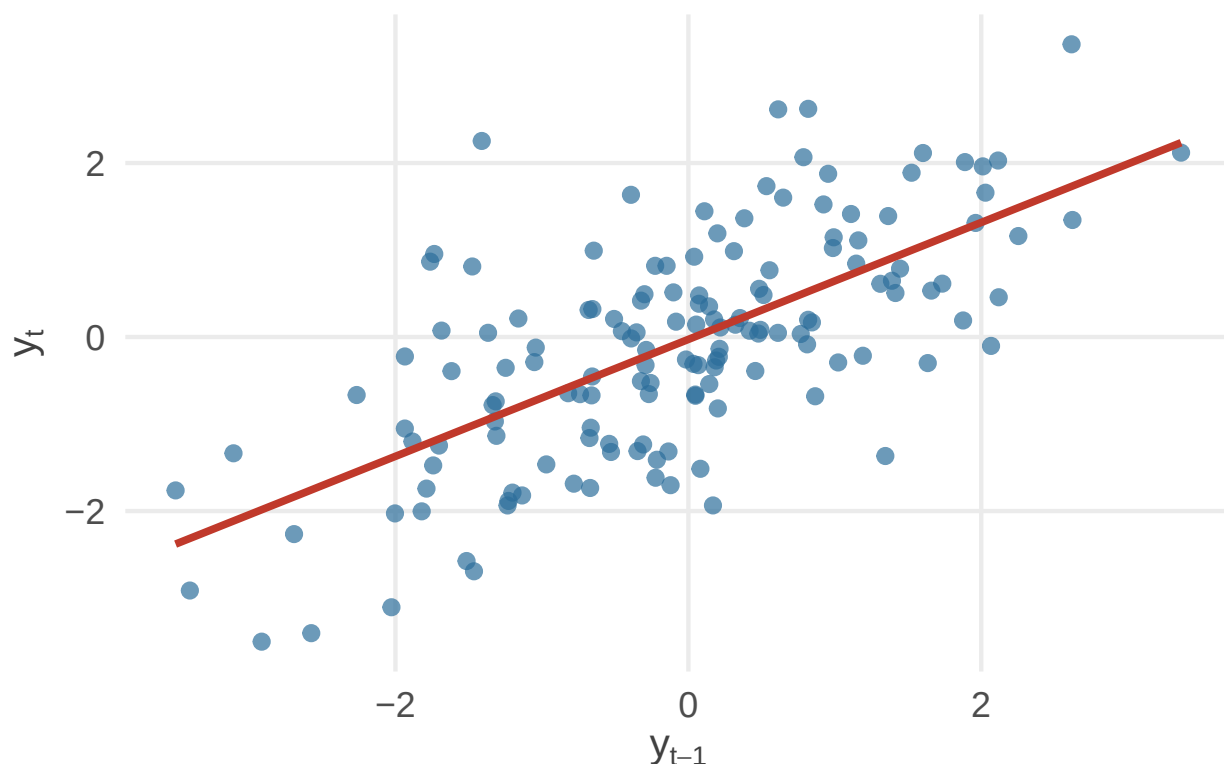


Figure 13: A stationary AR(1) series with $\phi = 0.7$, plotted against its own previous value. The slope of the fitted line estimates ϕ .

Observations that were large in one period tend to remain large in the next. That is the persistence which lagged models are built to capture, and the lag operator introduced shortly is the notation for it. A stationary series is used deliberately here: on a series with a unit root this scatterplot still slopes upward, but the fitted slope is not a consistent estimate of anything useful, which is one reason such series are differenced before modelling.

3.1.2 The frequency domain

Rather than asking what happened yesterday, the frequency domain asks which cycles are present, and how much of the variation each of them accounts for.

The **period** m is the number of observations in one complete cycle. The **frequency** is $f = 1/m$, the number of cycles per observation.

For a pattern repeating every twelve months, $m = 12$ and $f = 1/12 \approx 0.083$, so the series completes about 8.3 per cent of a cycle each month. Throughout this chapter, **frequency** refers to **cycles per observation**, which is the standard convention for discrete-time series. In later topics, particularly spectral analysis, frequency is often expressed as an angular frequency, $\omega = 2\pi/m$, measured in radians per observation. The two representations describe the same periodic behaviour using different scales. Recall from the conventions above that R's **frequency** argument is m , not f .

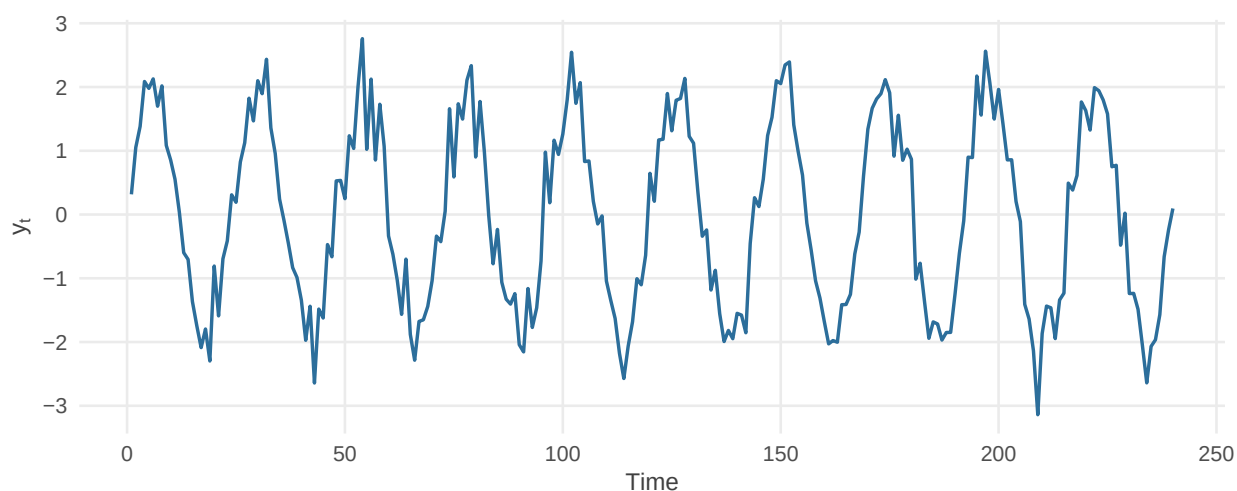


Figure 14: A cycle of period 24 buried in noise. Recovering it is the business of the frequency domain.

3.1.3 Comparing the two

Characteristic	Time domain	Frequency domain
Focus	Dependence	Cycles
Typical question	Does the past predict the future?	Which oscillations exist?
Common tools	ACF, PACF, ARIMA	Periodogram, spectrum
Direct forecasting	Yes	Indirect

Table 8: What each perspective is good at

It is worth being clear about what this contrast is not. The two domains are not different data, nor competing theories about the series; the periodogram is built by re-expressing the same n observations as a sum of sines and cosines of different frequencies, a change of representation rather than a change of subject, made precise in the Spectral Analysis chapter later in these notes. Nothing is gained or lost in the switch, only which questions become easy to read off directly.

Daily electricity demand contains short term persistence, weekly and annual seasonality, and a long run trend all at once. No single perspective tells the whole story, which is why seasonal decomposition, spectral analysis and dynamic harmonic regression routinely draw on both.

3.2 The components of a series

Never fit a model before you have looked at the data.

Because ordering carries the information, graphical exploration comes before any formal model. Many series can usefully be decomposed as

$$y_t = T_t + S_t + C_t + I_t,$$

with the pieces set out in Table 9. Note that T_t here is the trend component, not the forecast origin T .

Symbol	Component	Description
T_t	Trend	Long run direction of movement
S_t	Seasonality	Repeating pattern of fixed, known length
C_t	Cycle	Long oscillation of variable length
I_t	Irregular	What is left once the others are removed

Table 9: The classical decomposition

Where the seasonal swing grows with the level of the series, a multiplicative form $y_t = T_t \times S_t \times C_t \times I_t$ fits better. Taking logarithms turns that back into the additive case, which is why log transformations appear so often in practice.

3.2.1 Trend

A **trend** is the long run direction of movement in a series, once short term fluctuation is set aside.

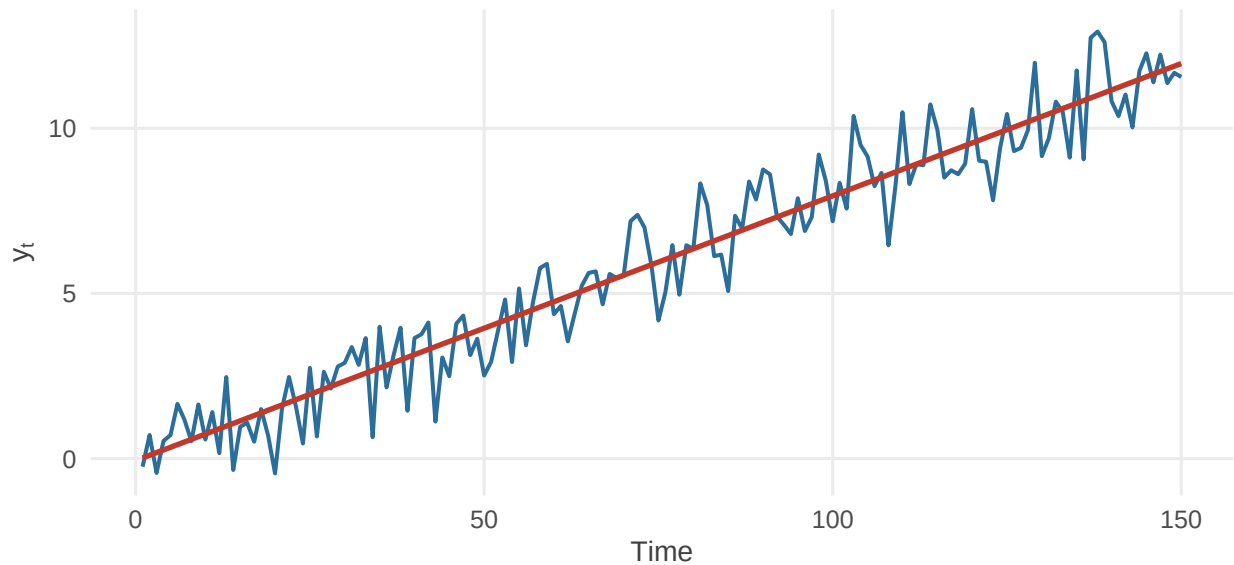


Figure 15: A linear trend with noise around it.

3.2.2 Seasonality

Seasonality is a pattern that repeats at a fixed and known interval, such as the month of the year or the day of the week.

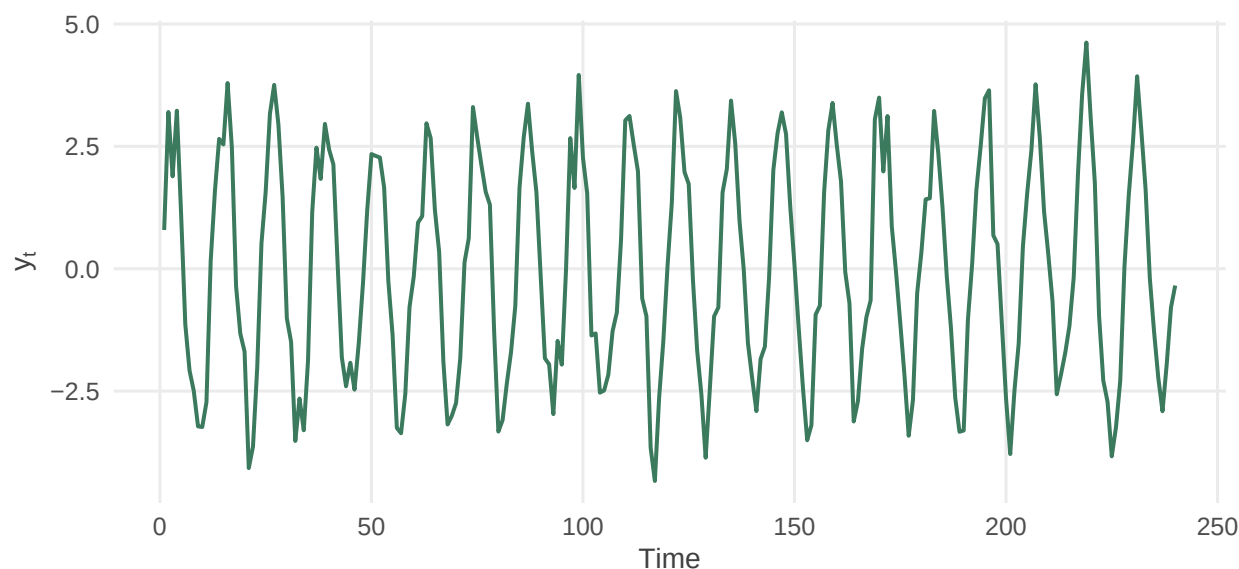


Figure 16: Seasonality of period 12. Successive peaks are equally spaced.

The constant spacing between peaks is the defining feature, and it is what separates seasonality from a cycle. A weekly pattern in daily data is therefore seasonality with $m = 7$, not a cycle, because its length is fixed by the calendar.

3.2.3 Cycles

A **cycle** is a long oscillation whose length is not fixed. The business cycle is the classic example.

Seasonality	Cycle
Fixed period	Variable period
Driven by the calendar	Driven by the process
Highly regular	Much less regular

Table 10: Seasonality against cycles

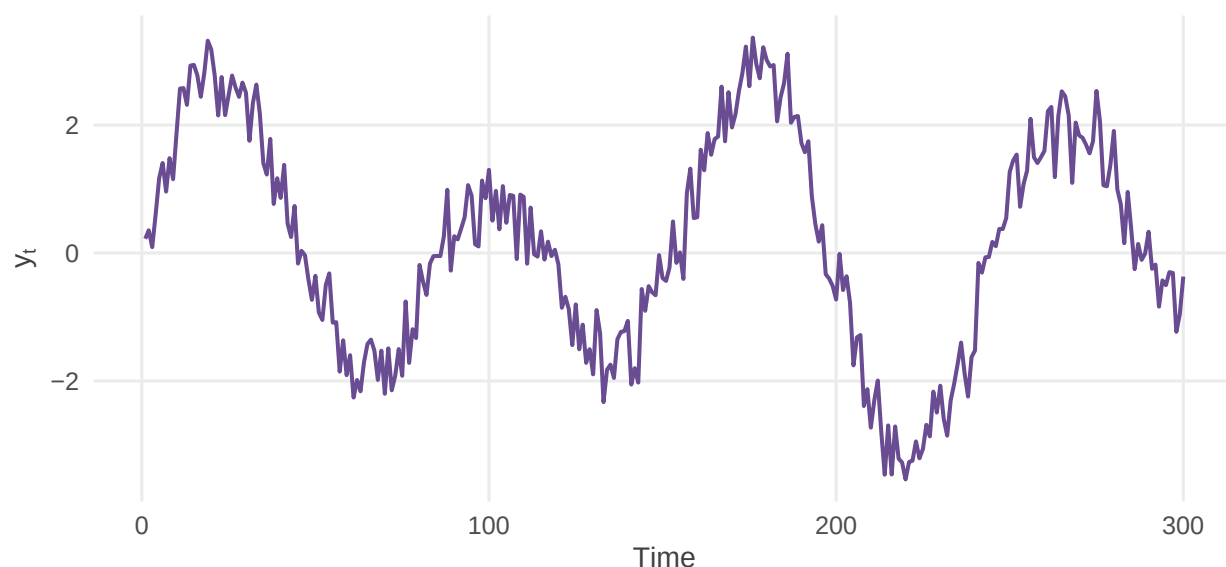


Figure 17: Two oscillations of different lengths superimposed. The peaks are no longer evenly spaced.

3.2.4 Irregular variation

The **irregular component** is the random variation left unexplained by trend, seasonality and cycle. It is usually called noise.

Irregular variation is treated at length in the section on white noise below, since it turns out to be the building block from which every model in the chapter is assembled rather than merely a residue.

3.2.5 The components together

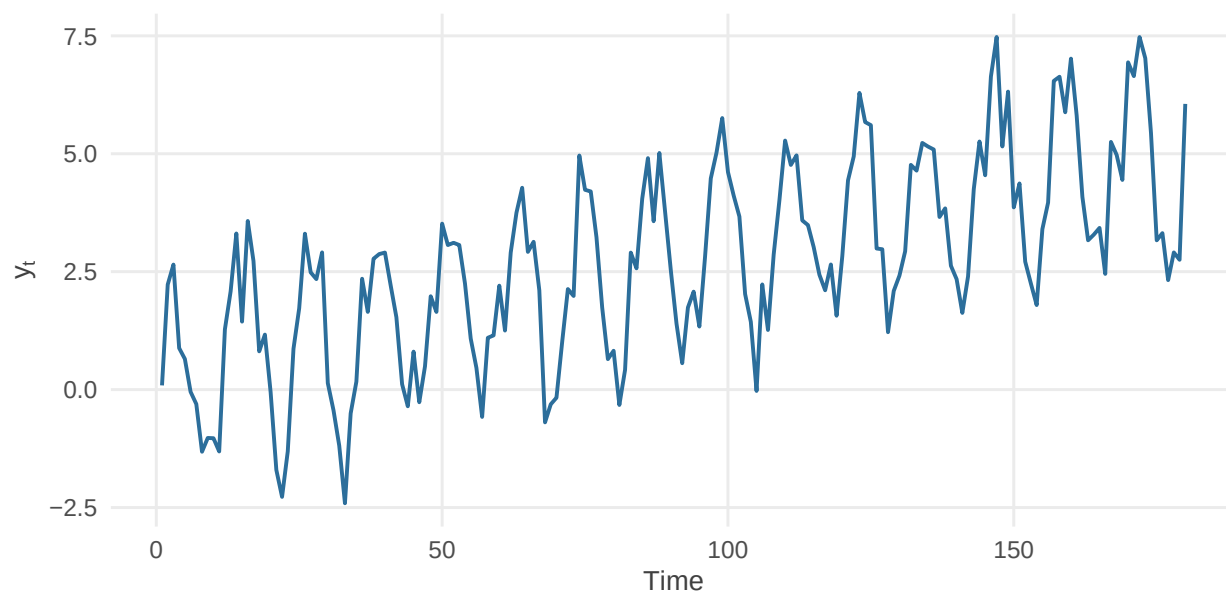


Figure 18: Trend, seasonality and noise superimposed, which is what most real series look like.

3.2.6 Common trend forms and mechanisms

Recognising both the **shape** of a trend and the **mechanism** that generates it is important because they influence the choice of model. The visual appearance of a trend often suggests an appropriate modelling strategy, while the underlying mechanism determines whether detrending alone is sufficient or whether differencing is required. The first table classifies trends by their observable form. The second distinguishes between deterministic and stochastic trends, which often appear similar over short periods but have fundamentally different statistical properties.

Trend form	Characteristic	Typical examples
Flat	Mean remains approximately constant	Resting heart rate, machine operating temperature
Linear	Constant additive rate of change	Straight line depreciation, fixed rate savings
Exponential	Rate of change proportional to the current level	Early epidemic spread, compound interest
Damped	Growth slows progressively towards a ceiling	Market penetration, enzyme reactions
Polynomial	Curvature changes direction over time	Product life cycles, technology adoption and decline
Trend with waves	Long term movement combined with cyclical fluctuations	Property markets, macroeconomic activity

Table 11: Common trend forms encountered in practice

Trend mechanism	Description	Typical examples
Deterministic trend	The trend follows a specified mathematical function with random noise around it. Once the trend is removed, the remaining series is typically stationary.	Linear regression trend, exponential growth with random error
Stochastic trend	The trend evolves through the accumulation of random shocks. Removing a fitted trend is generally insufficient; differencing is usually required to achieve stationarity.	Random walk, random walk with drift, many financial price series

Table 12: Two fundamentally different mechanisms that generate trends

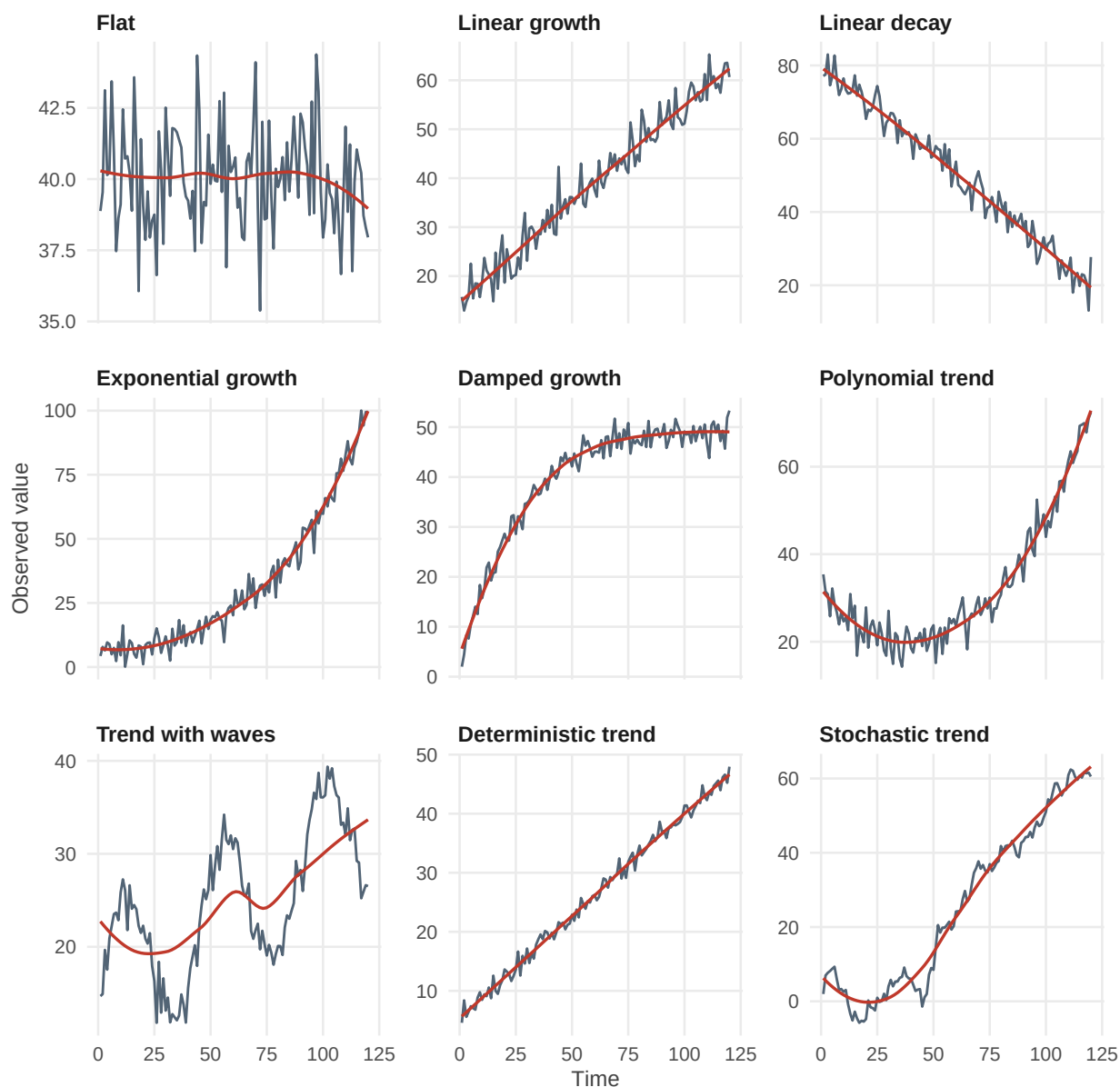


Figure 19: Examples of common trend forms together with deterministic and stochastic trend mechanisms. The red line is a loess smooth.

The first seven panels illustrate **observable trend forms** that are commonly encountered in practice. These are classified according to the way the series evolves over time and provide an initial guide to the type of model that may be appropriate.

The final two panels illustrate **trend mechanisms** rather than additional trend forms. A deterministic trend is generated by a known mathematical function with random deviations around it. Once the underlying function is removed, the remaining series is typically stationary. In contrast, a stochastic trend is driven by the accumulation of random shocks. Although it may resemble a deterministic trend over a limited time period, its statistical behaviour is fundamentally different, and differencing is usually required before models such as ARIMA can be fitted.

Distinguishing between deterministic and stochastic trends is one of the central objectives of stationarity analysis. The unit root tests mentioned in the stationarity section below, and treated in detail later in the course, are designed specifically to determine which of these mechanisms is

generating the observed series.

3.2.7 Seasonality shapes met in practice

Whether seasonality is additive, multiplicative or drifting decides whether a log transformation, a seasonal difference, or both are required.

Shape	Characteristic	Examples
Additive	Amplitude constant whatever the level	Residential electricity use, commuter rail volumes
Two peaks	Two maxima per cycle	University enrolments, restaurant trade
Single spike	One abrupt event each cycle	Black Friday sales, festival travel
Multiplicative	Amplitude grows with the level	Airline passengers, subscription billings
Shifting	Timing of the peak moves over time	Harvest dates, solar generation curves
Fading	Seasonal effect weakens over time	Novelty product sales

Table 13: Seasonality shapes and where they occur

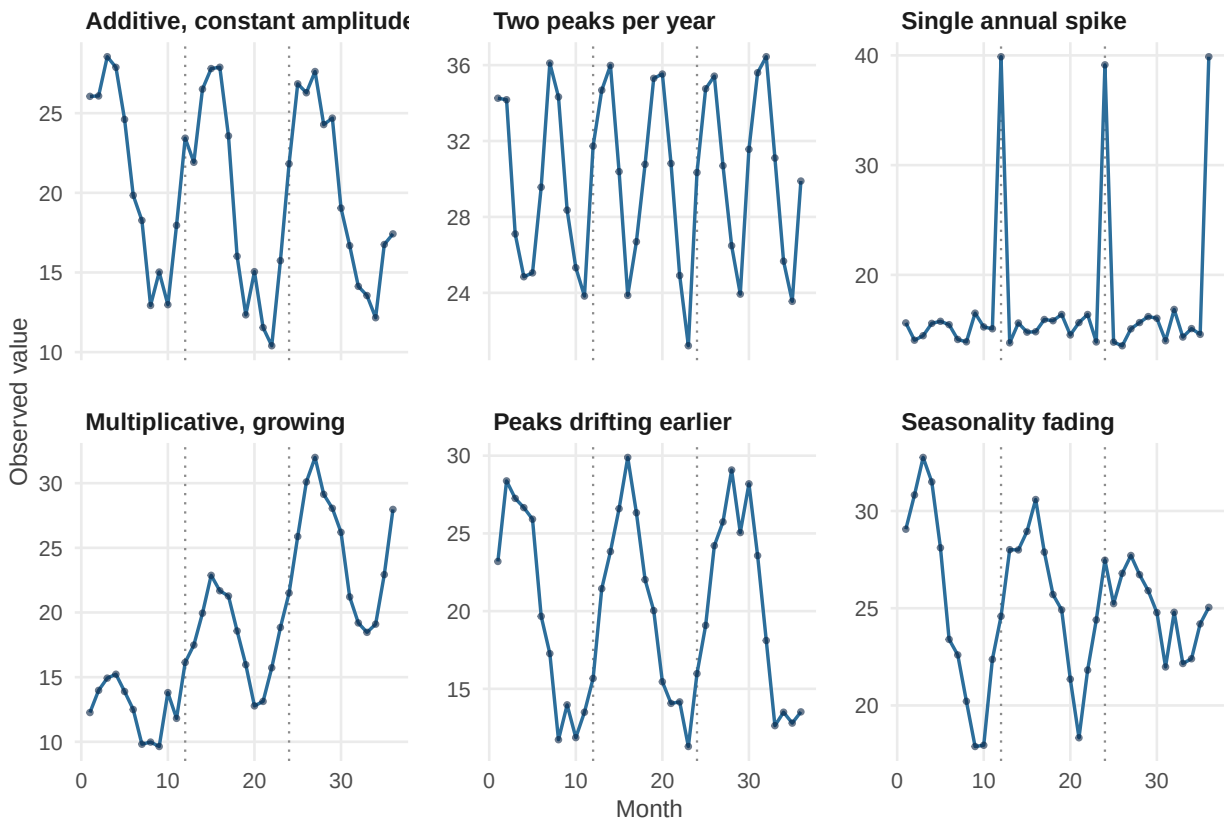


Figure 20: Six seasonal shapes over three years of monthly data. Dotted lines mark the year boundaries.

3.3 Reading a time plot

A time plot shows (t, y_t) joined by line segments. Before any model is fitted it should answer four questions: is the series trending, is it seasonal, is the variability stable, and are there observations that do not belong with the rest?

3.3.1 Changing variability

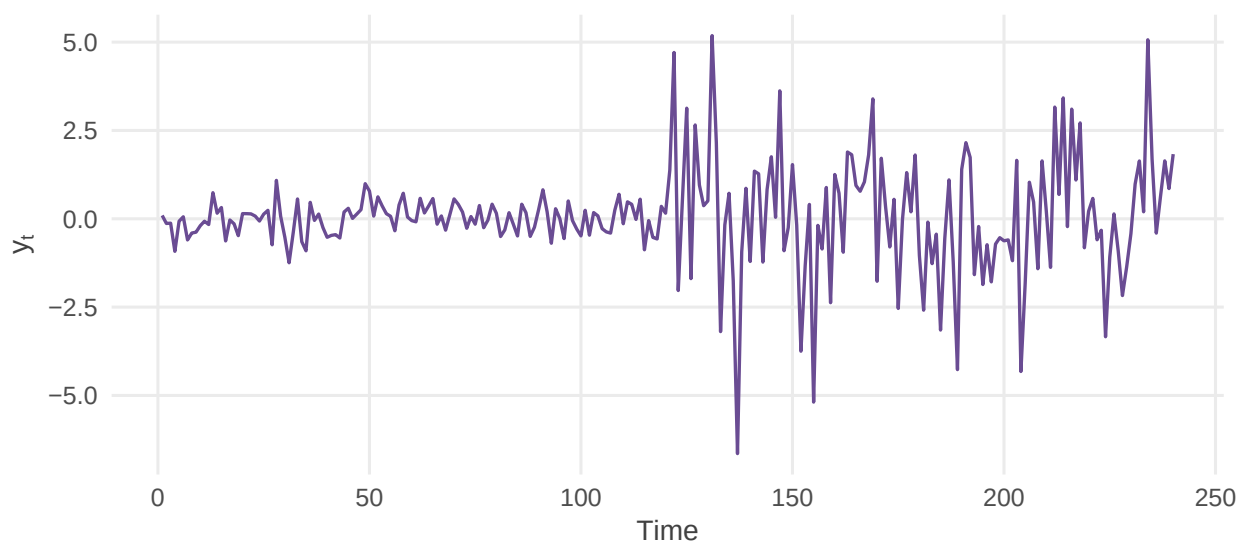


Figure 21: The mean is stable but the spread widens sharply after observation 120.

This is heteroscedasticity, and it is common in financial data. A log or square root transformation often stabilises it. When the variance moves in a persistent, forecastable way rather than simply jumping once, the ARCH and GARCH models at the end of the chapter are the proper response.

3.3.2 Outliers

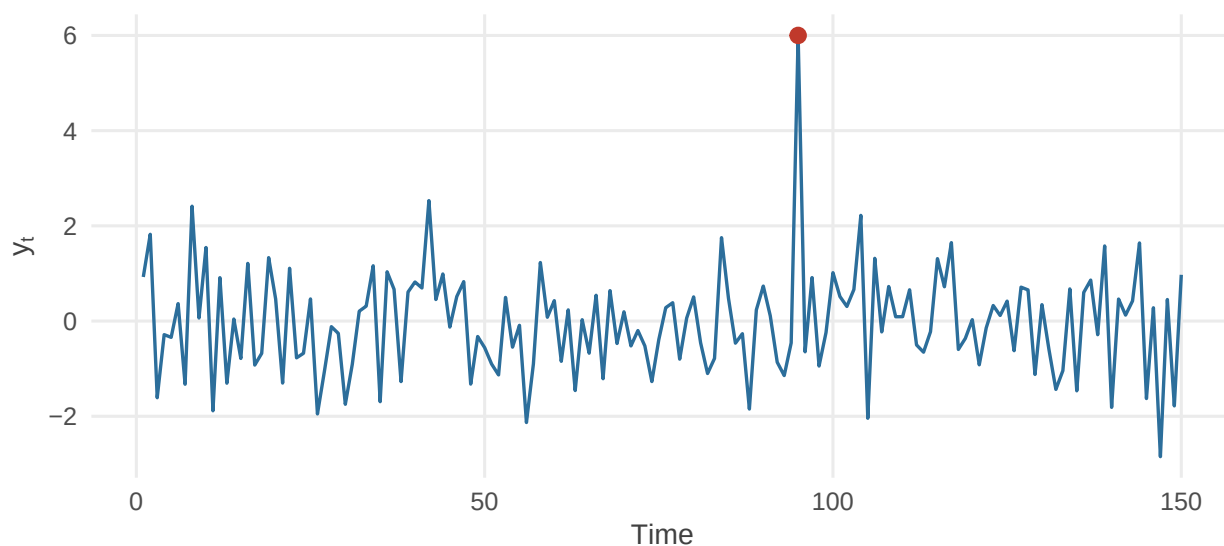


Figure 22: A single observation far outside the range of its neighbours.

Recording errors, equipment failure and genuine rare events all look like this on the page. Which of the three it is cannot be settled from the plot, so an outlier is a prompt to go back to the source rather than a licence to delete a point.

3.3.3 Structural breaks

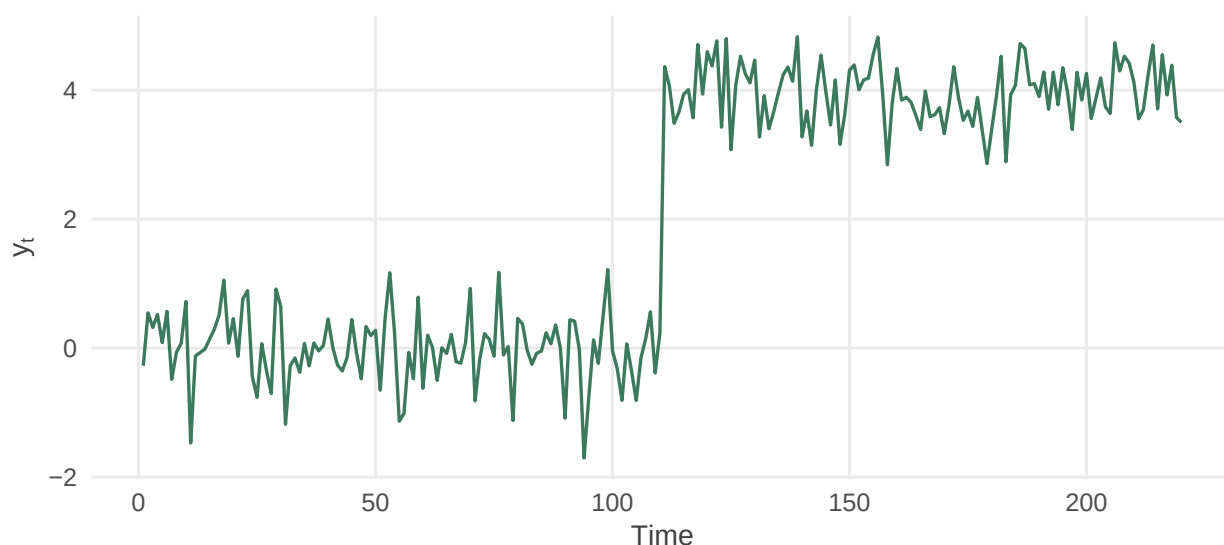


Figure 23: The level shifts abruptly after observation 110 and does not return.

Policy changes, crises and technological shifts produce breaks of this kind. Fitting a single model across a break is a reliable way to obtain a model that describes neither regime.

3.3.4 Four archetypes side by side

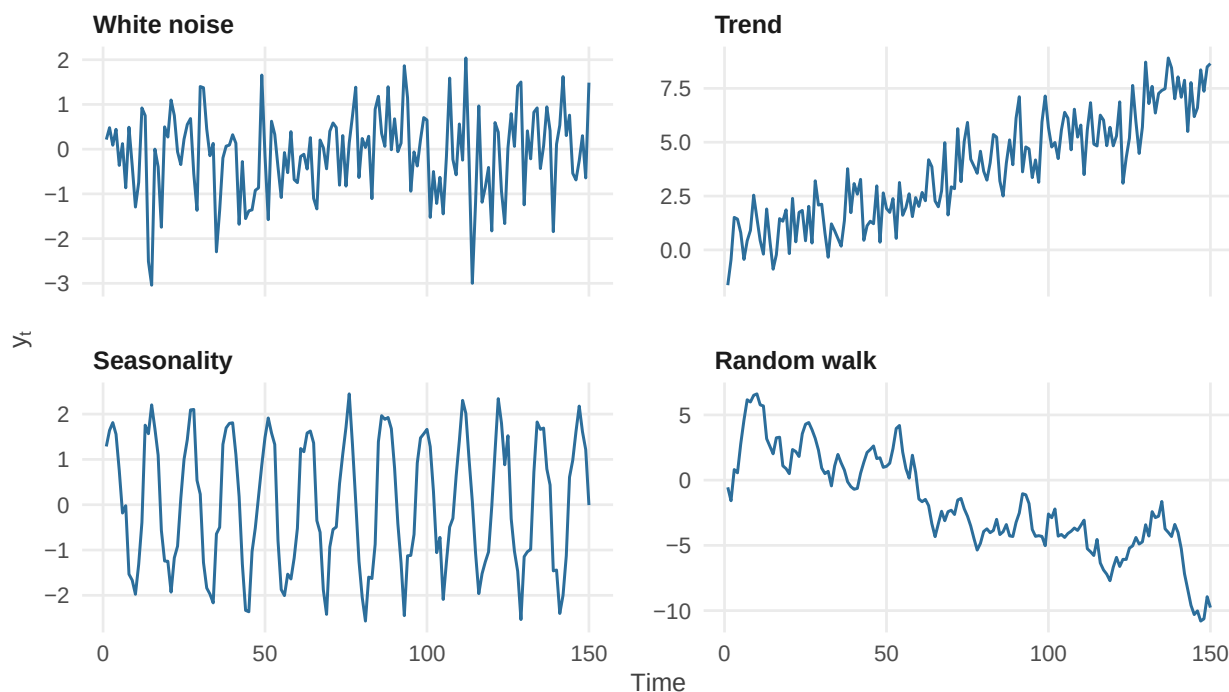


Figure 24: Four patterns worth being able to recognise instantly.

White noise has no systematic pattern. A trend moves gradually in one direction. Seasonality repeats on a fixed schedule. A random walk wanders without returning to any particular level, which is what makes it look as though it has a trend when it has none.

3.3.5 A working routine

The rest of this course follows one recurring cycle: **visualise, transform, check assumptions, identify, estimate, diagnose, forecast, evaluate, iterate**. At this stage only the first few links in that chain are available, and they are enough to be worth applying properly. Produce a time plot. Identify the dominant features. Describe them numerically rather than only in words. Decide whether a transformation or differencing is needed. Only then fit a model; the remaining links, estimating, diagnosing, forecasting and evaluating against the benchmarks of the next chapter, close the loop and feed directly into the next iteration.

Monthly retail sales rise over ten years, peak every December, and fall sharply for one pandemic year.

Trend: steady growth. *Seasonality*: a December peak of fixed timing. *Irregular*: the pandemic year, which is better treated as a structural episode than as an outlier. The plot alone already tells you the model must handle trend and seasonality together, which points to the seasonal models later in this chapter.

4 Benchmarks and Forecast Accuracy

4.1 Benchmark forecasting methods

Any proposed model has to earn its place against forecasts that require no modelling at all. Each benchmark is simply a different, explicit guess at $E(y_{T+h} | \mathcal{I}_T)$, built on a particular simplifying assumption about how y_{T+h} relates to the history $\mathcal{I}_T$. These four benchmarks are the floor, and a model that cannot clear it has told you something useful about itself.

Method	Forecast for all h	Suited to
Mean	$\hat{y}_{T+h T} = \bar{y}$	Series with no trend or season
Naive	$\hat{y}_{T+h T} = y_T$	Random walks, prices
Seasonal naive	$\hat{y}_{T+h T} = y_{T+h-m(k+1)}$	Strongly seasonal series
Drift	$\hat{y}_{T+h T} = y_T + \frac{h(y_T - y_1)}{T - 1}$	Series with a steady trend

Table 14: The four benchmark methods

In the seasonal naive formula, m is the length of the season and

$$k = \left\lfloor \frac{h-1}{m} \right\rfloor$$

is the number of complete seasons that fall strictly before the target period, so the method simply repeats the most recent observed season. The floor is what makes the formula behave correctly at $h = m$: with quarterly data and $h = 4$, $k = 0$ and the forecast is y_T , the same quarter one year earlier, not y_{T-4} .

The drift method is the naive forecast with a slope attached, equal to the average change per period over the observed history: drawing a straight line from the first observation to the last and extending it gives the same answer. Since forecasts are made from the end of the sample, $T = n$ and the denominator is the number of changes observed.

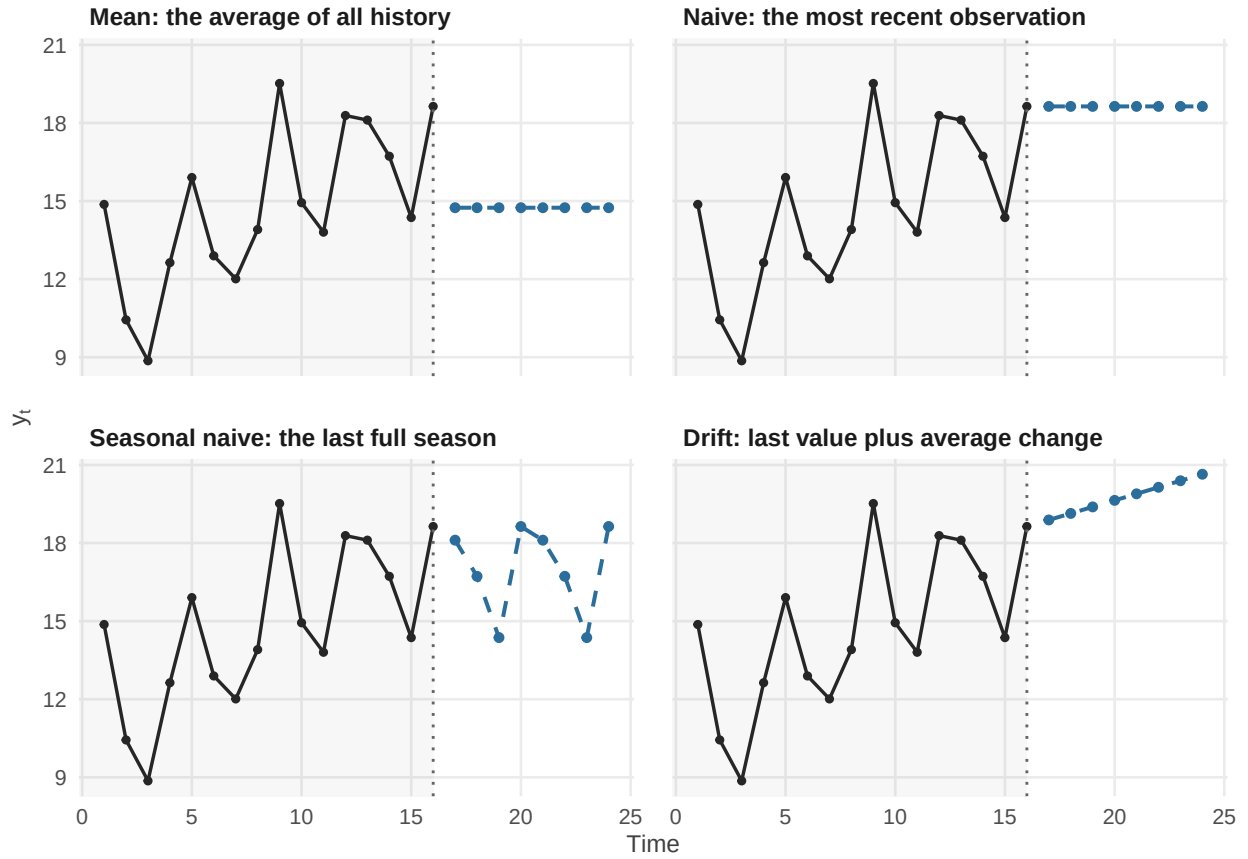


Figure 25: The four benchmarks applied to one series. Solid black is the history, dashed blue the forecast, and the dotted line marks the forecast origin.

The shaded region is the training sample and the unshaded region the forecast horizon. Notice how differently the four methods extrapolate the same sixteen observations, and that only the seasonal naive method carries the cycle forward. For a series with both trend and season, none of the four is adequate on its own, which is the gap the rest of the chapter fills.

4.2 Evaluating forecast accuracy

Saying that a model must beat the benchmarks is empty until we say beaten on what, and measured how. Two distinctions do most of the work.

4.2.1 Residuals are not forecast errors

A **residual** is an in-sample quantity, $y_t - \hat{y}_t$, where $\hat{y}_t$ is the one step fitted value computed with the model estimated on all the data. A **forecast error** is an out-of-sample quantity, $e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$, where the model never saw y_{T+h} .

A model with enough parameters can drive its residuals close to zero and still forecast badly, because small residuals may only mean the model has memorised the training sample. Residuals are used for *diagnostics*, to check that no structure has been left behind. Forecast errors are used for *evaluation*, to check that the model is useful. The two questions are separate, and a model can pass one and fail the other.

4.2.2 Training and test sets

The standard arrangement holds out the last portion of the series as a test set, fits every candidate on the training set only, and compares forecasts against the held out values. The test set should be at least as long as the largest horizon of interest, and never used to choose between models more than once, since repeated peeking turns it into a second training set.

Note that the split must respect time. Randomly sampling observations into training and test sets, as in cross-sectional work, lets future observations influence the fitted model before the forecasts they are meant to test are even made. The resulting accuracy figures look better than they have any right to; this failure has a name, **temporal leakage**, and it is examined properly once the rolling-origin schemes below give us the vocabulary to state the rule in general.

4.2.3 Measures

Measure	Definition	Notes
MAE	$\text{mean}(e_t)$	Scale dependent; optimal forecast is the median
RMSE	$\sqrt{\text{mean}(e_t^2)}$	Scale dependent; optimal forecast is the mean; penalises large errors
MAPE	$\text{mean}(100 e_t/y_t)$	Scale free, but undefined at $y_t = 0$ and asymmetric
MASE	$\text{mean}(e_t)/Q$	Scale free; Q is the in-sample MAE of the naive or seasonal naive forecast

Table 15: Common accuracy measures

RMSE is never smaller than MAE, and the gap between them widens as the error distribution becomes more uneven. MAPE is popular and troublesome: it cannot be used on series that pass through zero, and it penalises forecasts that are too high more heavily than those that are too low, which quietly biases method selection downward. MASE avoids both problems, and has the convenient reading that a value below one means the method beat the naive benchmark on the training data.

Method	RMSE	MAE	MAPE	MASE
Mean	6.51	5.86	18.33	3.37
Naive	3.60	3.02	9.38	1.74
Seasonal naive	2.23	1.96	6.30	1.13
Drift	3.31	2.89	9.13	1.66

Table 16: Out of sample accuracy of the four benchmarks on the last two years of a simulated monthly series.

The seasonal naive method wins here, which is exactly what the construction of the series would

lead you to expect. Any ARIMA or SARIMA model proposed later in the chapter has to beat that number on this test set before it is worth the extra parameters.

4.2.4 Rolling-origin evaluation: time series cross-validation

A single train and test split uses one forecast origin, so the comparison in Table 16 rests on one draw: shift the split point by even a few months and the ranking of methods could change. **Time series cross-validation**, also called **rolling-origin evaluation**, removes this dependence on a single split by repeating the fit, forecast and compare cycle at many origins and pooling the errors. It is the forecasting analogue of k -fold cross-validation, but it cannot be built the same way. Ordinary k -fold cross-validation assigns observations to folds at random, which is harmless when observations are exchangeable. Time series observations are not exchangeable: shuffling them before splitting would routinely place future values in a training fold and past values in a validation fold, so a model could be fitted using information that would not yet exist at the point a genuine forecast is made. This is temporal leakage stated in general form, and the rule that prevents it is easy to state and easy to forget under time pressure.

A validation scheme exhibits **temporal leakage** (equivalently, **look-ahead bias**) whenever any observation used to fit a model was recorded later than an observation the model is judged on. A scheme is valid only if, for every model fitted during the exercise,

$$t_{\text{train}} < t_{\text{validate}}$$

holds for every observation in the training set and every observation in the corresponding validation set. Forecast accuracy computed from a leaking scheme is optimistic in a way that never survives contact with real forecasting, because the future is never actually available when a forecast is made.

Rolling-origin evaluation satisfies this rule by construction. For a sequence of origins $T = T_0, T_0 + 1, \dots, n - h$, a model is fitted using only observations up to time T , used to forecast h steps ahead, and the resulting error

$$e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$$

is stored before the origin advances by one period and the cycle repeats. Averaging e_{T+h} , or a function of it such as its square, over every origin gives a considerably less noisy picture of accuracy than any single split, at the cost of fitting the model many times rather than once.

Two ways to grow the training set. Every origin follows the recipe above; origins differ only in what happens to the training set as T advances.

Under an **expanding window**, every observation collected since $t = 1$ is retained, so the training set at origin T is $\{y_1, \dots, y_T\}$ and simply grows by one observation each time the origin moves forward. This is what practitioners mean by *walk-forward validation*: the two names describe the same procedure, one emphasising what is kept, the other how the origin moves. Under a **sliding (rolling) window** of fixed length w , the training set is $\{y_{T-w+1}, \dots, y_T\}$: it always contains exactly w observations, and each step drops the oldest observation as it admits the newest.

Neither scheme dominates the other; the choice is a bias-variance trade-off. An expanding window uses the complete history, so parameter estimates become steadily more stable as the sample grows,

and it suits a process that is stable or changes only slowly, since old and new observations are equally informative about the parameters. A sliding window deliberately discards old information, which costs some estimation efficiency, in exchange for a fitted model that tracks structural change: in a market, or any system subject to regime shifts, observations from several years back may describe a world that no longer exists more than the one being forecast. The width w is itself a choice along the same trade-off, narrower windows adapting faster but more variably, wider windows more stable but slower to respond, in direct parallel to the smoothing span of a moving average.

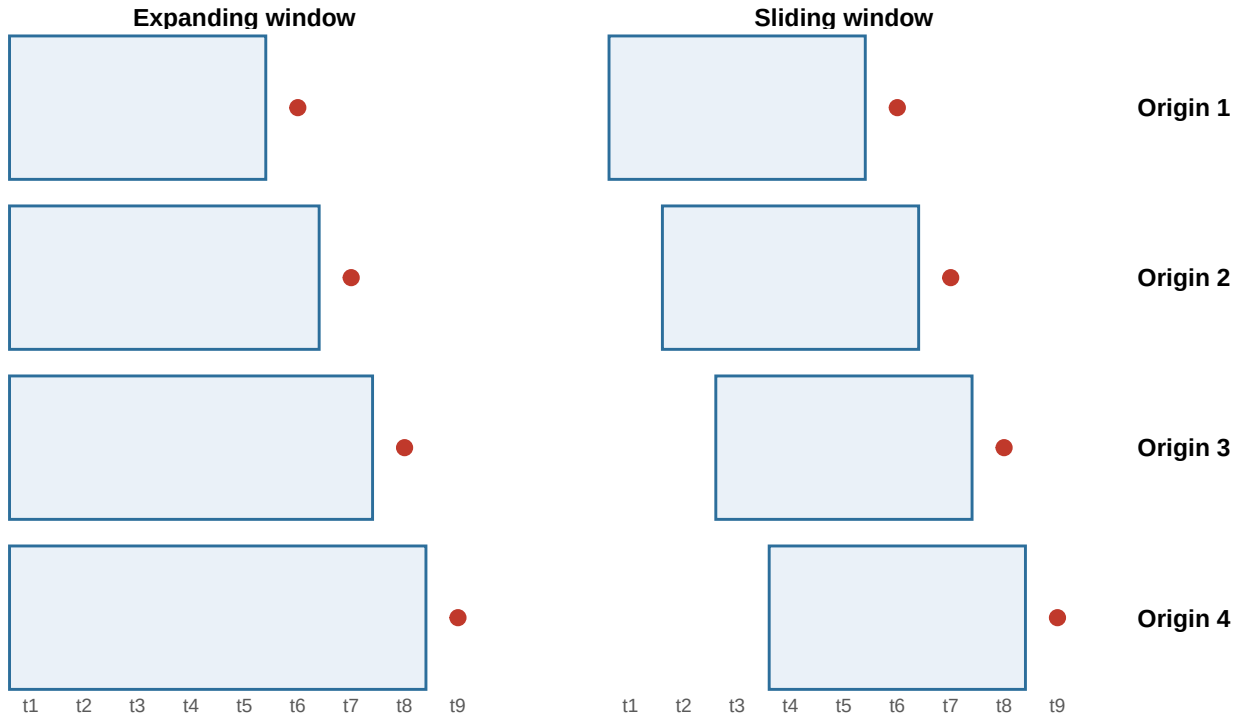


Figure 26: Four successive origins under an expanding window (left) and a sliding window of width four (right). The shaded block is the training sample; the point immediately to its right is the observation being forecast.

Under both schemes the training block always ends immediately before the point being forecast; only the left edge differs, fixed at the start of the series for the expanding window and moving forward in step with the origin for the sliding window. Walk-forward validation, in the sense used above, is exactly the expanding-window panel on the left.

Implementation. The `tsCV()` function in the `forecast` package carries out rolling-origin evaluation for a user-supplied forecast function, returning a matrix of errors with one row per origin and one column per horizon. By default the training set expands; passing `window = w` fixes it at length w and produces a sliding window instead. Applying it to the seasonal naive method on the series used throughout this section,

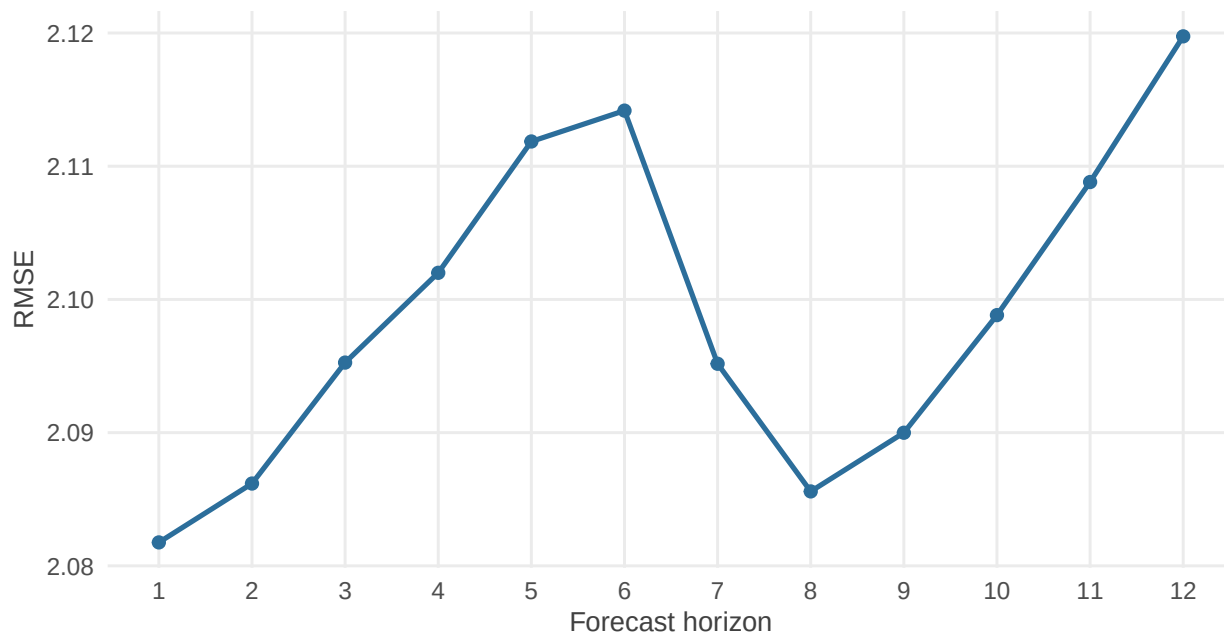


Figure 27: Expanding-window cross-validated RMSE of the seasonal naive method, by forecast horizon, for the series used in Table Table 16.

RMSE rises with horizon, as it must once a process has any autocorrelation to lose: predicting one month ahead uses recent information that thirteen months ahead cannot. The same call with `window = 60` fixes the training sample at the most recent sixty months at every origin rather than letting it grow, giving a sliding window comparison against the figure above:

```
cv_errors_slide <- tsCV(y_all, f_snaive, h = 1, window = 60)
rbind(
  Expanding = sqrt(mean(cv_errors[, 1]^2, na.rm = TRUE)),
  Sliding   = sqrt(mean(cv_errors_slide^2, na.rm = TRUE))
)
#>           [,1]
#> Expanding 2.081761
#> Sliding   1.875638
```

For a series this stable, the two give similar one-step RMSE; the gap would widen on a series with a genuine structural break, where the sliding window's willingness to forget old observations becomes an advantage rather than a cost.

Question to ask	Favours
Is the process stable, with a long, informative history?	Expanding window
Are structural breaks or regime changes plausible?	Sliding window
Are observations arriving continuously and forecasts updated every period?	Expanding window (walk-forward)
Is computation cheap enough to refit at every origin?	Either; otherwise, a single split

Table 17: Practical guidance for choosing a rolling-origin scheme

Limitations. Rolling-origin evaluation is not free. Refitting at every origin multiplies computational cost roughly by the number of origins, which matters once the model being evaluated, an ARIMA order chosen by `auto.arima()` at every step, for instance, takes appreciable time to fit on its own. Errors from neighbouring origins are also computed from overlapping stretches of data and are consequently correlated, so the usual formula for the standard error of a mean overstates how much independent information the exercise has actually gathered; treat cross-validated accuracy as considerably more reliable than a single split, without treating the origins as independent replicates. Finally, cross-validation answers the question of how a method has performed across the available history, which is not automatically the same question as how it will perform going forward if the underlying process is itself changing; a sliding window with a deliberately short w is one way of asking a more forward-looking version of the question. A single split remains a reasonable, cheap first screen when a shortlist of candidate models is long; rolling-origin evaluation earns its cost once that shortlist, and the decision resting on it, is short and important enough to justify the extra fitting.

5 Operators, Stationarity and White Noise

5.1 Lag and lead operators

5.1.1 Why an operator is worth the trouble

Most things measured over time carry some memory of their own past. Today's rainfall is easier to interpret once you know what fell yesterday. This quarter's GDP figure is read against last quarter's. A share price tends to sit near where it closed the day before.

Variable	Usually depends on
Rainfall	The previous day's rainfall
Temperature	Yesterday's temperature
Gross domestic product	The previous quarter
Inflation	The previous month
Exchange rate	The previous rate
Share prices	The previous close
Recorded crime	The previous month's level

Table 18: Variables and their usual dependence

Once models start connecting a variable to its own past, spelling out phrases like “the value two periods ago” becomes tiresome and error prone. The shorthand for it is the **lag operator**. When the question runs the other way, as it does in forecasting, the matching tool is the **lead operator**.

Neither is complicated. What takes getting used to is that they do not touch the numbers in a series at all. They change which time label a number sits under. Almost every mistake made with this notation comes from losing hold of that distinction.

5.1.2 A series to work with

Everything below uses the same five numbers,

$$y_t = \{3, 7, 4, 8, 5\}, \quad t = 1, \dots, 5.$$

Time t	1	2	3	4	5
Original y_t	3	7	4	8	5
Lagged Ly_t	NA	3	7	4	8
Led Fy_t	7	4	8	5	NA

Table 19: The same five numbers under three sets of time labels

5.1.3 The lag operator

The **lag operator** is defined by $Ly_t = y_{t-1}$: at time t , use the observation recorded at time $t - 1$.

Reading along the middle row of Table 19, every value has moved one column to the right. The first slot has nothing to draw on, since there is no observation at $t = 0$, so it comes out missing. Checking the definition against the table, $Ly_2 = y_1 = 3$, and indeed the value 3, originally recorded at $t = 1$, appears under $t = 2$.

It is worth pausing on why a formula containing $t - 1$ produces a series that shifts to the right rather than the left. The subscript records where a value came from, not where it is displayed. Those are two different things.

5.1.4 The lead operator

The **lead operator** is defined by $Fy_t = y_{t+1}$: at time t , use the observation recorded at time $t + 1$. It is the inverse of the lag operator, $F = L^{-1}$.

In the last row of Table 19 every value has moved one column left, and it is the final slot that runs out of information, since there is no observation at $t = 6$. So $Fy_2 = y_3 = 4$. Lag and lead are exact mirror images: one borrows from the past, the other from the future.

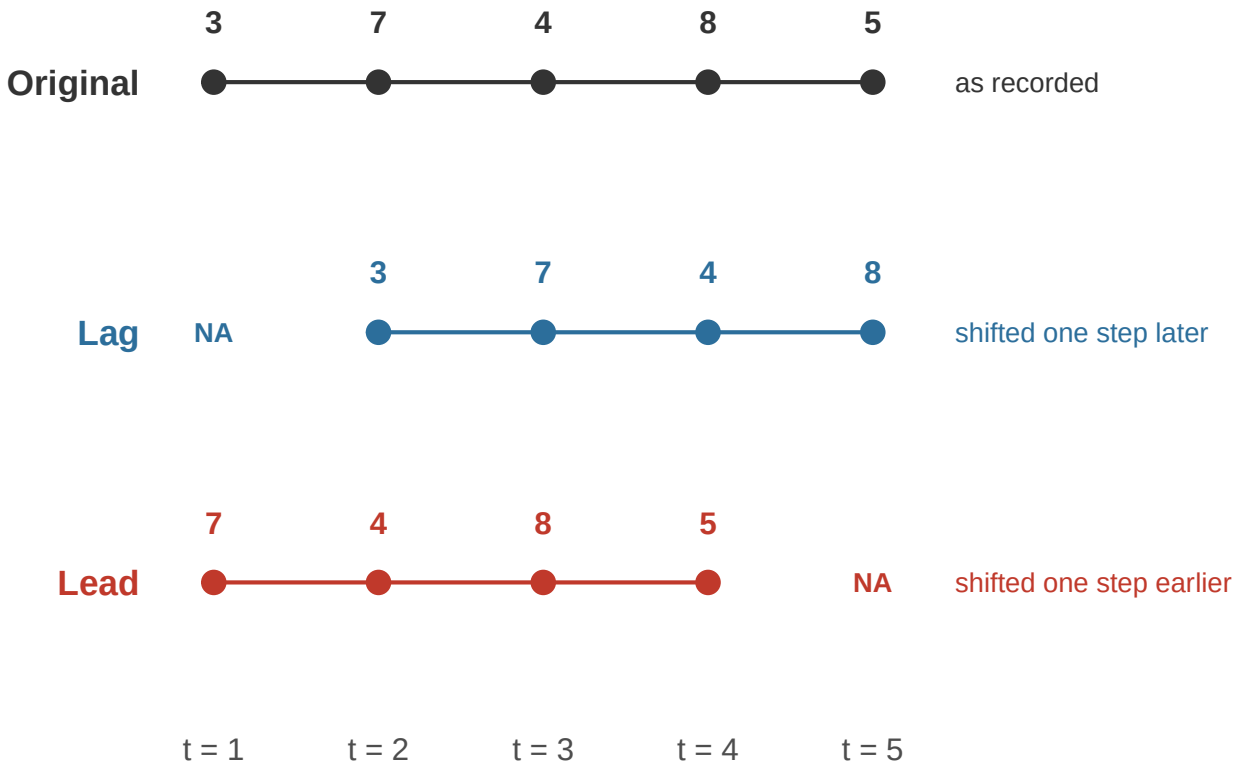


Figure 28: The same five observations under three sets of time labels. The values never change; only the label above each one does.

5.1.5 Doing it in R

Lagging is nothing more than shifting a vector along one position and padding the gap with NA. You can write it by hand or use a function that does it for you.

```
y <- c(3, 7, 4, 8, 5)

# By hand in base R.
c(NA, y[1:(length(y) - 1)])
#> [1] NA 3 7 4 8

# The same shift, using dplyr.
dplyr::lag(y, n = 1)
#> [1] NA 3 7 4 8

# Leading, again both ways.
c(y[2:length(y)], NA)
#> [1] 7 4 8 5 NA
dplyr::lead(y, n = 1)
#> [1] 7 4 8 5 NA
```

The package version is shorter and less error prone once you start lagging by more than one period, but there is no hidden machinery in it.

One warning about names. Base R already supplies `stats::lag()`, which is always attached, and it does something quite different: it shifts the *time base* of a `ts` object rather than inserting NA into a vector, and `stats::lag(x, k = 1)` moves the series **forward** in time, the opposite of `dplyr::lag()`. The `zoo` package supplies a method for the same generic with its own sign handling. Because `dplyr` masks the base generic when it is attached, the safe habit is to write `dplyr::lag()` or `stats::lag()` in full rather than trusting whichever package attached last.

5.1.6 The algebra of L

Once L is defined it behaves enough like ordinary algebra to be stacked, scaled and combined. That is what earns it a place in model notation rather than leaving it a labelling trick.

Repeated application. Applying L twice steps back two periods,

$$L^2 y_t = L(Ly_t) = Ly_{t-1} = y_{t-2},$$

and in general $L^k y_t = y_{t-k}$. This is what lets a twelve month seasonal lag be written L^{12} instead of twelve nested statements.

Lead as inverse. Lagging steps the index back and leading steps it forward, so the two cancel,

$$L(Fy_t) = Ly_{t+1} = y_t,$$

which is the sense in which $F = L^{-1}$. Some texts drop F entirely and write $L^{-1}y_t$.

Linearity. L distributes over sums and pulls out constants, in the same way differentiation does,

$$L(ay_t + bx_t) = ay_{t-1} + bx_{t-1}.$$

Compound operators. Because L can be scaled, added and raised to powers, an expression such as $1 - \phi_1 L - \phi_2 L^2$ is a well formed operator in its own right. It is not shorthand for something simpler. It is a new object built out of L , applied to y_t exactly as L itself was.

```
# L squared, checked two ways.
lag_twice <- dplyr::lag(dplyr::lag(y, n = 1), n = 1)
lag_twice
#> [1] NA NA 3 7 4
dplyr::lag(y, n = 2)
#> [1] NA NA 3 7 4
```

5.1.7 Putting the operator to work

The operator earns its keep as soon as a model references more than one past value. Take the autoregression with two lags,

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t.$$

Moving both lag terms across gives

$$y_t - \phi_1 y_{t-1} - \phi_2 y_{t-2} = \varepsilon_t,$$

which, using $L^k y_t = y_{t-k}$, is exactly

$$(1 - \phi_1 L - \phi_2 L^2)y_t = \varepsilon_t.$$

The same trick collapses the two most common data preparation steps into single symbols. A first difference is

$$\nabla y_t = y_t - y_{t-1} = (1 - L)y_t,$$

and a seasonal difference at lag twelve is

$$\nabla_{12} y_t = y_t - y_{t-12} = (1 - L^{12})y_t.$$

```
# First differences, by hand and with diff().
y2 <- c(10, 6, 9, 12, 7)
y2 - dplyr::lag(y2, 1)
#> [1] NA -4 3 3 -5
diff(y2)
#> [1] -4 3 3 -5
```

Note that `diff()` returns a vector one element shorter rather than padding with NA. Both expressions represent $(1 - L)y_t$; they differ only in whether the missing first value is kept as a placeholder or dropped.

Stack enough of these together and you arrive at the general form every ARIMA model is written in,

$$\phi(L)y_t = \theta(L)\varepsilon_t,$$

where $\phi(L)$ and $\theta(L)$ are polynomials in L built exactly as above. None of this is new mathematics. It is the same substitution you would do by hand, but writing it in L makes the structure of a model visible at a glance: how many lags it has, whether it is differenced, whether it is seasonal.

The lead operator has a smaller but real role. Event studies and forecast evaluation often need $Fy_t - y_t$, the change between the next period and this one, which is the natural object when the question is about what is about to happen rather than what just did.

5.1.8 Where this trips people up

Five mistakes account for most of the confusion.

Forgetting where the value is displayed. The subscript in y_{t-1} records where the value came from. The column it is written under in a lagged series is a different matter.

Losing track of which end goes missing. Lagging always leaves a gap at the start, leading always at the end. Each lag or lead costs exactly one usable observation, which is an easy thing to check.

Treating $\phi(L)$ as a number. An expression such as $1 - \phi_1 L - \phi_2 L^2$ cannot be evaluated on its own. It means something only once applied to a series, in the same way d/dx means nothing until applied to a function.

Reading the exponent as a power. $L^2 y_t = y_{t-2}$, not $(y_{t-1})^2$ and not $2y_{t-1}$. The exponent counts repeated applications of the operator.

Confusing lagging with differencing. Ly_t and $(1 - L)y_t$ are different objects. The first is a shifted copy of the data; the second is a new series of period to period changes. In R, `lag()` and `diff()` correspond to exactly this distinction.

5.2 Stationarity

Every model from here onward assumes, at some point, that the statistical character of the series does not drift. That assumption has a precise name and a precise content, and it is worth stating before it is used rather than after.

A process $\{y_t\}$ is **weakly stationary**, also called second-order or covariance stationary, if for all t and h :

1. $E(y_t) = \mu$, a constant that does not depend on t ;
2. $\text{Var}(y_t) = \gamma(0) < \infty$, also constant;
3. $\text{Cov}(y_t, y_{t+h}) = \gamma(h)$ depends on the separation h only, not on where in time the pair is taken.

A process is **strictly stationary** if the joint distribution of $(y_{t_1}, \dots, y_{t_k})$ is the same as that of $(y_{t_1+h}, \dots, y_{t_k+h})$ for every choice of times and every shift h .

The two notions are not equivalent in either direction. Strict stationarity with finite second

moments implies weak stationarity, but not otherwise, since a strictly stationary process need not have finite variance. Weak stationarity constrains only the first two moments and so says nothing about higher ones; a process can have a constant mean and a constant autocovariance function while its conditional variance moves around dramatically. The GARCH processes at the end of this chapter are exactly that case, and they are the reason the distinction is worth making at this level.

For a Gaussian process the two coincide, because a multivariate normal distribution is completely determined by its mean vector and covariance matrix.

5.2.1 Why the assumption is needed

Three things depend on it. First, the autocorrelation function $\rho(h)$ is only well defined if $\gamma(h)$ does not depend on t ; otherwise there is no single number to estimate. Second, we estimate μ , $\gamma(0)$ and $\gamma(h)$ from time averages over one realisation, which is only legitimate if the process is stationary and, strictly, ergodic, meaning that time averages converge to the corresponding ensemble averages. Third, a forecast projects the observed structure forward, and there is nothing to project if the structure is different tomorrow.

The second point is worth seeing rather than only stating. Imagine thirty weather stations, each recording the same stationary process independently. At any fixed time, averaging *across* the thirty stations gives the **ensemble mean**. A single forecaster, however, has only one station and one long history, and estimates μ by averaging *along* it instead. Ergodicity is the property that guarantees these two very different averaging exercises, one across realisations and one along a single realisation, converge to the same number.

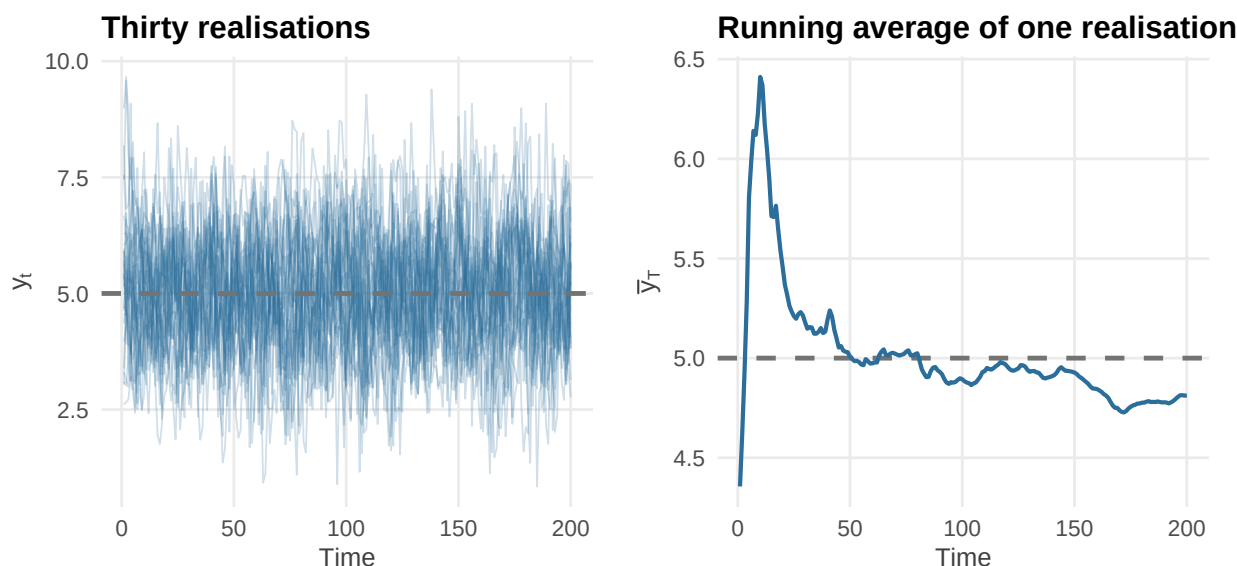


Figure 29: Left: thirty realisations of a stationary AR(1) process with mean 5. Right: the running time average of a single one of those realisations. Both settle on the same value, which is what it means for the process to be ergodic.

The dashed line marks the true mean, 5, and it is the same line in both panels. On the left, every realisation wanders around it independently. On the right, the running average of just one of those realisations, computed from nothing but its own history, homes in on the identical value as more

of the series is observed. This is why a single long series is usable evidence about a process at all: without ergodicity, one history would only ever tell you about itself, and the entire enterprise of estimating μ , $\gamma(0)$ and $\gamma(h)$ from one realisation, which every method in this chapter does, would have no justification. Not every stationary process is ergodic, but the distinction rarely bites in practice and is treated as a technical regularity condition from here on rather than as something to verify separately for every model.

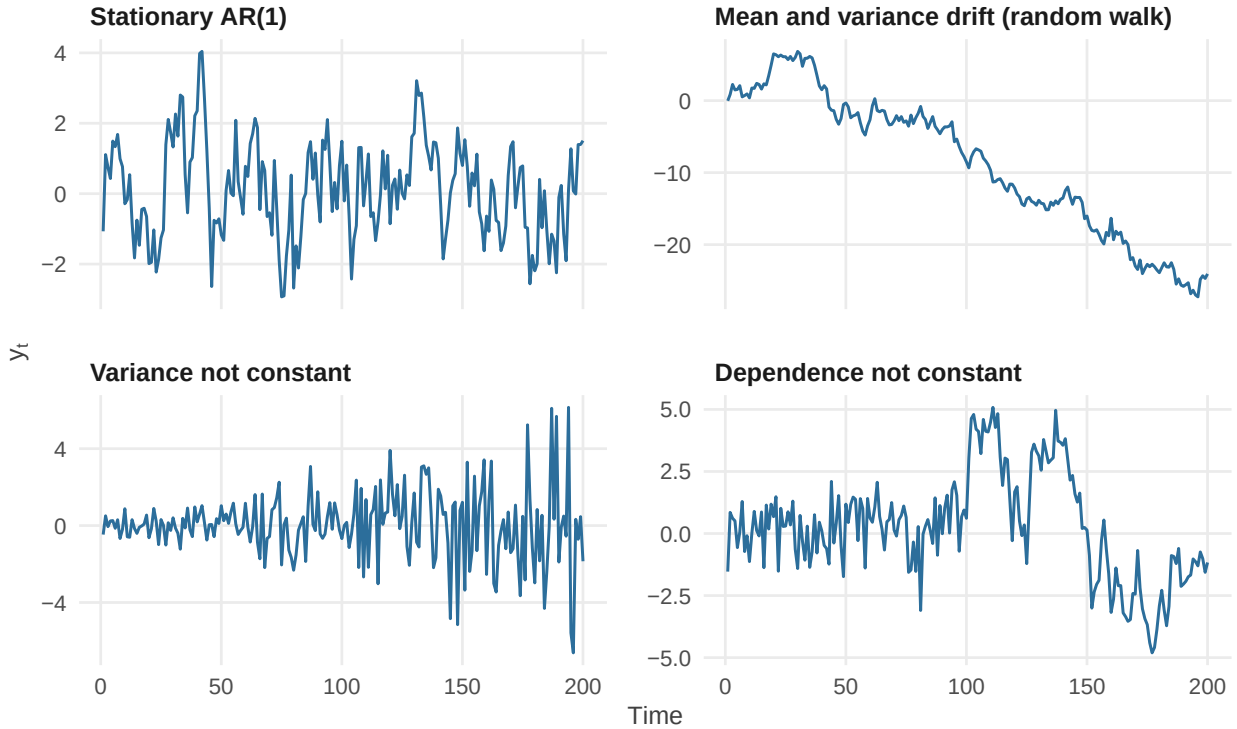


Figure 30: One stationary series and three violations, one for each condition in the definition.

A random walk fails conditions two and three: $\text{Var}(y_t) = t\sigma^2$ grows without bound, and $\text{Cov}(y_t, y_{t+h}) = t\sigma^2$ depends on t as well as h . A series with a deterministic linear trend fails condition one instead, and is called trend stationary, because subtracting the fitted trend leaves a stationary remainder. That difference is precisely the deterministic against stochastic distinction of Table 12, and it decides whether the correct remedy is detrending or differencing.

5.2.2 Restoring stationarity

Symptom	Usual remedy
Variance grows with the level	Log or Box–Cox transformation
Deterministic trend in the mean	Fit and subtract the trend
Stochastic trend, unit root	First differencing
Seasonal pattern that does not fade	Seasonal differencing at lag m
Structural break	Split the sample or add an indicator

Table 20: Common departures from stationarity and their remedies

Transformations are applied before differencing, since stabilising the variance often reduces the amount of differencing needed. Formal tests exist to help choose d : the augmented Dickey and Fuller test takes a unit root as its null hypothesis, while the KPSS test takes stationarity as its null, so the two are usefully read together. Both are treated in detail later in the course.

5.3 White noise

If today's coin toss tells you nothing about tomorrow's, the process is white noise. Each observation is random variation carrying no *linear* information about the next.

A process $\{\varepsilon_t\}$ is **white noise**, written $\varepsilon_t \sim WN(\mu, \sigma^2)$, if its observations are uncorrelated with constant mean and constant variance. In this chapter the mean is zero unless stated otherwise.

For white noise, $E(\varepsilon_t) = \mu$, $\text{Var}(\varepsilon_t) = \sigma^2$ and $\text{Cov}(\varepsilon_t, \varepsilon_{t+h}) = 0$ for $h \neq 0$, so that

$$\rho(h) = \begin{cases} 1, & h = 0, \\ 0, & h \neq 0. \end{cases}$$

5.3.1 Weak and strict white noise

The definition above is **weak white noise**: it requires only that the observations be *uncorrelated*. **Strict white noise** requires that they be *independent* and identically distributed, which is a substantially stronger claim.

The distinction is not pedantry, and it becomes essential at the end of this chapter. A GARCH process is weak white noise, since its autocorrelations are all zero at every non-zero lag, and yet it is plainly not independent: a large value today makes a large value tomorrow more likely, because the conditional variance carries over. Uncorrelated means there is no *linear* relationship to exploit; it does not mean there is no relationship at all.

So when we say white noise is unpredictable, the accurate statement is that it is *linearly* unpredictable beyond its mean: the best linear forecast is μ at every horizon. Whether some non-linear function of the past helps is a separate question, and for volatility models the answer is yes.

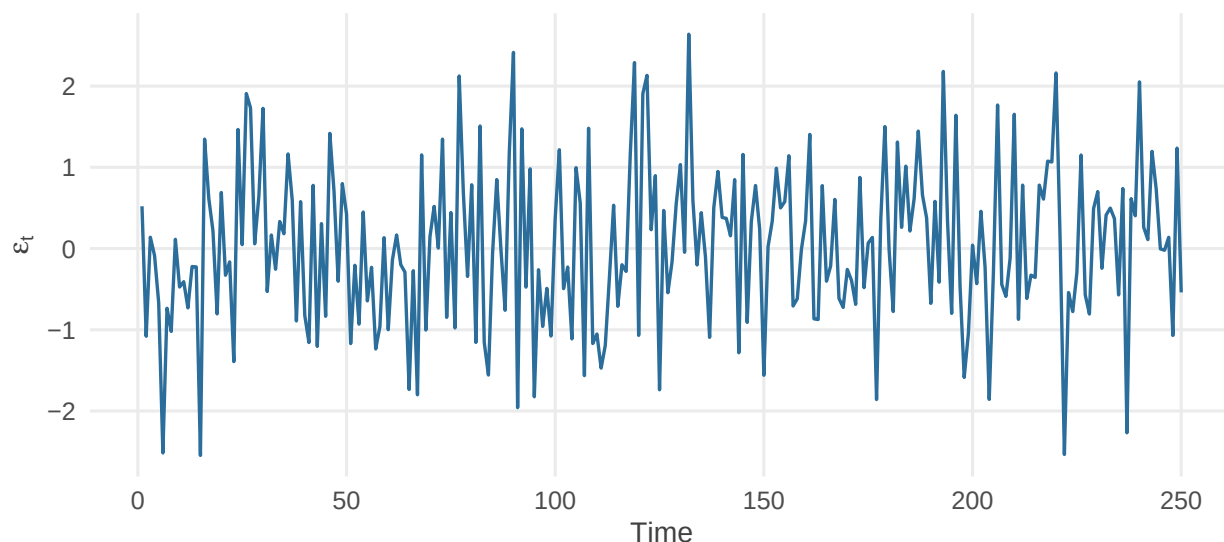


Figure 31: White noise. No trend, no season, no persistence.

White noise matters for a reason that is easy to miss on first encounter. It is not an interesting model of anything. It is the standard against which fitted models are judged. If a model has captured all the systematic linear structure in a series, its residuals should be indistinguishable from white noise, and any pattern left in them is structure the model failed to use.

Note also that white noise does not mean small variation or constant values. It means unpredictable variation, whatever its variance.

Monthly sales follow $y_t = 50 + \varepsilon_t$ with $\varepsilon_t \sim WN(0, 4)$.

Then $E(y_t) = 50$ and $\text{Var}(y_t) = 4$. Because successive observations are uncorrelated, no linear forecast can improve on the constant 50, and the mean benchmark of Table 14 is already optimal.

6 Linear Models for the Conditional Mean

6.1 Moving average processes

6.1.1 From white noise to dependence

White noise is uncorrelated, but a weighted combination of neighbouring white noise terms is not. Consider the two-sided average

$$y_t = \frac{1}{3}(\varepsilon_{t-1} + \varepsilon_t + \varepsilon_{t+1}).$$

Successive values of y_t share random components, so they must be correlated. This is **filtering**: building a new series as a weighted combination of the terms of another,

$$y_t = \sum_j a_j \varepsilon_{t-j},$$

where j runs over whichever set of integers the filter uses.

The example above is two-sided, because j takes the value -1 as well as 0 and 1 , so y_t depends on a shock that has not happened yet. That is fine for smoothing a series after the fact, which is what `stats::filter(..., sides = 2)` does below, but it is useless for forecasting. The MA models defined next therefore restrict the sum to $j \geq 0$: current and past shocks only. The mechanism by which dependence is created is identical; the one-sided restriction is what makes the result usable.

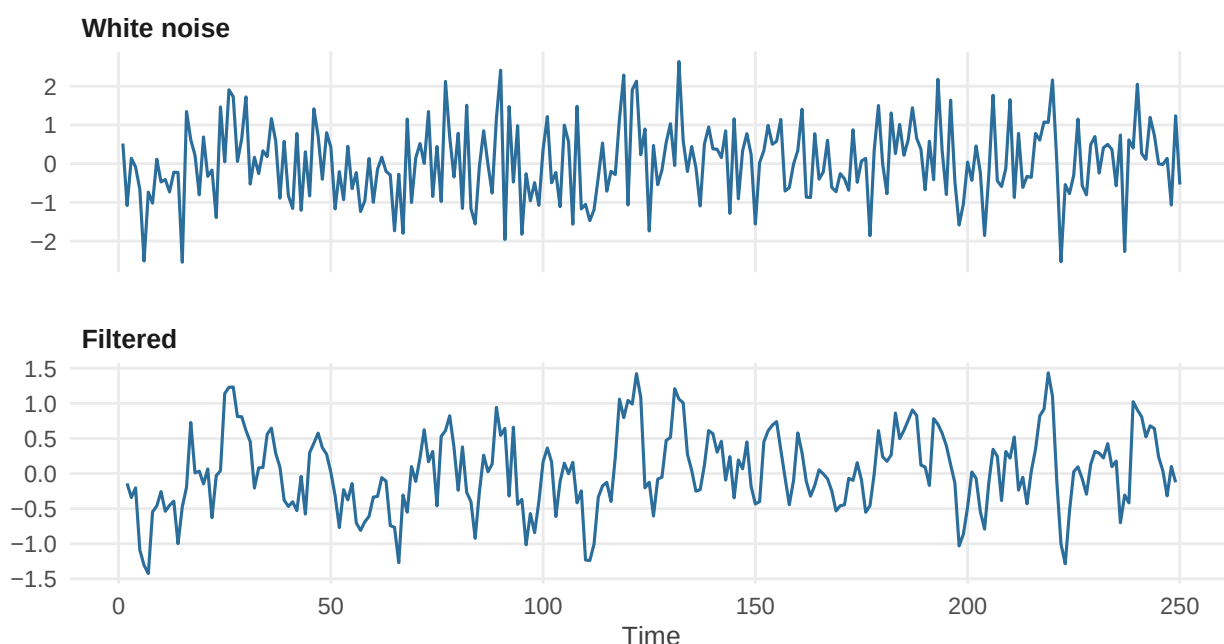


Figure 32: White noise before and after a three term two-sided moving average filter. Smoothing has introduced dependence.

6.1.2 The MA(q) model

$$y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}, \quad \varepsilon_t \sim WN(0, \sigma^2).$$

This is a distinct idea from the moving average smoother used in decomposition, and the shared name causes needless confusion. Here the model is driven by q lagged random shocks, not by lagged observations.

An $MA(q)$ process is stationary for any values of the coefficients, since it is a finite weighted sum of white noise terms with fixed weights. What the coefficients do have to satisfy, for estimation to be well posed, is invertibility, which is taken up in the ARMA section.

6.1.3 Properties of the MA(1) process

Consider $y_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1}$ with $\varepsilon_t \sim WN(0, \sigma^2)$.

MA(1) moments and autocorrelation.

$$E(y_t) = \mu, \quad \text{Var}(y_t) = (1 + \theta^2)\sigma^2, \quad \text{Cov}(y_t, y_{t-1}) = \theta\sigma^2,$$

so that

$$\rho(1) = \frac{\theta}{1 + \theta^2}, \quad \rho(h) = 0 \text{ for } h > 1.$$

The MA(1) process therefore has short memory: dependence vanishes completely beyond a single lag. That sharp cut off is the signature used to identify it from a sample ACF.

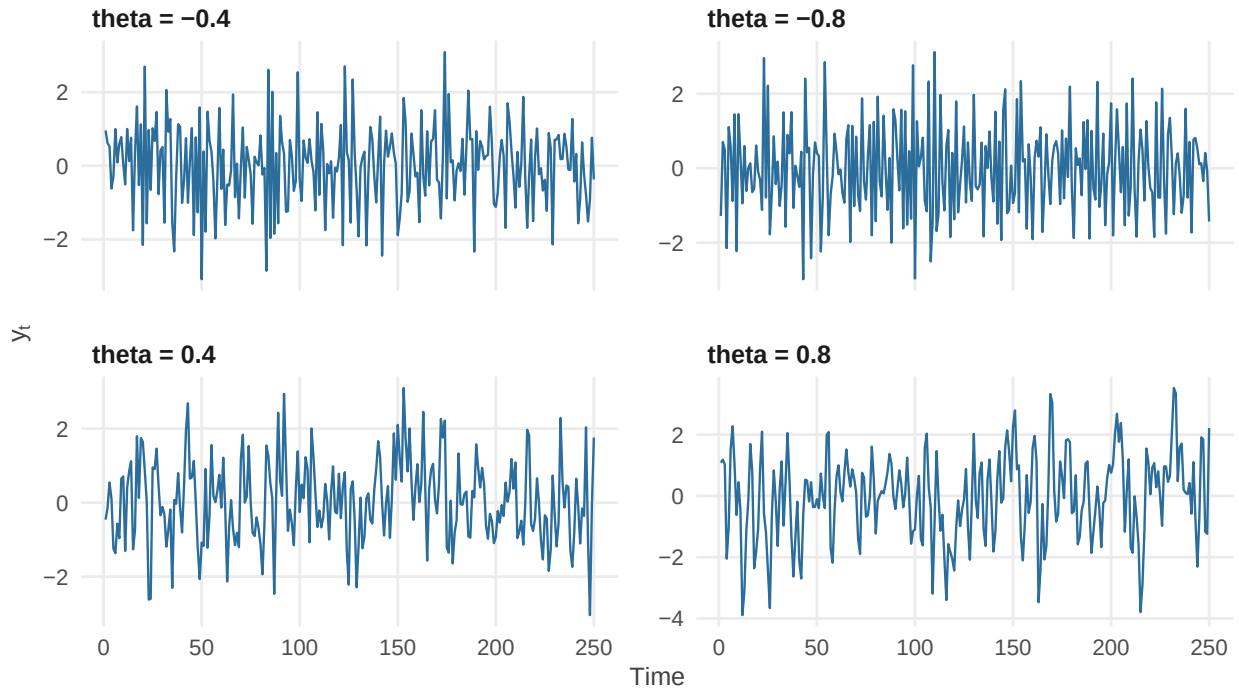


Figure 33: MA(1) processes for four values of θ . Negative values alternate, positive values smooth.

For $y_t = 5 + \varepsilon_t + 0.6\varepsilon_{t-1}$ with $\varepsilon_t \sim WN(0, 9)$:

$E(y_t) = 5$, $\text{Var}(y_t) = (1 + 0.36)(9) = 12.24$, and $\text{Cov}(y_t, y_{t-1}) = 0.6 \times 9 = 5.4$, giving $\rho(1) = 5.4/12.24 \approx 0.44$.

6.2 Autoregressive processes

If a river is running high today it will probably still be running high tomorrow. Temperature, unemployment and inflation all behave this way, which suggests regressing a series on its own past.

$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t, \quad \varepsilon_t \sim WN(0, \sigma^2).$$

6.2.1 The AR(1) process

Take $y_t = c + \phi y_{t-1} + \varepsilon_t$.

Value of ϕ	Behaviour
$\phi = 0$	White noise
$0 < \phi < 1$	Positive persistence
$-1 < \phi < 0$	Alternation between high and low
$ \phi $ near 1	Very strong persistence
$\phi = 1$	Random walk: a unit root, non-stationary
$\phi = -1$	Unit root with alternation, non-stationary
$ \phi > 1$	Explosive

Table 21: How the AR(1) coefficient governs behaviour

Assume for the derivations below that $|\phi| < 1$, so that $\{y_t\}$ is stationary and, in particular, $E(y_t) = E(y_{t-1}) = \mu$ and $\text{Var}(y_t) = \text{Var}(y_{t-1})$; this is what is used to solve for the moments in closed form.

Mean. Taking expectations of both sides gives $\mu = c + \phi\mu$, so $\mu(1 - \phi) = c$ and $\mu = c/(1 - \phi)$.

Variance. Subtracting the mean gives $y_t - \mu = \phi(y_{t-1} - \mu) + \varepsilon_t$. Since y_{t-1} and ε_t are uncorrelated, $\text{Var}(y_t) = \phi^2 \text{Var}(y_t) + \sigma^2$, so $\text{Var}(y_t) = \sigma^2/(1 - \phi^2)$.

AR(1) mean and variance. For the stationary AR(1) process $y_t = c + \phi y_{t-1} + \varepsilon_t$, $\varepsilon_t \sim WN(0, \sigma^2)$, $|\phi| < 1$,

$$\mu = \frac{c}{1 - \phi}, \quad \text{Var}(y_t) = \frac{\sigma^2}{1 - \phi^2}.$$

Outside $|\phi| < 1$ these expressions are not merely inconvenient but meaningless, since the variance formula would return a negative value.

Autocorrelation. The same substitution trick gives the ACF. Multiplying $y_t - \mu = \phi(y_{t-1} - \mu) + \varepsilon_t$ through by $(y_{t-h} - \mu)$ and taking expectations, the innovation term drops out because ε_t is uncorrelated with anything dated before t , leaving $\gamma(h) = \phi \gamma(h-1)$ for $h \geq 1$. This is a first order difference equation in h , with solution $\gamma(h) = \phi^h \gamma(0)$.

AR(1) autocorrelation function. For the stationary AR(1) process above,

$$\rho(h) = \phi^{|h|}, \quad h = 0, \pm 1, \pm 2, \dots,$$

the absolute value restoring the symmetry $\rho(h) = \rho(-h)$ that any autocorrelation function must have.

This is exactly the gradual, geometric decay visible in Figure 35 and Figure 36 below: a positive ϕ decays smoothly towards zero, a negative one decays with the same speed while alternating in sign, and the closer $|\phi|$ is to one, the slower the decay, which is why strongly persistent series take so long to forget a shock.

Stationarity and causality. In operator form the model is $(1 - \phi L)y_t = c + \varepsilon_t$, and the condition is that the root of $1 - \phi z = 0$, namely $z = 1/\phi$, lies outside the unit circle. That is the general statement, and it extends directly to $\text{AR}(p)$: every root of $\phi(z) = 0$ must lie outside the unit circle.

Strictly, this root condition is one of **causality** rather than stationarity alone. When $|\phi| > 1$ a stationary solution does still exist, but it expresses y_t in terms of *future* innovations, which is of no use for forecasting and is not identifiable from data. Every text in this area, and every software package, adopts the causal solution by convention, and this chapter follows suit. The case $|\phi| = 1$ admits no stationary solution at all.

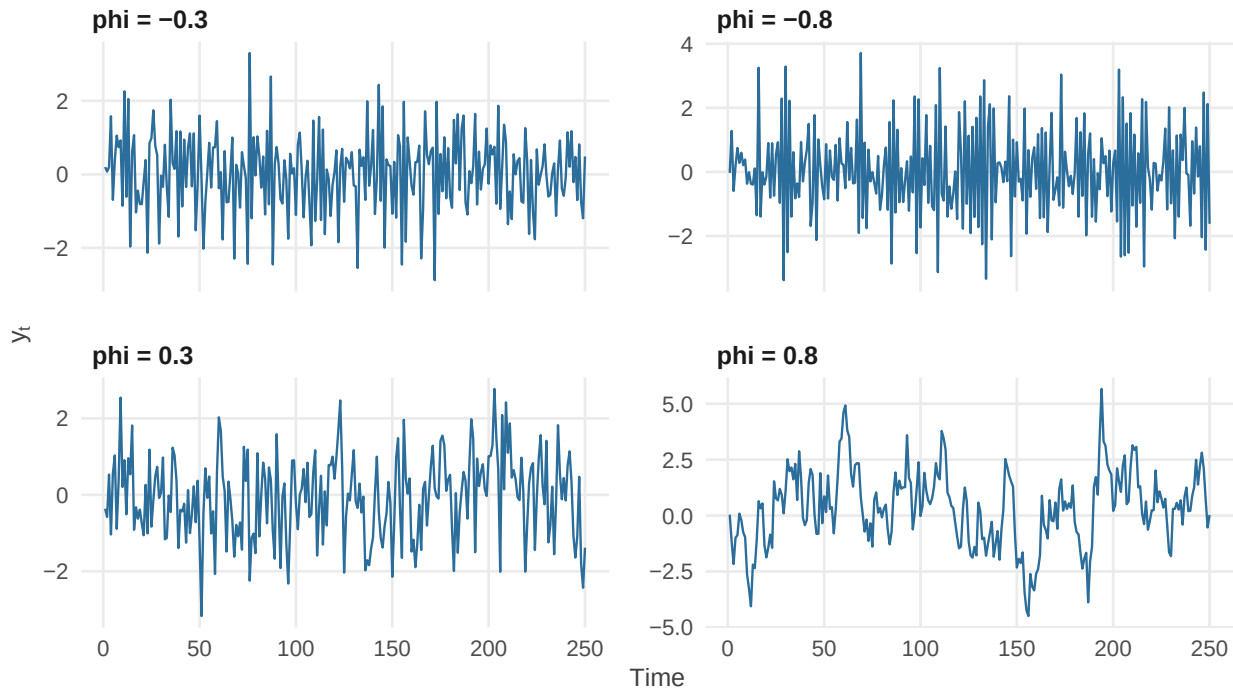


Figure 34: AR(1) processes for four values of ϕ . Larger values give longer runs in one direction.

Comparing this with the MA figure above is instructive. The dependence in an AR process decays gradually over many lags, whereas in an MA process it stops dead after lag q . That contrast is exactly what the ACF and PACF are built to detect.

For $y_t = 3 + 0.6y_{t-1} + \varepsilon_t$ with $\varepsilon_t \sim WN(0, 4)$:

$\mu = 3/(1-0.6) = 7.5$ and $\text{Var}(y_t) = 4/(1-0.36) = 6.25$. Since $|0.6| < 1$ the process is stationary.

6.3 Autocovariance and autocorrelation

The **autocovariance function** is $\gamma(h) = \text{Cov}(y_t, y_{t-h})$ for $h = 0, \pm 1, \pm 2, \dots$. For a stationary process, $\gamma(0) = \text{Var}(y_t)$ and $\gamma(h) = \gamma(-h)$.

Both statements in that definition need stationarity. Without it, γ would carry a t as well as an h and there would be nothing to plot.

Why $\gamma(h) = \gamma(-h)$. Covariance is symmetric in its two arguments for any pair of random variables, and relabelling $t+h$ as the time origin invokes stationarity: the covariance between two observations h apart depends on the separation h alone, not on which one happens to come first,

$$\gamma(-h) = \text{Cov}(y_t, y_{t+h}) = \text{Cov}(y_{t+h}, y_t) = \gamma(h).$$

Symmetry of the autocovariance function. For a stationary process, $\gamma(h) = \gamma(-h)$ for every h . The autocovariance function is therefore even, which is why a correlogram is only ever drawn for $h \geq 0$; the negative lags carry no further information.

Because $\gamma(h)$ carries the units of the measurement, it is standardised into the **autocorrelation function**

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}, \quad -1 \leq \rho(h) \leq 1, \quad \rho(0) = 1.$$

The ACF is therefore dimensionless, which is what makes it comparable across series measured in different units.

Why $|\rho(h)| \leq 1$. This is the Cauchy–Schwarz inequality, applied to y_t and y_{t-h} as two random variables:

$$|\text{Cov}(y_t, y_{t-h})| \leq \sqrt{\text{Var}(y_t) \text{Var}(y_{t-h})} = \text{Var}(y_t),$$

the right-hand equality using stationarity, $\text{Var}(y_t) = \text{Var}(y_{t-h}) = \gamma(0)$. Dividing through by $\gamma(0)$ gives $|\gamma(h)| \leq \gamma(0)$.

Cauchy–Schwarz bound for the autocorrelation function. For a stationary process, $|\rho(h)| \leq 1$ for every h , with equality only when y_{t-h} is an exact linear function of y_t .

A correlation of ± 1 therefore signals a deterministic relationship rather than merely a strong one, and the bound is worth knowing rather than taking on faith: a computed value outside $[-1, 1]$ is not a surprising result, it is a coding error.

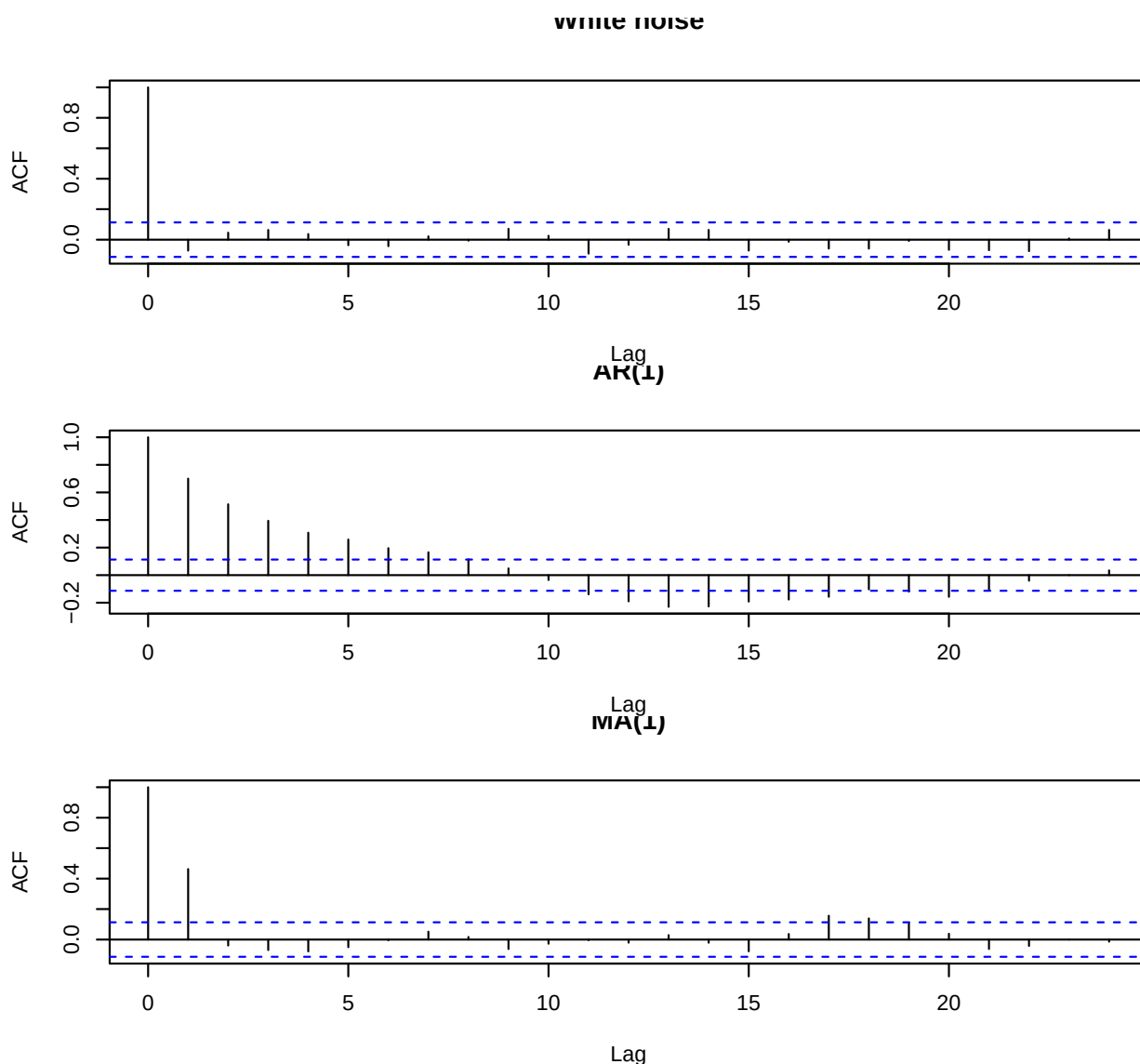


Figure 35: Sample ACFs for three processes. The shapes differ enough to be used for identification.

White noise shows no significant autocorrelation at any lag. The AR(1) correlation decays gradually towards zero. The MA(1) cuts off sharply after lag one. The dashed lines are approximate significance bounds at $\pm 1.96/\sqrt{n}$, so with 300 observations a correlation beyond about 0.11 is worth attention. Remember that these bounds are pointwise: with 25 lags plotted, roughly one spike outside the band is expected even under white noise, which is one motivation for the portmanteau test introduced later.

6.4 Partial autocorrelation

If y_{t-2} influences y_{t-1} , which in turn influences y_t , then y_t and y_{t-2} will appear correlated even with no direct link between them. The ACF cannot separate these two routes.

The **partial autocorrelation** ϕ_{hh} is the correlation between y_t and y_{t-h} once the linear effect of the intervening observations $y_{t-1}, \dots, y_{t-h+1}$ has been removed.

For an AR(1) process y_t depends directly only on y_{t-1} , so $\phi_{11} = \phi$ and $\phi_{hh} = 0$ for $h > 1$: a sharp

cut off. For an MA(1) process the PACF decays gradually instead, the mirror image of its ACF.

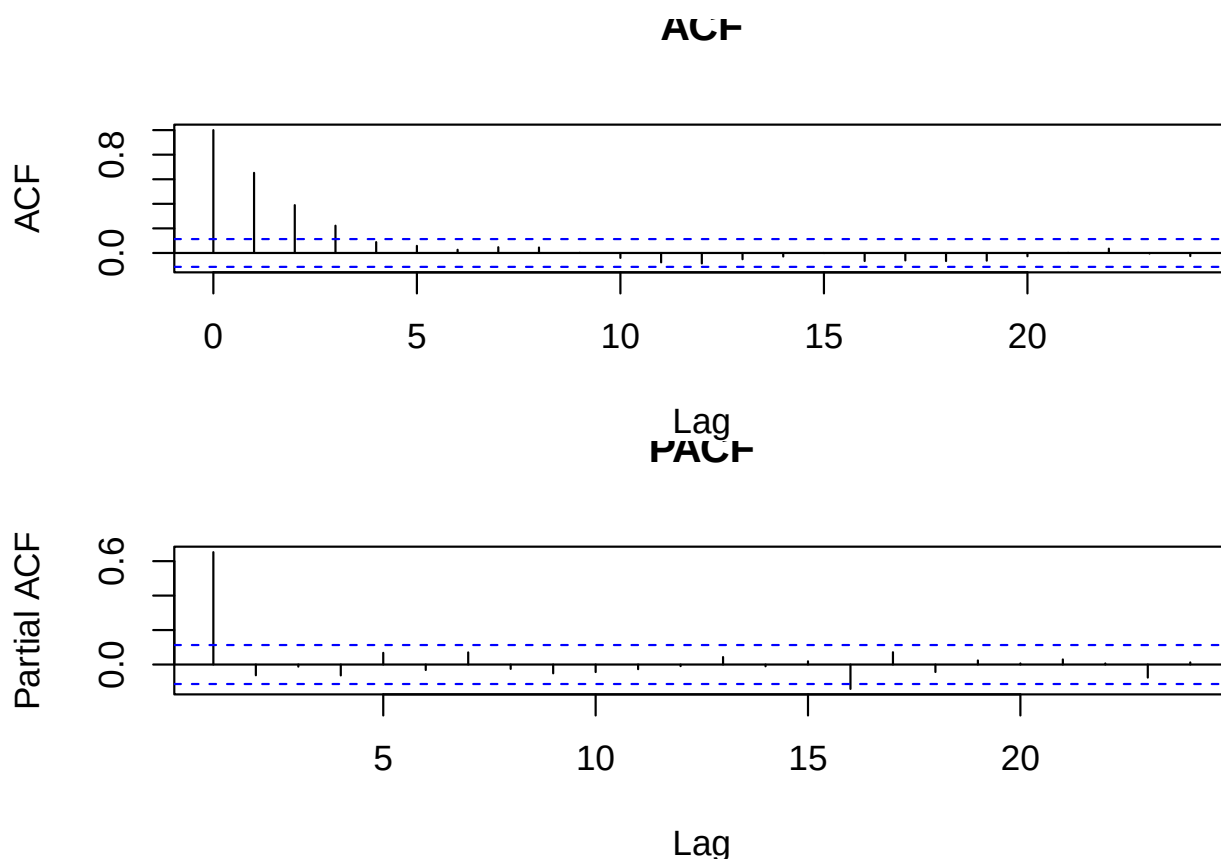


Figure 36: ACF and PACF of an AR(1) process. Gradual decay above, a single spike below.

Process	ACF	PACF
White noise	No significant spikes	No significant spikes
AR(p)	Decays gradually	Cuts off after lag p
MA(q)	Cuts off after lag q	Decays gradually
ARMA(p, q)	Decays gradually	Decays gradually

Table 22: Identification from the ACF and PACF

This table is the heart of the identification stage in the workflow of Box and Jenkins. It is a guide rather than a rule: sample ACFs are noisy, and the final choice between candidate models rests on information criteria and residual diagnostics as well.

A sample PACF cuts off after lag one, with $\phi_{11} = 0.71$ and later lags near zero, while the sample ACF decays gradually.

Read against Table 22, this is an AR(1) process, with $\hat{\phi} \approx 0.71$.

6.5 ARMA models

Many processes show persistence and short lived shocks at the same time. Electricity demand depends on yesterday's demand, but an unexpected cold snap also leaves a trace that fades after a few periods. The ARMA model allows both.

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j}, \quad \varepsilon_t \sim WN(0, \sigma^2),$$

or in operator form, $\phi(L)y_t = c + \theta(L)\varepsilon_t$.

The simplest useful case is ARMA(1,1),

$$y_t = c + \phi y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}.$$

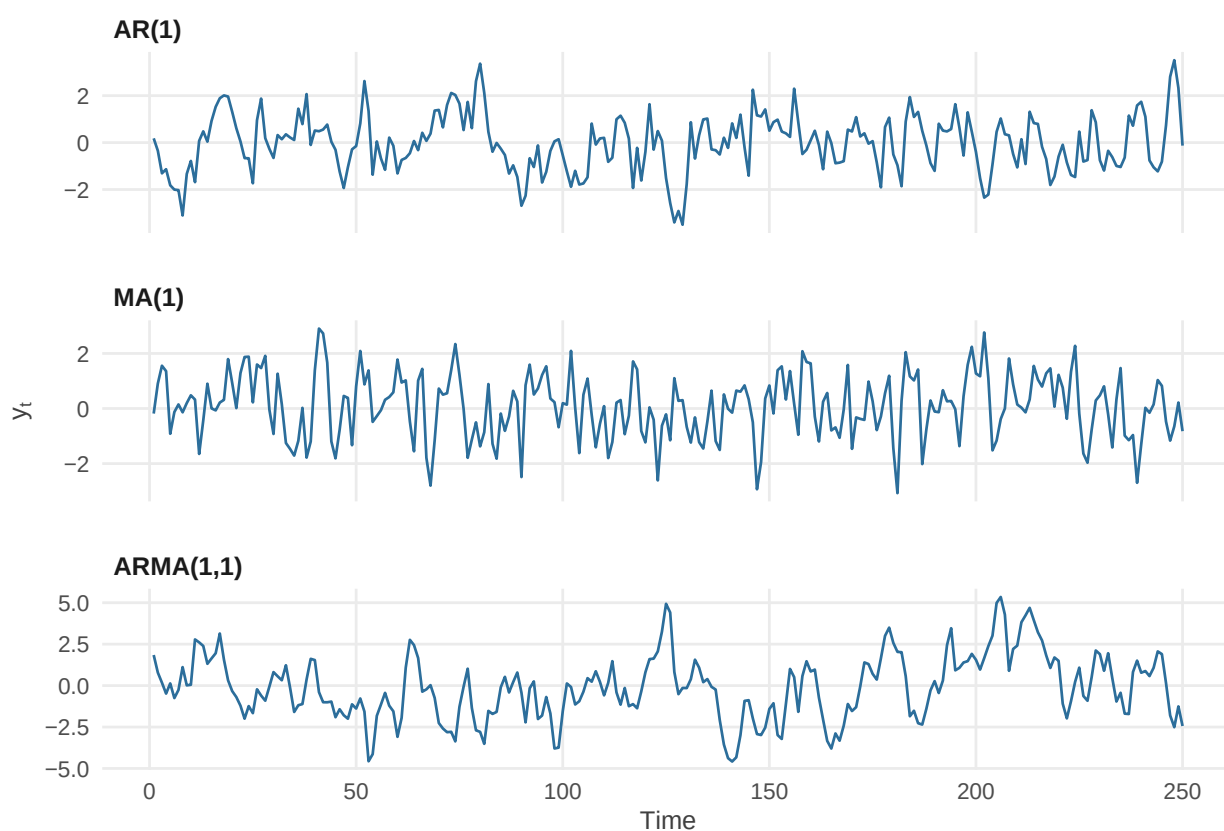


Figure 37: AR(1), MA(1) and ARMA(1,1) simulated from the same generator.

Feature	AR	MA	ARMA
Depends on past observations	yes	no	yes
Depends on past innovations	no	yes	yes
Memory	Long	Short	Both

Table 23: What each family can represent

An ARMA process shows neither of the clean signatures of a pure AR or a pure MA model, since its autocorrelation structure mixes the two. That is why identification of a mixed model needs the ACF and PACF read together, supported by AIC or BIC and by residual diagnostics.

Stationarity and invertibility. The autoregressive part must satisfy the root condition already given, so that the influence of past observations dies away. The moving average part must in addition be invertible, meaning it can be rewritten as an infinite autoregression; this requires the roots of $\theta(z) = 0$ to lie outside the unit circle. Without invertibility, two different sets of parameters describe the same second-order structure and estimation has no unique answer. For an MA(1), θ and $1/\theta$ give the same autocorrelation function, and only the value inside the unit circle is retained. Modern software checks both conditions during fitting.

Estimation. ARMA models are fitted by maximum likelihood, not by ordinary least squares. The reason is the moving average part: the terms $\varepsilon_{t-1}, \dots, \varepsilon_{t-q}$ are unobserved, so there is no design matrix to regress on. The likelihood is instead evaluated recursively, usually through a state space representation and the Kalman filter, with the innovations reconstructed as the recursion proceeds. Invertibility is what makes that reconstruction stable. This likelihood is ordinarily built under a working assumption that ε_t is Gaussian: the resulting point estimates of ϕ and θ remain consistent under much weaker conditions, but the AIC and BIC comparisons of the next section, and the prediction intervals of the forecasting section, are most directly justified when the innovations are approximately normal. A pure AR model *can* be estimated by least squares, and conditional least squares is sometimes used as a starting value, but `Arima()` reports maximum likelihood estimates by default.

6.6 ARIMA models

Before extending ARMA to non-stationary series, it is worth being explicit about what a model of this kind is for. An ARIMA model is not a claim that the process generating the data literally obeys a linear difference equation driven by Gaussian shocks; it is a deliberately simplified approximation, adopted because it captures enough of the structure in the series to be useful. Usefulness, not literal truth, is the standard a model is held to: residual diagnostics decide whether an approximation is adequate for the structure it claims to have captured, and forecasting performance against the benchmarks of an earlier chapter is the standard against which the whole exercise is ultimately judged.

6.6.1 Non-stationarity and its symptoms

Everything so far has assumed a constant mean, variance and dependence structure, in the sense set out in the stationarity section above. A great many practical series break that assumption through a persistent trend.

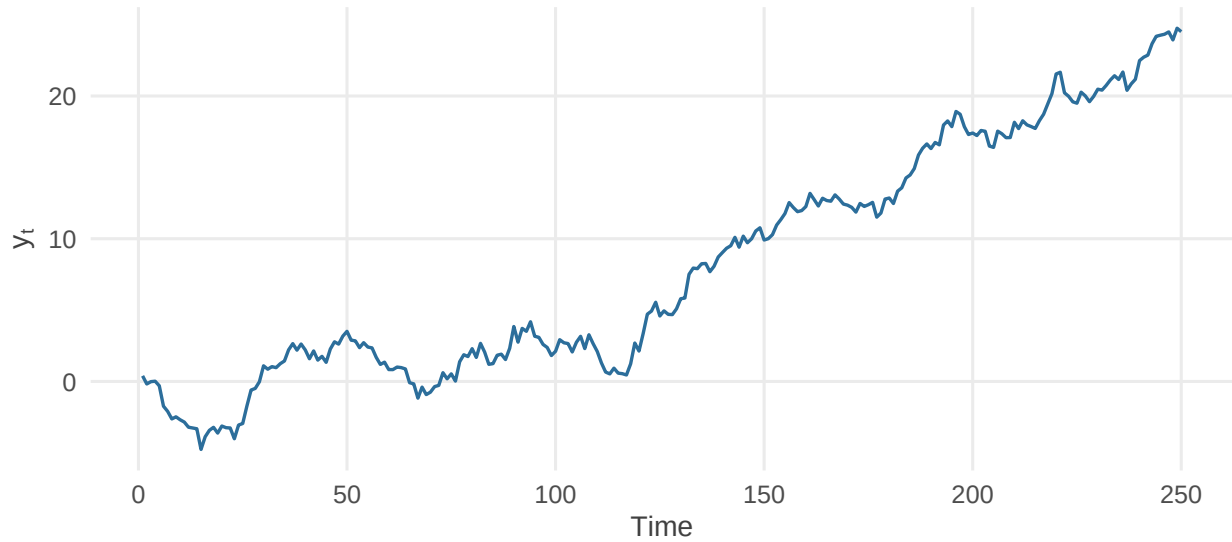


Figure 38: A series with both drift and a random walk component. Its mean is not constant.

Fitted to data like these, the ARMA results derived above simply do not hold, so the series is transformed first.

6.6.2 The difference operator

The **first difference** is $\nabla y_t = y_t - y_{t-1} = (1 - L)y_t$. The **second difference** applies the operator twice, $\nabla^2 y_t = (1 - L)^2 y_t = y_t - 2y_{t-1} + y_{t-2}$.

Differencing works on a trend for a reason that is easy to see. For $y_t = 5 + 0.4t + \varepsilon_t$,

$$\nabla y_t = (5 + 0.4t + \varepsilon_t) - (5 + 0.4(t-1) + \varepsilon_{t-1}) = 0.4 + \varepsilon_t - \varepsilon_{t-1}.$$

The trend is gone, leaving a constant and noise. A second useful special case: for the random walk $y_t = y_{t-1} + \varepsilon_t$, differencing returns $\nabla y_t = \varepsilon_t$ exactly.

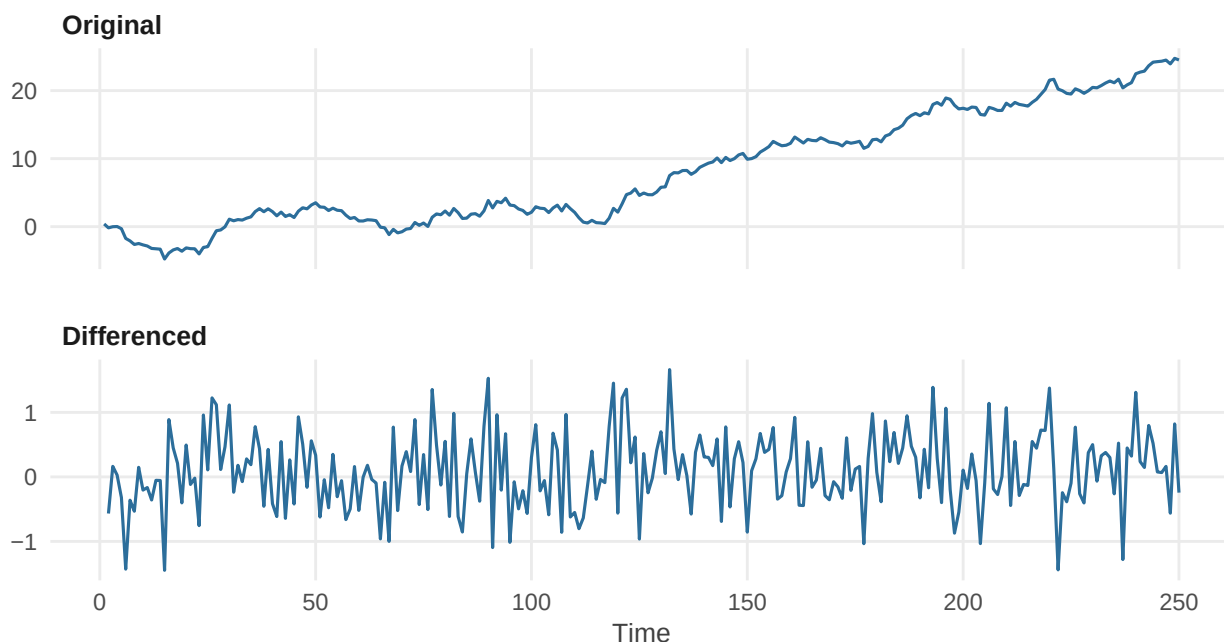


Figure 39: The series from above, before and after first differencing.

6.6.3 Over-differencing and how to spot it

Differencing should be used sparingly. Notice what the first calculation above actually produced: differencing a deterministic trend left $\varepsilon_t - \varepsilon_{t-1}$, which is an MA(1) with $\theta = -1$, exactly on the non-invertible boundary. The same thing happens whenever a series is differenced more often than it needs.

That gives a clean diagnostic. If white noise is differenced once, the result has

$$\rho(1) = \frac{-1}{1 + (-1)^2} = -0.5,$$

so an over-differenced series betrays itself through a large negative autocorrelation at lag one, typically near -0.5 , together with an inflated sample variance. If differencing a series makes its variance go up rather than down, you have gone one step too far. Choose the smallest d that produces approximate stationarity. Almost no series in practice needs $d > 2$.

6.6.4 The ARIMA(p, d, q) model

An ARIMA(p, d, q) model applies d differences to y_t and fits an ARMA(p, q) model to the result. In operator form,

$$\phi(L)(1 - L)^d y_t = c + \theta(L)\varepsilon_t.$$

Specification	Equivalent to
ARIMA(1,0,0)	AR(1)
ARIMA(0,0,1)	MA(1)
ARIMA(1,0,1)	ARMA(1,1)
ARIMA(0,1,0)	Random walk

Specification	Equivalent to
ARIMA(0,1,0) with $c \neq 0$	Random walk with drift
ARIMA(1,1,1)	One difference, then ARMA(1,1)

Table 24: Familiar models as special cases of ARIMA

One warning about model selection. AIC and BIC compare models fitted to the *same* data by likelihood, so they may be used to choose p and q with d held fixed. They may **not** be used to choose d : an ARIMA($p, 1, q$) model is fitted to $n - 1$ differenced values and an ARIMA($p, 2, q$) to $n - 2$, so the likelihoods refer to different datasets and the numbers are not comparable. Choose d from plots and unit root tests first, then use information criteria within that choice.

Exactly the same principle applies to the seasonal differencing order D . Models with different values of D are fitted to different transformed series, so their likelihoods are likewise not directly comparable. Determine D from seasonal diagnostics and subject-matter knowledge before using AIC or BIC to compare models with common values of d and D .

6.6.5 From ARMA(1,1) to the ARIMA(1,1,1) equation

The most instructive case is ARIMA(1,1,1), because the derivation shows exactly how differencing and the ARMA structure fit together. Write the differenced series as $w_t = y_t - y_{t-1}$ and model it as ARMA(1,1),

$$w_t = c + \phi w_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}.$$

Substituting $w_t = y_t - y_{t-1}$ and $w_{t-1} = y_{t-1} - y_{t-2}$,

$$y_t - y_{t-1} = c + \phi(y_{t-1} - y_{t-2}) + \varepsilon_t + \theta \varepsilon_{t-1}.$$

Expanding the bracket and moving $-y_{t-1}$ across gives the unified forecasting equation,

$$y_t = c + (1 + \phi)y_{t-1} - \phi y_{t-2} + \varepsilon_t + \theta \varepsilon_{t-1}.$$

Term	What it contributes
c	A constant, which after differencing acts as drift
$(1 + \phi)y_{t-1}$	The previous level, adjusted for momentum
$-\phi y_{t-2}$	The correction that turns levels into changes
ε_t	The current shock, unknowable in advance
$\theta \varepsilon_{t-1}$	The previous forecast error, fed back

Table 25: Reading the ARIMA(1,1,1) equation term by term

The coefficient on y_{t-1} is $1 + \phi$ rather than ϕ , and there is a negative coefficient on y_{t-2} that no

ARMA model would have. Both are artefacts of writing a differenced model back on the original scale, and neither should be read as evidence of strong dependence at lag one and two.

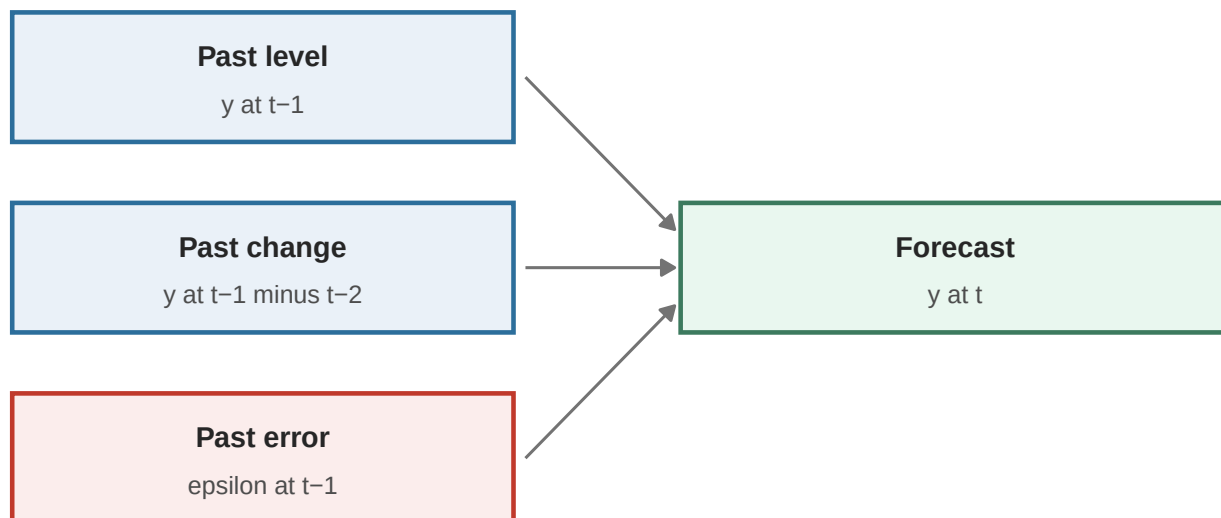


Figure 40: Three sources of information feed the ARIMA forecast. The last of them is what distinguishes it from a pure autoregression.

Take $c = 1$, $\phi = 0.6$, $\theta = 0.3$, $y_{t-1} = 120$, $y_{t-2} = 115$ and $\varepsilon_{t-1} = 2$. The current shock ε_t is not known when the forecast is made, so it is replaced by its expectation, zero: formally, $E(\varepsilon_t \mid \mathcal{I}_{t-1}) = 0$, the same conditional-expectation principle behind every forecast in this chapter.

Substituting into the unified equation,

$$\hat{y}_t = 1 + 1.6(120) - 0.6(115) + 0 + 0.3(2) = 1 + 192 - 69 + 0.6 = 124.6.$$

The forecast is therefore about 125. The rise of 4.6 units above the previous level decomposes cleanly: the constant contributes 1, the momentum term $\phi(y_{t-1} - y_{t-2}) = 0.6 \times 5$ contributes 3, and the error feedback $\theta\varepsilon_{t-1} = 0.3 \times 2$ contributes 0.6.

6.6.6 Fitting and forecasting in R

```
#> Series: y_int
#> ARIMA(1,1,1)
#>
#> Coefficients:
#>          ar1      ma1
#>       0.6444  0.2784
#> s.e.  0.0642  0.0857
#>
#> sigma^2 = 0.9241: log likelihood = -342.95
#> AIC=691.91  AICc=692.01  BIC=702.46
```

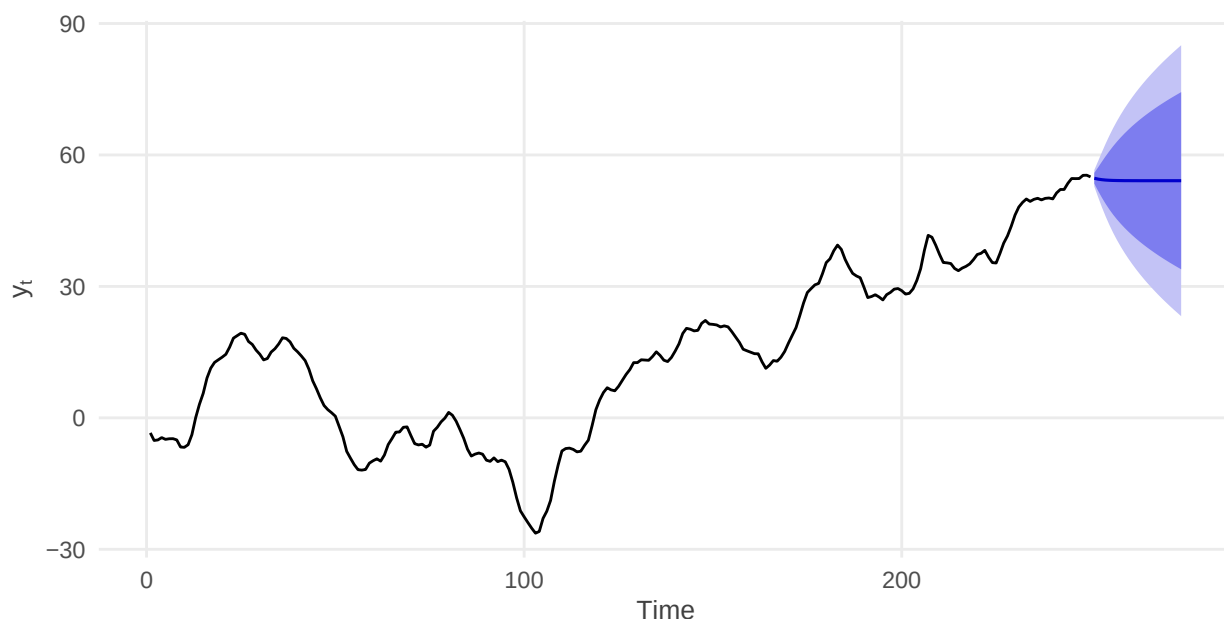


Figure 41: An ARIMA(1,1,1) model fitted to a simulated ARIMA(1,1,1) series, with forecasts and prediction intervals for 24 periods.

The point forecasts settle quickly onto a constant level, and the prediction intervals continue to widen as the horizon increases. Both features follow from the unit root. Because the differenced series is stationary with mean zero, the expected future *change* decays geometrically to zero, so the expected future *level* converges to a constant. Future innovations are unpredictable, so the best available statement about the level far ahead is simply where the series is now. Meanwhile the forecast variance accumulates, growing roughly in proportion to h , which is the $\sqrt{h}$ widening visible in the figure and derived in the section on forecast horizons.

These prediction intervals are conditional on the fitted ARIMA model and its estimated parameters. If the model is misspecified or the distributional assumptions are poor, the realised coverage may differ from the nominal confidence level, which is why careful residual diagnostics are an essential part of the forecasting workflow.

Notice that the forecast is not saying the series will stop changing. It says that positive and negative future innovations are equally likely, which makes the current level the best unbiased prediction.

A note on software. This chapter uses the `forecast` package, which remains a robust and widely taught standard. The same models are also available through the newer `fable`, `tsibble` and `feasts` packages, which fit the tidy data conventions used elsewhere in this document: a `tsibble` is a data frame with a time index, and `fable` fits and forecasts many models at once with `model()` and `forecast()`. Students comfortable with `dplyr` may find that workflow more natural, and the statistical content is identical.

6.6.7 Residual diagnostics

Estimation gives a fitted model; diagnostics decide whether to keep it. The standard is the one set out in the white noise section: if the model has captured the linear structure, nothing linear should remain in the residuals.

Checking individual autocorrelations against the $\pm 1.96/\sqrt{n}$ bounds tests one lag at a time, and with twenty or thirty lags on the plot some will breach the band by chance. A portmanteau test asks the joint question instead.

The **Ljung and Box** statistic for H lags is

$$Q_{\text{LB}} = n(n+2) \sum_{k=1}^H \frac{\hat{\rho}_k^2}{n-k},$$

where $\hat{\rho}_k$ is the sample autocorrelation of the residuals at lag k . Under the null hypothesis that the residuals are white noise, Q_{LB} is approximately χ^2 with $H - K$ degrees of freedom, where K is the number of estimated ARMA parameters.

The degrees of freedom correction matters: fitting parameters uses up some of the residual autocorrelation, and ignoring K makes the test too conservative. A common choice is $H = 10$ for non-seasonal data and $H = 2m$ for seasonal data. A small p-value is evidence against the model; a large one is the outcome you want, though as always it is absence of evidence rather than proof of adequacy.

The Ljung–Box test assesses whether the residuals behave as weak white noise, meaning they are approximately uncorrelated. It does not test whether the residuals are independent. For ARIMA modelling this weak-white-noise property is the essential diagnostic. When modelling conditional heteroscedasticity with GARCH, the same idea is applied to the squared standardised residuals, since remaining autocorrelation in the squared residuals indicates that volatility clustering has not yet been adequately modelled.

```
#>
#>  Ljung-Box test
#>
#> data:  Residuals from ARIMA(1,1,1)
#> Q* = 6.4885, df = 8, p-value = 0.5927
#>
#> Model df: 2.    Total lags used: 10
```

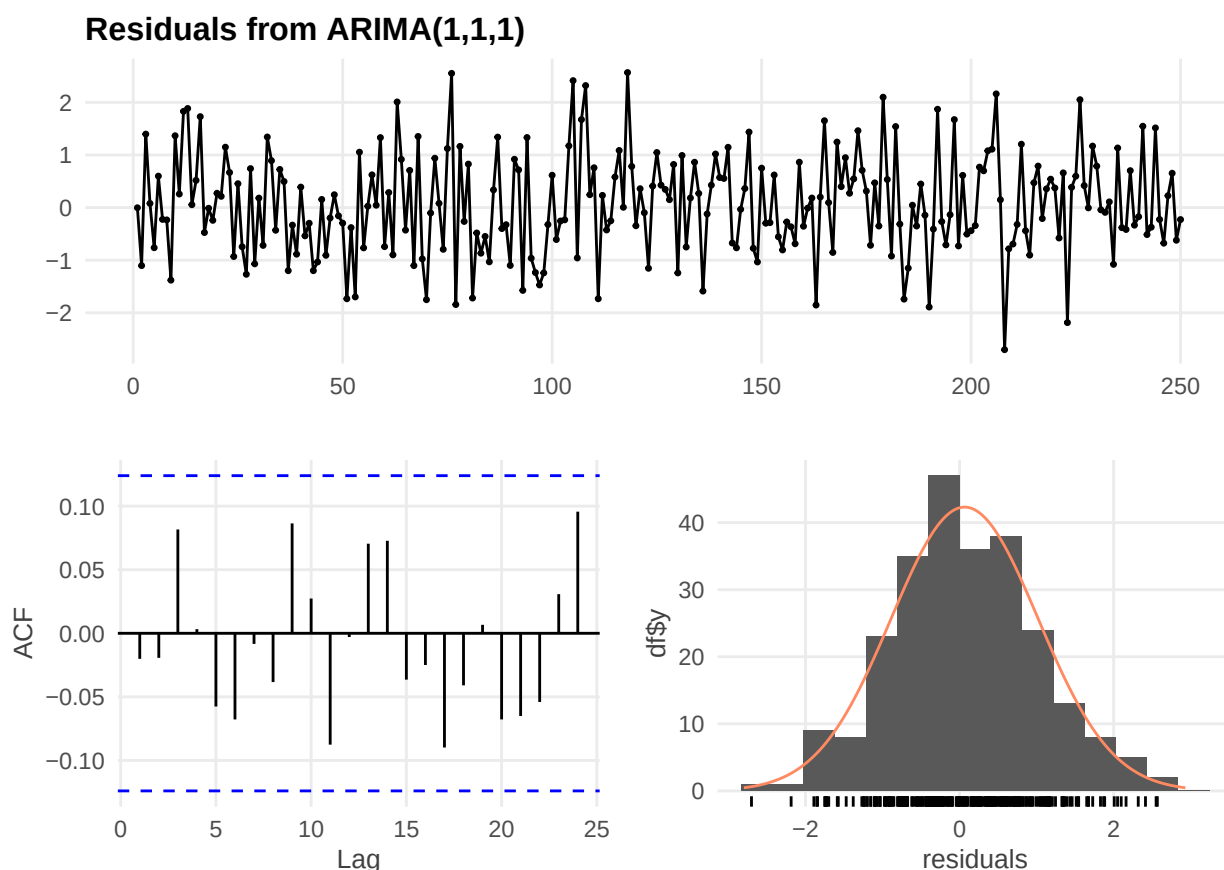


Figure 42: Residual diagnostics for the fitted ARIMA(1,1,1) model: the residual series, its ACF, and a histogram against a normal density.

```
#>
#> Box-Ljung test
#>
#> data: residuals(fit_arima)
#> X-squared = 6.4885, df = 8, p-value = 0.5927
```

Three things are being checked at once here. The residual plot should show no remaining trend, seasonality or change in variance; the ACF should show no significant spikes; and the histogram bears on whether the normality assumption behind the prediction intervals is reasonable. Failure of the first two sends you back to identification. Failure of the third leaves the point forecasts usable but the intervals suspect.

6.6.8 The workflow of Box and Jenkins

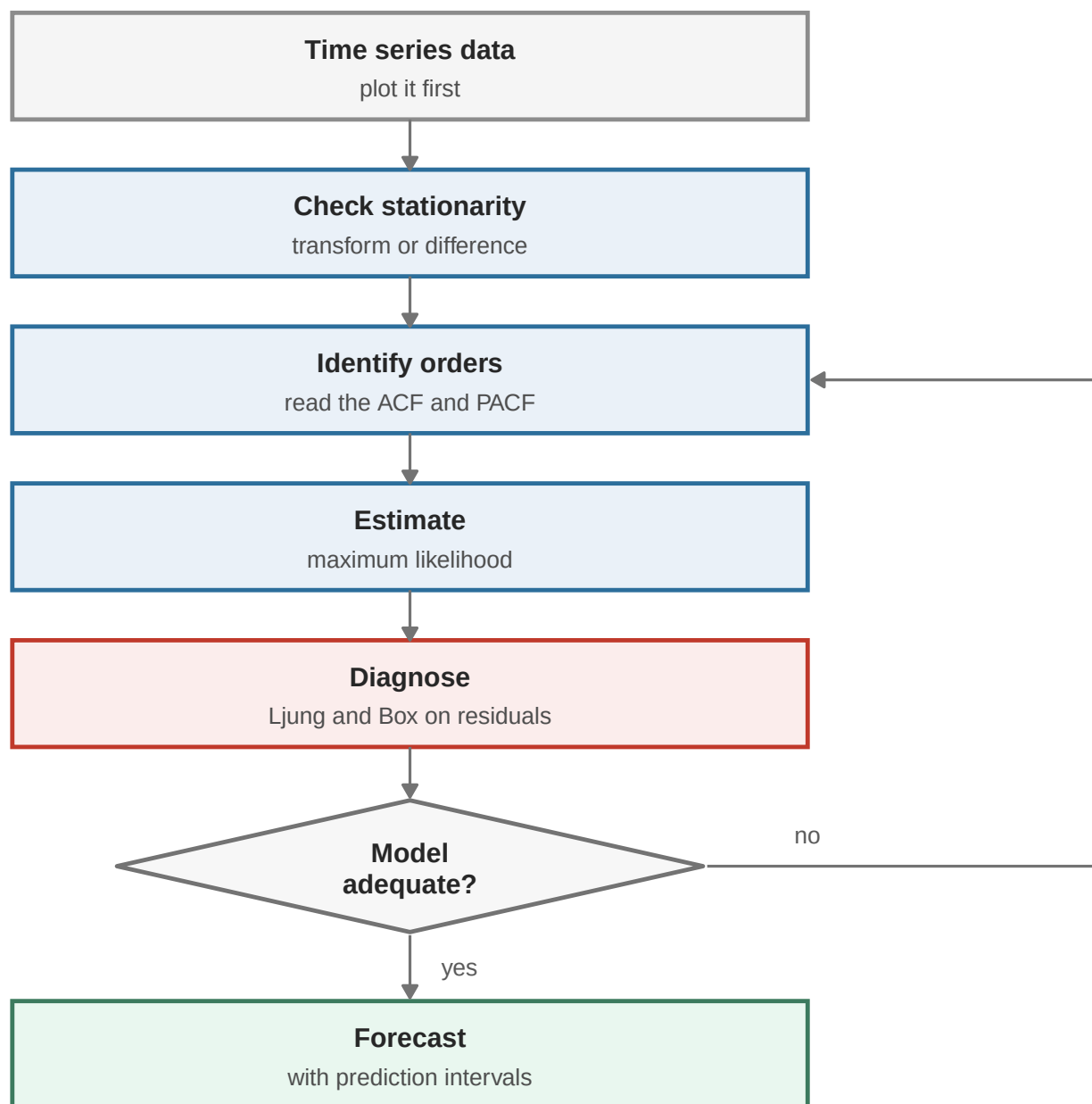


Figure 43: The model building cycle. Diagnostic failure sends you back to identification rather than forward to forecasting.

In words: plot the data, difference until the series is approximately stationary, read the ACF and PACF to choose candidate orders, estimate the parameters by maximum likelihood, check the residuals against the white noise standard, and only then forecast. If the residuals fail, the loop returns to identification. Once a model survives the loop, it still has to beat the benchmarks on the test set before it earns its place. This is the general visualise, transform, identify, estimate, diagnose, forecast, evaluate, iterate cycle introduced earlier in the course, specialised here to ARIMA models.

6.7 Seasonal series and SARIMA models

Seasonality that survives ordinary differencing needs its own treatment.

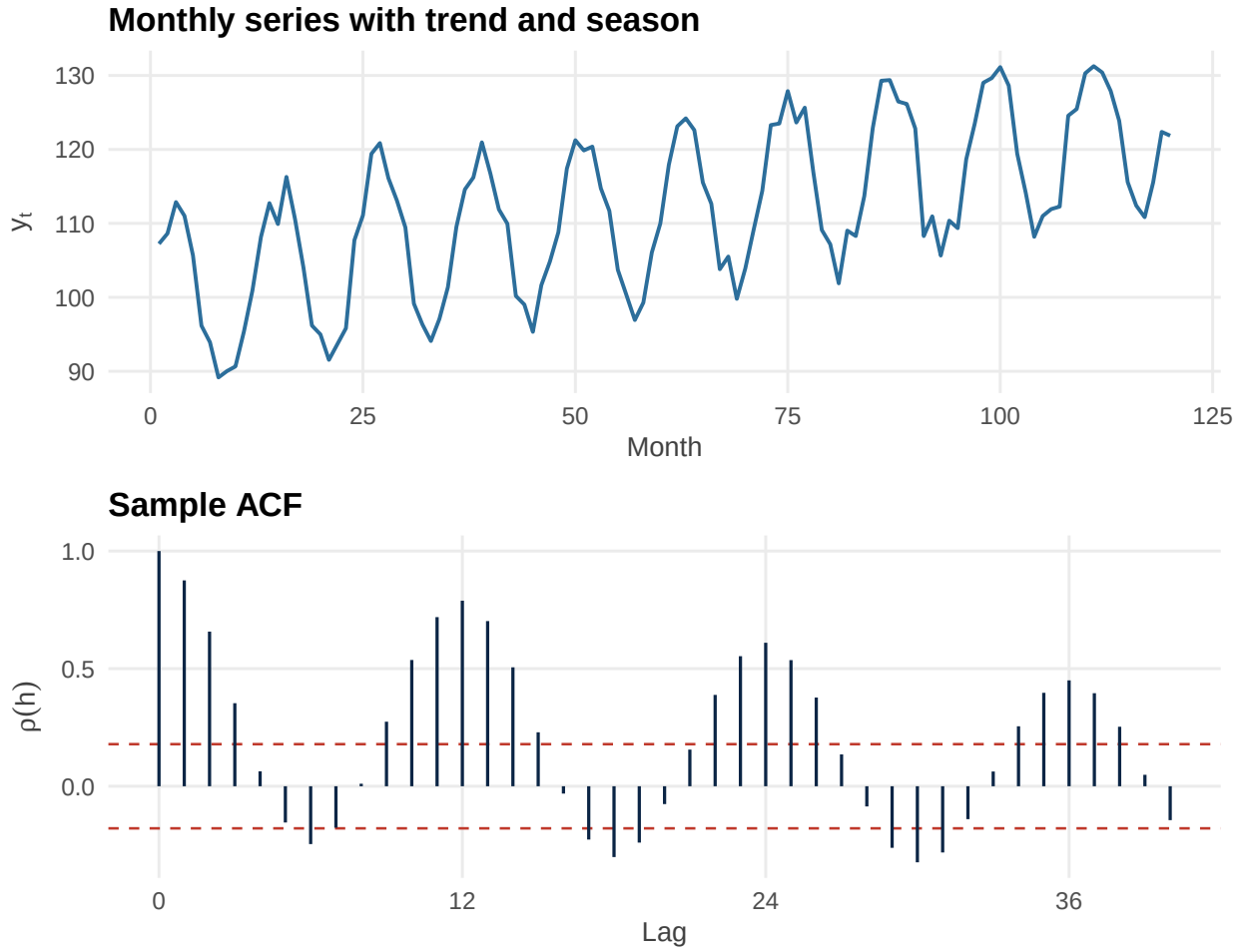


Figure 44: A seasonal series and its ACF. The recurring peaks at lags 12, 24 and 36 are the seasonal signature.

The **seasonal difference** operator is

$$\nabla_m y_t = y_t - y_{t-m} = (1 - L^m)y_t,$$

where m is the number of observations in one season, for example $m = 12$ for monthly data or $m = 4$ for quarterly data.

Ordinary differencing removes trend and seasonal differencing removes the repeating pattern. Series with both need both, giving $(1 - L)(1 - L^m)y_t$.

A seasonal difference is warranted when the seasonal pattern itself evolves, that is, when this December's level is genuinely informative about next December's. Where the seasonal pattern is fixed and deterministic, seasonal dummy variables or harmonic regressors are the better tool, and seasonal differencing over-differences in exactly the sense described earlier, leaving a seasonal MA on the non-invertible boundary.

A **SARIMA** $(p, d, q)(P, D, Q)_m$ model combines non-seasonal orders (p, d, q) with seasonal orders

(P, D, Q) operating at lag m :

$$\Phi(L^m)\phi(L)(1-L^m)^D(1-L)^d y_t = \Theta(L^m)\theta(L)\varepsilon_t.$$

So SARIMA(1,1,1)(0,1,1)₁₂ means one autoregressive term, one ordinary difference, one moving average term, no seasonal autoregressive term, one seasonal difference and one seasonal moving average term, at a monthly period.

One nearby specification is common enough to have a name. The **airline model** is SARIMA(0,1,1)(0,1,1)₁₂, with no autoregressive terms at all, and it takes its name from the international airline passenger series that Box and Jenkins used to introduce it. It remains a remarkably strong default for monthly data with trend and evolving seasonality, and it is a sensible benchmark to fit alongside anything more elaborate.

```
#> Series: seasonal_series
#> ARIMA(1,1,1)(0,1,1)[12]
#>
#> Coefficients:
#>          ar1          ma1          sma1
#>          0.0188   -0.6686   -0.8675
#> s.e.    0.1182    0.0871    0.0743
#>
#> sigma^2 = 1.982:  log likelihood = -301.12
#> AIC=610.24   AICc=610.49   BIC=622.71
```

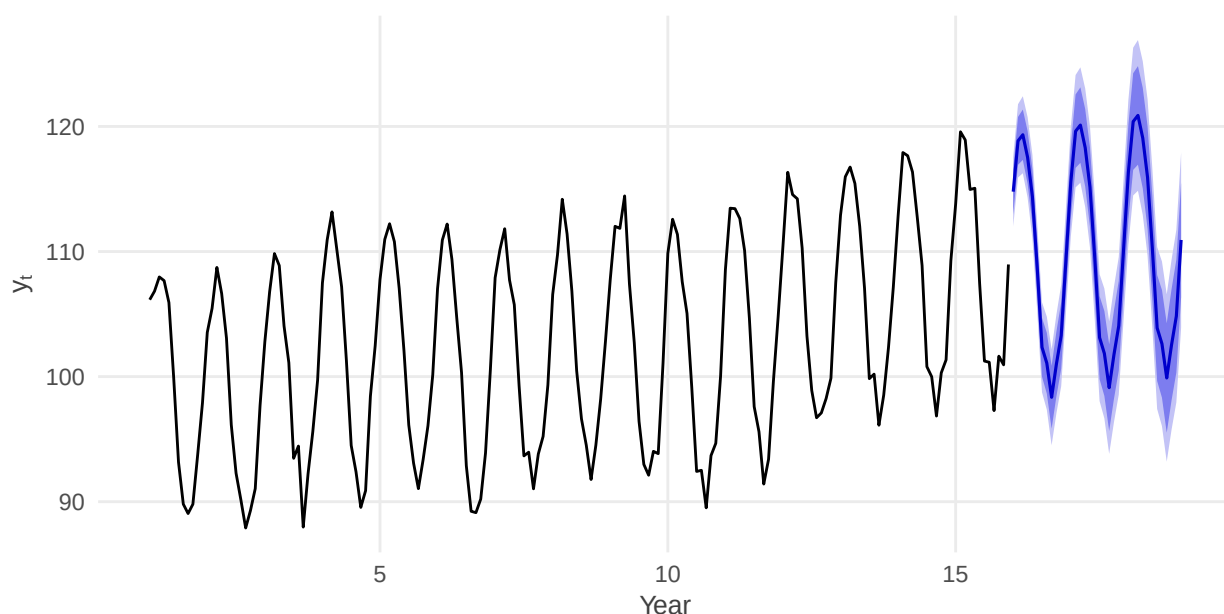


Figure 45: A SARIMA(1,1,1)(0,1,1) model fitted to a series with a stochastic level and an evolving seasonal pattern, forecast three years ahead.

The seasonal shape is carried forward while the intervals widen, which is the behaviour a seasonal model is fitted for. Compare the estimated seasonal MA coefficient with -1 : values close to the boundary are a warning that the seasonal difference was not needed.

Monthly rainfall, electricity demand, airline passengers, retail sales, tourism arrivals and disease incidence are all standard applications. Note that seasonal differencing, like ordinary differencing, costs observations: a single seasonal difference on monthly data removes the first twelve, which matters when the series is short.

7 Conditional Variance Models

Everything up to this point has modelled the conditional mean. An ARMA model says where a series is going next; the spread around that statement was assumed fixed, a single σ^2 applying equally to every period in the sample.

For a great many series that assumption is harmless. For financial returns it is not, and it fails in a particular way: it understates risk during exactly the periods when risk is highest, and overstates it when markets are quiet. This chapter builds a second equation to describe that movement.

The material assumes the sections on white noise, stationarity and the Box and Jenkins workflow. It leans particularly on the distinction between weak and strict white noise, which until now has looked like a technicality and is about to become the whole point.

7.1 Learning outcomes

By the end of this chapter you should be able to:

1. Explain why a constant variance assumption fails for financial returns, and recognise volatility clustering in a plot.
2. Explain why returns rather than prices are modelled, and compute them.
3. Test a series for ARCH effects and interpret the result.
4. Write down the ARCH(q) and GARCH(p, q) models and explain what each term contributes.
5. Interpret α_0 , α_1 and β_1 , and compute the persistence, the half life and the long run variance from a fitted model.
6. Produce and interpret volatility forecasts, and explain why they revert to a long run level.
7. Fit ARCH and GARCH models in R and assess their adequacy using standardised residuals.
8. Explain why asymmetric models exist and when they are appropriate.

7.2 Additional notation

Symbol	Meaning
p_t	The price or level of an asset at time t
r_t	The log return, $r_t = \nabla \ln p_t$
z_t	Standardised innovation, ε_t/σ_t
α_0	Constant in the variance equation
α_1	Coefficient on the lagged squared shock
β_1	Coefficient on the lagged conditional variance
δ	Asymmetry or leverage coefficient
$\bar{\sigma}^2$	Unconditional, long run variance
ν	Degrees of freedom of a Student t innovation

Table 26: Symbols added in this chapter

Two warnings about software naming. The **rugarch** package calls α_0 by the name **omega**, and many textbooks write it ω ; this document reserves ω for the angular frequency of the spectral chapter, so

the constant is written α_0 throughout. The same package calls the asymmetry coefficient `gamma1`, which cannot be used here because $\gamma(h)$ is already the autocovariance function. It is written δ instead.

7.3 Why a second equation is needed

7.3.1 Returns, not prices

Asset prices are not modelled directly. Prices wander, their variance grows without bound, and the quantity of interest is in any case not the price but the change in it. The standard transformation is the log return,

$$r_t = \ln p_t - \ln p_{t-1} = \nabla \ln p_t = (1 - L) \ln p_t,$$

which introduces nothing new: it is the log transformation of Table 20 followed by the first difference of the ARIMA chapter, applied in that order and for the reasons given there.

The simple return $R_t = (p_t - p_{t-1})/p_{t-1}$ is what an investor actually receives, and the two are related by $r_t = \ln(1 + R_t)$, so they agree closely for small movements and diverge for large ones. Log returns are preferred for modelling because they add across time: the return over h periods is $r_{t+1} + \dots + r_{t+h}$, with no compounding to track. Simple returns are preferred for portfolio arithmetic, because they add across assets. Neither is more correct; they answer different questions, and mixing them is a common source of small persistent errors.

7.3.2 The stylised facts

Return series from completely different markets, sampled at completely different intervals, share a list of properties so consistent that they have a name.

Fact	What is observed
Near zero autocorrelation	The ACF of r_t shows almost nothing at any lag
Strong autocorrelation in squares	The ACF of r_t^2 decays slowly over many lags
Volatility clustering	Large movements follow large movements, of either sign
Heavy tails	Far more extreme returns than a normal distribution allows
Aggregational normality	Monthly returns look far more normal than daily ones
The leverage effect	Falls raise future volatility more than rises of equal size

Table 27: Stylised facts of financial return series

The first two facts, taken together, are the entire motivation for this chapter. A series can be uncorrelated and still be highly predictable in its magnitude. That is precisely the gap between weak and strict white noise opened up earlier, and the models below live inside it.



Figure 46: A simulated return series with its two correlograms and a normal quantile plot. The returns carry no linear structure, their squares carry a great deal, and the tails are far too heavy for a normal.

The middle row is the diagnosis. An ARMA model reads the left panel, finds nothing, and correctly reports that no linear forecast beats the mean. The right panel is where the structure has been sitting the whole time.

7.3.3 What goes wrong if the variance is left constant

Suppose a constant variance model is fitted to the series above and used to produce prediction intervals. The intervals have one width, computed from the average volatility over the whole sample. During the calm stretches they are far too wide, so positions are sized too conservatively and opportunities are passed over. During the turbulent stretches they are far too narrow, so the model reports that a loss is almost impossible on precisely the days it is most likely.

The point forecasts are unaffected. Nothing in this chapter improves them. What changes is the honesty of the uncertainty attached to them, which for risk work is the whole product.

7.4 Testing for ARCH effects

Before fitting a variance model it is worth establishing that one is needed. Both standard tests work on the residuals of the fitted conditional mean model, since it is the innovations, not the raw series, whose variance is in question.

The first is the Ljung and Box statistic of the ARIMA chapter, applied to the **squared** residuals rather than the residuals. It asks whether any of the first H autocorrelations of $\hat{\varepsilon}_t^2$ differ from zero.

The second is Engle's Lagrange multiplier test, aimed directly at the ARCH alternative.

Engle's LM test. Regress $\hat{\varepsilon}_t^2$ on a constant and its own first q lags,

$$\hat{\varepsilon}_t^2 = a_0 + a_1 \hat{\varepsilon}_{t-1}^2 + \cdots + a_q \hat{\varepsilon}_{t-q}^2 + u_t,$$

and form nR^2 from that auxiliary regression. Under the null hypothesis of no ARCH effect, nR^2 is approximately χ^2 with q degrees of freedom.

The logic is worth stating in words, because it is the whole idea of the chapter in one sentence: if yesterday's squared shock helps predict today's squared shock, then the variance is not constant.

```
#>
#> ARCH LM-test; Null hypothesis: no ARCH effects
#>
#> data: e_sf
#> Chi-squared = 68.818, df = 5, p-value = 1.805e-13
#>
#> Box-Ljung test
#>
#> data: e_sf^2
#> X-squared = 231.05, df = 10, p-value < 2.2e-16
```

A small p-value says the squared residuals are serially correlated, which is the signature of time varying conditional variance. It does not say which model to fit.

7.5 The ARCH model

7.5.1 The idea

Engle's insight was that the variance could be given its own equation, of exactly the kind already familiar from the conditional mean. If a large shock today makes a large shock tomorrow more likely, then let today's variance depend on yesterday's squared shock, and the clustering follows automatically.

The **ARCH**(q) model of Engle (1982) writes the series as a conditional mean plus a scaled innovation,

$$y_t = \mu_t + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \text{ iid with } E(z_t) = 0, \text{ Var}(z_t) = 1,$$

with the conditional variance driven by past squared shocks,

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_q \varepsilon_{t-q}^2, \quad \alpha_0 > 0, \alpha_i \geq 0.$$

Three features of that specification deserve attention.

There are two equations, not one. The conditional mean μ_t can be a constant, an ARMA model from the previous chapter, or a regression on explanatory variables. Nothing in the variance equation constrains it. This is a second equation added to the first, not a replacement for it.

There are two noise terms, doing different jobs. The standardised innovation z_t is *strict* white noise: independent, identically distributed, carrying no information of any kind about its own future. The innovation ε_t is only *weak* white noise: uncorrelated, but with a magnitude that is partially predictable from yesterday. Everything interesting in this chapter happens in the gap between those two statements.

The constraints are not decoration. A variance cannot be negative, and since σ_t^2 is a linear combination of squares, the only way to guarantee positivity for every possible history is to require $\alpha_0 > 0$ and $\alpha_i \geq 0$.

7.5.2 Properties of ARCH(1)

Take the simplest case, $\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2$.

The mean is zero, conditionally and unconditionally. Since $E(z_t) = 0$ and z_t is independent of σ_t , which depends only on information dated $t-1$ and earlier,

$$E(\varepsilon_t \mid \mathcal{J}_{t-1}) = \sigma_t E(z_t) = 0, \quad E(\varepsilon_t) = 0.$$

So adding a variance equation does not disturb the conditional mean, and the two parts of the model can be read separately.

The innovations are uncorrelated. For $h \neq 0$, $\text{Cov}(\varepsilon_t, \varepsilon_{t+h}) = 0$. The process therefore satisfies the definition of white noise from the earlier chapter, exactly as claimed there, which is why an ACF of returns shows nothing.

The long run variance. Taking expectations of the variance equation and writing $\bar{\sigma}^2 = E(\sigma_t^2) = E(\varepsilon_t^2)$, which assumes stationarity in the same way the AR(1) derivation did,

$$\bar{\sigma}^2 = \alpha_0 + \alpha_1 \bar{\sigma}^2 \quad \implies \quad \bar{\sigma}^2 = \frac{\alpha_0}{1 - \alpha_1}, \quad 0 \leq \alpha_1 < 1.$$

The parallel with the AR(1) mean $\mu = c/(1 - \phi)$ is not a coincidence. Squared shocks behave, to a close approximation, like an autoregression, and α_1 plays the part of ϕ .

The tails come out heavy on their own. Suppose z_t is normal. The observed series ε_t is then a mixture of normal distributions with different variances, and a mixture of normals is not a normal. Its kurtosis exceeds three, and the larger α_1 is, the heavier the tails become. A model built from normal parts explains one of the stylised facts it was never asked to explain, which is a good sign about the mechanism.

7.5.3 Interpreting an ARCH(1) fit

An ARCH(1) model is fitted to daily returns and gives $\hat{\alpha}_0 = 0.04$, $\hat{\alpha}_1 = 0.45$.

The long run variance is $0.04/(1 - 0.45) \approx 0.0727$, so the long run standard deviation is about 0.27 per cent per day. That is the level volatility returns to when nothing is happening.

Yesterday's squared return receives a weight of 0.45. A shock four times the usual size contributes $\varepsilon_{t-1}^2 = 16\bar{\sigma}^2 = 1.164$, so today's conditional variance becomes

$$\sigma_t^2 = 0.04 + 0.45(1.164) \approx 0.564,$$

nearly eight times its resting level. One large day therefore has a very large effect on the next.

But only on the next. Two days later, if the intervening day is ordinary, the variance is back near $0.04 + 0.45\bar{\sigma}^2 \approx 0.073$. ARCH(1) has a memory of exactly one period, which is its central weakness.

7.5.4 Why ARCH(q) is rarely enough

Fitted to real returns, ARCH models need long lag structures. The squared returns of a daily series typically show significant autocorrelation out to twenty or thirty lags, and matching that with a pure ARCH model means estimating twenty or thirty coefficients, all of them constrained to be non-negative, most of them poorly determined.

The problem is the one that motivated ARMA over pure AR in the conditional mean: a slowly decaying correlation structure is expensive to represent with lags alone and cheap to represent with feedback. The solution takes the same form.

7.6 The GARCH model

The **GARCH**(p, q) model of Bollerslev (1986) lets the conditional variance depend on its own past as well as on past squared shocks,

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2,$$

with $\alpha_0 > 0$, $\alpha_i \geq 0$ and $\beta_j \geq 0$.

In practice $p = q = 1$ is almost always the answer. GARCH(1,1) fits an enormous range of financial series adequately, and higher orders rarely repay the parameters they cost. A useful working rule is to fit GARCH(1,1) first and to treat any apparent need for more as a sign that the conditional mean is misspecified or that the sample spans a structural break.

7.6.1 GARCH(1,1) is an ARCH process with infinite memory

The single most useful way to understand the β_1 term is to unroll the recursion. Substituting for σ_{t-1}^2 , then for σ_{t-2}^2 , and so on,

$$\sigma_t^2 = \frac{\alpha_0}{1 - \beta_1} + \alpha_1 (\varepsilon_{t-1}^2 + \beta_1 \varepsilon_{t-2}^2 + \beta_1^2 \varepsilon_{t-3}^2 + \dots).$$

GARCH(1,1) is therefore an ARCH model of infinite order whose weights decline geometrically. Every past shock contributes, with older shocks counting less, and the rate at which their influence fades is set by β_1 . Three parameters generate an unbounded sequence of ARCH weights, which is exactly the bargain ARMA struck against AR in the conditional mean.

This also explains why β_1 is usually large in fitted models. Volatility has long memory, and a small β_1 would truncate that memory after a few days.

7.6.2 What the three parameters mean

This is the part to remember.

Parameter	Governs	How to read it
α_0	The level	Sets $\bar{\sigma}^2$ together with the other two; on its own it is not interpretable
α_1	Reaction	How sharply volatility jumps on a single large day
β_1	Memory	How long that jump takes to fade
$\alpha_1 + \beta_1$	Persistence	Overall speed of return to the long run level

Table 28: Reading a fitted GARCH(1,1)

Two markets can share a persistence and behave quite differently. A market with $\alpha_1 = 0.05$ and $\beta_1 = 0.91$ has a smooth volatility path: single days barely move the estimate, and what movement there is lasts a long time. A market with $\alpha_1 = 0.25$ and $\beta_1 = 0.71$ has the same persistence and a spiky path, jumping hard on news and then decaying over the same horizon. Reading only the sum discards that distinction.

7.6.3 Long run variance and persistence

Taking expectations as before,

$$\bar{\sigma}^2 = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}, \quad \alpha_1 + \beta_1 < 1.$$

The condition $\alpha_1 + \beta_1 < 1$ is the stationarity condition for the variance, and it plays the same role that $|\phi| < 1$ played for the AR(1). If it fails, there is no long run level for volatility to return to.

A convenient summary of the persistence is the **half life**, the number of periods for a deviation

from $\bar{\sigma}^2$ to decay by half,

$$h_{1/2} = \frac{\ln 0.5}{\ln(\alpha_1 + \beta_1)}.$$

Persistence	Half life	Character
0.80	About 3 periods	Volatility settles within the week
0.95	About 14 periods	Typical of weekly data
0.99	About 69 periods	Typical of daily equity returns
1.00	Infinite	Integrated: shocks never decay

Table 29: Persistence and the half life of a volatility shock

Estimated persistence on daily equity returns is routinely between 0.98 and 0.995. That is a substantive finding rather than an estimation artefact: volatility shocks take a long time to fade, and a market disturbed today is still slightly disturbed two months later.

A GARCH(1,1) fitted to daily index returns gives $\hat{\alpha}_0 = 0.012$, $\hat{\alpha}_1 = 0.085$, $\hat{\beta}_1 = 0.905$.

Is it stationary? Persistence is $0.085 + 0.905 = 0.990 < 1$, so yes, though only just.

What is the resting level? $\bar{\sigma}^2 = 0.012/0.010 = 1.20$, so the long run daily standard deviation is $\sqrt{1.20} \approx 1.10$ per cent. Annualising by $\sqrt{252}$ gives about 17.4 per cent, which is a plausible figure for an equity index.

How does it respond? Only 8.5 per cent of the weight sits on the most recent squared shock, so a single large day moves the forecast modestly. The remaining 90.5 per cent carries yesterday's level forward, so the effect of that day persists.

For how long? The half life is $\ln 0.5 / \ln 0.990 \approx 69$ trading days, roughly three months.

Reacting slowly and forgetting slowly is the characteristic GARCH signature for an index. An individual share would typically show a larger $\hat{\alpha}_1$.

7.6.4 A note on identification

The ACF and PACF table of Table 22 cannot be used to choose p and q here. The reason is that squared innovations from a GARCH(1,1) behave like an ARMA(1,1) rather than an AR, and Table 22 says an ARMA shows gradual decay in both correlograms with no cut off in either. The correlograms of the squares therefore establish that *some* volatility model is needed, and say nothing about which.

Orders are chosen instead by information criteria and residual diagnostics, and in practice by starting at GARCH(1,1) and stopping there unless the diagnostics object.

7.7 Forecasting volatility

7.7.1 One step ahead requires no forecasting

Look carefully at

$$\sigma_{T+1}^2 = \alpha_0 + \alpha_1 \varepsilon_T^2 + \beta_1 \sigma_T^2.$$

Every term on the right is known at time T . Tomorrow's conditional variance is not an estimate in the usual sense; given the parameters, it is a computed quantity.

This is what makes GARCH useful in practice. Setting a margin requirement, a position limit or a one day value at risk all need tomorrow's uncertainty before tomorrow happens, and the model supplies it directly.

7.7.2 Further ahead, forecasts revert

Beyond one step the squared shocks are unknown and are replaced by their expectations. The result is a recursion of exactly the form solved for the AR(1), whose solution is

$$\hat{\sigma}_{T+h}^2 = \bar{\sigma}^2 + (\alpha_1 + \beta_1)^{h-1} (\sigma_{T+1}^2 - \bar{\sigma}^2).$$

Read the formula from left to right. The forecast starts wherever volatility currently sits and decays geometrically back towards the long run level, at a rate set by the persistence. Fitted after a crisis it forecasts a decline; fitted during a quiet spell it forecasts a rise.

That mean reversion is a real prediction and a testable one. It is also the property that distinguishes a volatility model from simply using recent volatility as a forecast of future volatility.

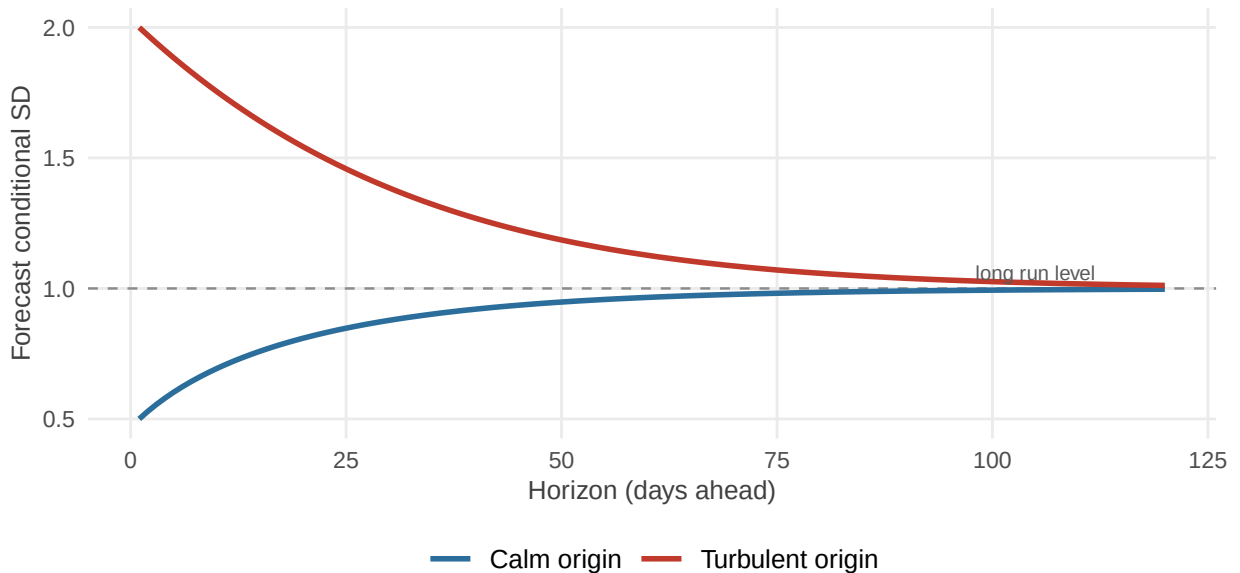


Figure 47: Volatility forecasts from a GARCH(1,1) with persistence 0.96, made from a turbulent origin and from a calm one. Both converge on the long run level, from opposite directions.

7.7.3 Over a whole horizon

Risk questions usually concern a period rather than a day. Since returns are uncorrelated, the variance of the cumulative return over h periods is the sum of the individual conditional variances,

$$\text{Var}\left(\sum_{k=1}^h r_{T+k} \mid \mathcal{J}_T\right) = \sum_{k=1}^h \hat{\sigma}_{T+k}^2.$$

Contrast this with the square root of time rule used widely in practice, which multiplies the current daily variance by h . That rule is the special case obtained by holding $\hat{\sigma}_{T+k}^2$ fixed at σ_{T+1}^2 , that is, by pretending volatility never reverts. Applied at a turbulent origin it overstates long horizon risk; applied at a calm one it understates it. The GARCH sum is the correction, and computing it is the main reason to bother fitting the model for anything beyond one day.

7.7.4 Intervals that breathe

With a conditional mean model and a GARCH variance, the h step prediction interval is

$$\hat{y}_{T+h|T} \pm q_{1-\alpha/2} \sqrt{\text{Var}(e_{T+h} \mid \mathcal{J}_T)},$$

where q is a quantile of the standardised innovation distribution and the variance now moves with the volatility forecast.

A closely related quantity is the value at risk, the loss exceeded with some stated small probability,

$$\text{VaR}_\alpha = \hat{\mu}_{T+1|T} + q_\alpha \hat{\sigma}_{T+1|T}.$$

Under a constant variance model this is the same number every day, which is less a description of risk than a refusal to describe it.

7.8 Estimation

There is no least squares route here, for the same reason there was none for the moving average part of an ARMA model: the conditional variances σ_t^2 are never observed. There is no column of data to regress on. Instead the variances are reconstructed by running the recursion forward from a starting value, usually the sample variance, and the parameters are chosen to maximise the resulting likelihood.

Three practical points follow, and none of them requires the theory behind them.

Estimation is numerical and constrained. The coefficients must be non-negative and the persistence below one, so the optimiser works on a bounded region and can fail to converge. Trying a different solver, or rescaling the returns to percentages, usually fixes it.

Report the robust standard errors. Fitting by Gaussian likelihood gives reliable coefficient estimates even when the innovations are not normal, but the ordinary standard errors are then wrong. Every reputable package reports a corrected set alongside, and those are the ones to read. `rugarch` labels them *robust standard errors* in the output.

Choose the innovation distribution deliberately. Normal innovations produce heavy tailed returns, as noted earlier, but on daily data usually not heavy enough. The standard remedy is a Student t with ν degrees of freedom, rescaled to unit variance so that σ_t keeps its meaning. Estimated $\hat{\nu}$ on daily equity returns is commonly between four and eight. Skewed variants exist and matter for value at risk, where the two tails are not interchangeable. Fit with the normal first, look at the quantile plot of the standardised residuals, and let it tell you whether something heavier is needed.

7.9 Diagnostics

The residuals to examine are the **standardised** residuals,

$$\hat{z}_t = \frac{\hat{\varepsilon}_t}{\hat{\sigma}_t},$$

because it is these, not $\hat{\varepsilon}_t$, that the model claims are independent and identically distributed. Checking $\hat{\varepsilon}_t$ for constant variance would be checking an assumption nobody made.

Check	Applied to	What failure means
Ljung and Box	$\hat{z}_t$	The conditional mean model is inadequate
Ljung and Box	$\hat{z}_t^2$	The variance model is inadequate
Engle LM test	$\hat{z}_t$	ARCH effects remain unmodelled
Quantile plot	$\hat{z}_t$	The innovation distribution is wrong
Refit on subsamples	Whole model	A structural break in the sample

Table 30: Diagnostics for a fitted volatility model

The distinction between the first two rows matters and is easy to get wrong. Correlation left in $\hat{z}_t$ sends you back to the mean equation; correlation left in $\hat{z}_t^2$ sends you back to the variance equation. Confusing them wastes a good deal of time.

The degrees of freedom correction of the ARIMA chapter applies to both portmanteau tests, and `rugarch` reports weighted versions with better properties for squared standardised residuals than the plain statistic.

7.10 A worked application

The workflow below is the one to reproduce on any return series: plot, test, fit, diagnose, interpret, forecast.

```
library(rugarch)

spec_fit <- ugarchspec(
  variance.model = list(model = "sGARCH", garchOrder = c(1, 1)),
  mean.model     = list(armaOrder = c(0, 0), include.mean = TRUE),
  distribution.model = "std"      # Student t, degrees of freedom estimated
```

```
)

fit_g <- ugarchfit(spec_fit, data = e_sf, solver = "hybrid")
round(coef(fit_g), 4)
#>      mu      omega alpha1  beta1  shape
#> -0.0431  0.0203  0.0776  0.8984  5.4360

sig_g <- as.numeric(sigma(fit_g))

tibble(Time = 1:n_sf, r = e_sf, s = sig_g) |>
  ggplot(aes(Time)) +
  geom_line(aes(y = r), colour = "grey65", linewidth = 0.25) +
  geom_line(aes(y = 2 * s), colour = col_pred, linewidth = 0.5) +
  geom_line(aes(y = -2 * s), colour = col_pred, linewidth = 0.5) +
  labs(x = "Day", y = expression(r[t]))
```

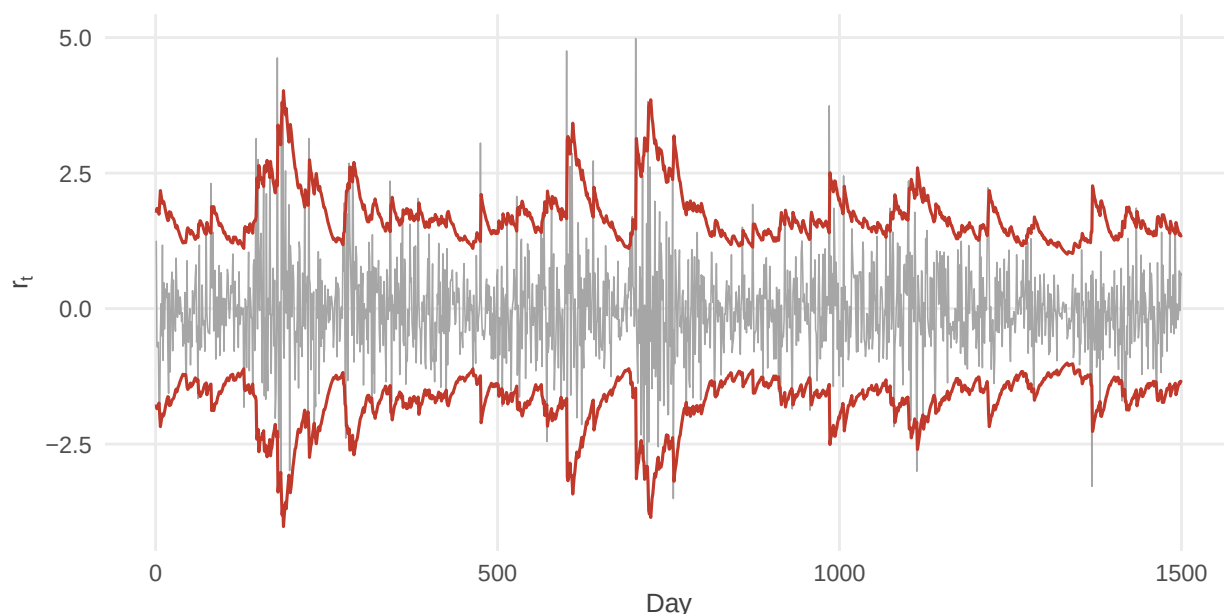


Figure 48: A GARCH(1,1) with Student t innovations fitted to the simulated returns, showing the returns against plus and minus two fitted conditional standard deviations.

The fitted band tracks the series: wide where the returns are wide, and narrow where they are calm. A constant variance model would draw two horizontal lines instead, and about five per cent of the observations would sit outside them regardless of the market conditions on the day.

```
#> persistence  half_life long_run_sd
#>  0.9760923  28.6446566  0.9205961
```

Note the name `omega` in the coefficient vector, which is the package's label for α_0 .

```
#>
```

```

#> Box-Ljung test
#>
#> data:  z_hat
#> X-squared = 4.0388, df = 10, p-value = 0.9456
#>
#> Box-Ljung test
#>
#> data:  z_hat^2
#> X-squared = 9.6307, df = 10, p-value = 0.4735
#>
#> ARCH LM-test; Null hypothesis: no ARCH effects
#>
#> data:  z_hat
#> Chi-squared = 2.6744, df = 5, p-value = 0.75

```

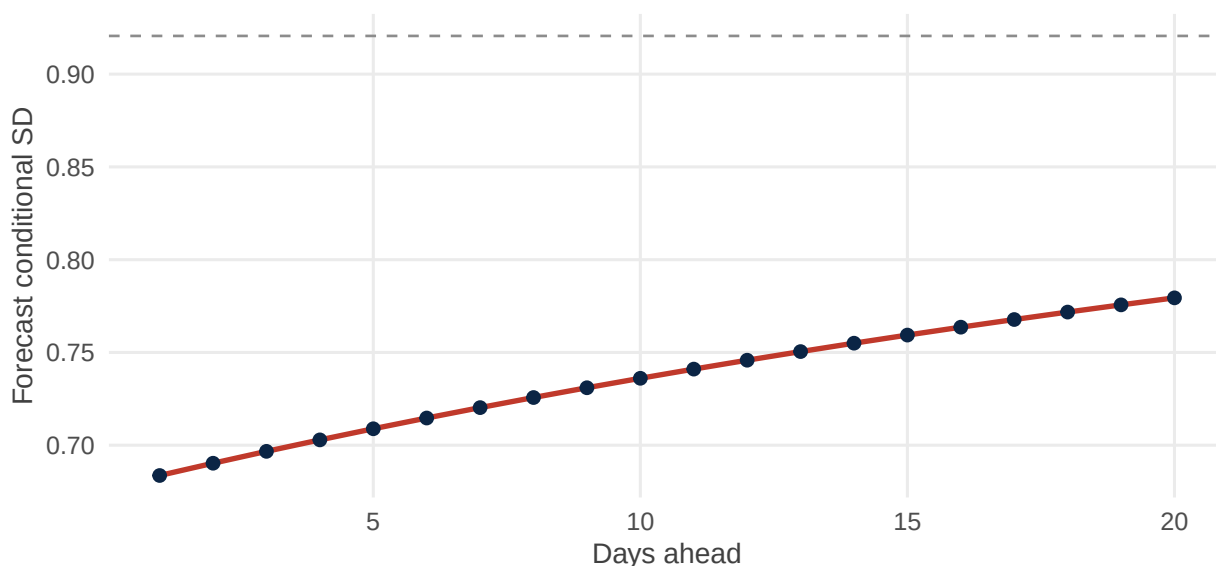


Figure 49: Twenty day ahead conditional standard deviation forecast from the fitted model, with the estimated long run level marked.

To fit an ARMA conditional mean alongside the GARCH variance, set `armaOrder` in the mean specification and both equations are estimated jointly. Fitting the ARMA first and a GARCH to its residuals afterwards is a common shortcut; it gives reasonable coefficients but the two stage standard errors are wrong.

For real data, `quantmod::getSymbols()` retrieves prices and `PerformanceAnalytics::CalculateReturns()` or a direct `diff(log(price))` produces the returns. Everything above then applies unchanged.

7.11 Asymmetry: when good and bad news differ

Both models so far depend on ε_{t-1}^2 , which discards the sign of the shock. A fall of three per cent and a rise of three per cent have identical consequences for tomorrow's variance.

Equity markets do not behave that way. Falls raise subsequent volatility considerably more than

risers of the same size. The name **leverage effect** comes from one proposed explanation, that a fall in equity value raises a firm's debt to equity ratio and so makes the remaining equity riskier, though the effect appears in markets where that story does not apply and the name has outlived the explanation.

Two models are in common use, and for an Honours course it is enough to know what each does and when to reach for it.

The **GJR-GARCH(1,1)** model of Glosten, Jagannathan and Runkle adds a term that switches on for negative shocks,

$$\sigma_t^2 = \alpha_0 + (\alpha_1 + \delta \mathbb{1}[\varepsilon_{t-1} < 0]) \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2,$$

where $\mathbb{1}[\cdot]$ is one when the condition holds and zero otherwise.

The interpretation is direct. A positive shock has effect α_1 ; a negative shock has effect $\alpha_1 + \delta$. Testing $\delta = 0$ is a test of the leverage effect, and the estimated $\hat{\delta}$ is often several times $\hat{\alpha}_1$ on equity data. Because the indicator is on about half the time, the persistence becomes $\alpha_1 + \delta/2 + \beta_1$, and it is that sum which must be below one.

The **EGARCH(1,1)** model of Nelson models the logarithm of the variance,

$$\ln \sigma_t^2 = \alpha_0 + \beta_1 \ln \sigma_{t-1}^2 + \alpha_1 (|z_{t-1}| - E|z_{t-1}|) + \delta z_{t-1},$$

with $z_t = \varepsilon_t / \sigma_t$.

Two features recommend it. Because the left hand side is a logarithm, σ_t^2 is positive whatever the coefficients, so no non-negativity constraints are needed and estimation is better behaved. And the asymmetry enters through the *signed* standardised shock δz_{t-1} , cleanly separated from the magnitude effect carried by α_1 , so a negative $\hat{\delta}$ means falls raise volatility.

The cost is interpretability: there is no simple long run variance, and parameterisations differ between texts and packages, so read the documentation before interpreting a fitted δ .

In `rugarch`, both are a change of one argument, `model = "gjrGARCH"` or `model = "eGARCH"`, and the fitted objects support the same extractors. The practical guidance is to fit the symmetric model first, then the asymmetric one, and keep the asymmetric version if $\hat{\delta}$ is significant and the information criteria agree.

7.12 Very high persistence

Estimated persistence close to one raises the same question that $\hat{\phi}$ close to one raised in the ARIMA chapter.

When $\alpha_1 + \beta_1 = 1$ exactly, the model is **IGARCH**. The long run variance does not exist, volatility forecasts do not revert, and the forecast function becomes a straight line in h . Setting $\alpha_0 = 0$ as well gives the exponentially weighted moving average used in the RiskMetrics methodology,

$$\sigma_t^2 = (1 - \lambda) \varepsilon_{t-1}^2 + \lambda \sigma_{t-1}^2,$$

with $\lambda = 0.94$ conventional for daily data. It is a one parameter model that performs respectably at short horizons and badly at long ones, for the obvious reason.

Before accepting near unit persistence at face value, consider the alternative explanation. A structural break in the level of volatility, of the kind Figure 23 showed for the mean, biases estimated persistence towards one: the model can only reconcile two different volatility regimes by concluding that shocks never die. Splitting the sample and refitting takes five minutes, and a persistence that drops substantially within each regime indicates a break rather than long memory.

7.13 Where this goes wrong

Fitting to prices. The variance equation describes innovations around a conditional mean, not levels. Transform to returns first.

Reading β_1 as unimportant because α_1 is small. The persistence is the sum. A model with $\alpha_1 = 0.05$ is not one in which volatility barely responds; it is one in which volatility responds smoothly and forgets slowly.

Diagnosing on the wrong residuals. The model's claim concerns $\hat{\varepsilon}_t/\hat{\sigma}_t$. Testing $\hat{\varepsilon}_t$ for constant variance rejects an assumption nobody made.

Ignoring the mean equation. Autocorrelation left in the returns contaminates the squared residuals and produces spurious ARCH effects. Get the conditional mean right first, even if right means a constant.

Using the square root of time rule after fitting GARCH. The model supplies the correct aggregation. Fitting it and then discarding it lends unearned authority to the wrong number.

Treating high persistence as established fact. Check for breaks before concluding that shocks last for months.

7.14 Summary

The conditional mean and the conditional variance are separate models of separate things, and a complete description of a financial series needs both.

The variance model is built from the same materials as the mean model. ARCH lets today's variance depend on yesterday's squared shock, which produces clustering but forgets too quickly. GARCH adds feedback from yesterday's variance, which turns the model into an ARCH process with geometrically fading infinite memory and makes it usable. Of its three parameters, α_1 measures how sharply volatility reacts, β_1 measures how long the reaction lasts, and their sum measures overall persistence and sets the half life.

Volatility forecasts revert geometrically to a long run level $\alpha_0/(1 - \alpha_1 - \beta_1)$ at a rate the persistence determines, which is what makes the model useful for horizons beyond one day and what distinguishes it from the rules of thumb it replaces.

The payoff is not a better point forecast, which GARCH does not provide, but an honest interval. And the reason the whole apparatus is needed is the gap between the two definitions of white noise given earlier: returns are uncorrelated, so nothing linear predicts them, and yet their magnitudes are highly predictable, so a model that stops at uncorrelatedness has stopped too early.

8 Spectral Analysis

The chapter on two perspectives set out the second question a time series can be asked. Rather than asking how yesterday influences today, the frequency domain asks which oscillations are present in a series, and how much of its variation each one accounts for. That question was posed there and set aside while the time domain machinery was built. This chapter answers it.

Nothing here competes with ARIMA modelling. The two domains describe the same stochastic process and contain the same information. What differs is what each makes obvious. A twelve month cycle buried in noise is a diffuse pattern in an ACF and a single spike in a periodogram, and several superimposed cycles of different lengths are close to unreadable in an ACF and immediately legible in a spectrum. Choosing a domain is choosing which facts you would like to be able to see at a glance.

8.1 Learning outcomes

By the end of this chapter you should be able to:

1. Convert between period, frequency and angular frequency, and state which of these R's `ts()` expects.
2. Explain the Nyquist frequency and what aliasing does to a series.
3. Explain what a periodogram displays and how it relates to the variance of the series.
4. Read a periodogram: locate the dominant frequencies, convert them to periods, and report them in the units of the data.
5. Distinguish a real peak from noise, and explain why a raw periodogram must be smoothed.
6. Recognise the spectral signature of white noise, an autoregression and a seasonal series.
7. Produce and interpret periodograms in R.
8. Say when a periodogram answers a question better than an ACF, and when it does not.

8.2 Additional notation

Symbol	Meaning
f	Frequency in cycles per observation, as in Table 1
ω	Angular frequency, $\omega = 2\pi f$, in radians per observation
f_N	Nyquist frequency, $f_N = 1/2$
$I(f)$	Periodogram ordinate at frequency f
$S(f)$	Spectral density at frequency f
$\hat{S}(f)$	Smoothed estimate of the spectral density

Table 31: Symbols added in this chapter

One notational point requires care. Many texts write the spectral density as $f(\omega)$, which is impossible here because f already denotes frequency. This chapter writes $S(f)$ for the spectral density and reserves f for the frequency itself.

Recall also, from the conventions of the first chapter, that R's `ts()` argument **frequency** is the period m , not the frequency $f = 1/m$. That collision causes more trouble in this chapter than anywhere else, because both quantities now appear in the same code.

8.3 Frequency, period and the sampling limit

8.3.1 Three ways of saying the same thing

A sinusoid of period m completes one cycle every m observations. Its frequency is $f = 1/m$ cycles per observation, and its angular frequency is $\omega = 2\pi f$ radians per observation. All three describe the same oscillation.

Data	Cycle	m	f	Interpretation
Monthly	Annual	12	0.0833	8.3 per cent of a cycle per month
Quarterly	Annual	4	0.25	A quarter cycle each quarter
Hourly	Daily	24	0.0417	One cycle per day
Hourly	Weekly	168	0.00595	One cycle per week
Daily	Weekly	7	0.1429	One cycle per week

Table 32: Periods and frequencies for common sampling intervals

Always convert an estimated frequency back to a period before reporting it. A peak at $f = 0.0833$ means nothing to a reader; a cycle of twelve months means something immediately. This is the single most common failing in student write-ups of spectral analysis.

8.3.2 The Nyquist frequency and aliasing

Sampled data cannot resolve arbitrarily fast oscillation. Two observations are the minimum needed to describe one cycle, one at the peak and one at the trough, so the fastest frequency a series can represent is

$$f_N = \frac{1}{2} \text{ cycle per observation,}$$

a period of two observations. This is the **Nyquist frequency**, and it is a property of how often you sampled, not of the process you sampled.

Oscillations faster than f_N do not disappear. They reappear at a lower frequency, folded back into the observable range: a true frequency f above the Nyquist appears at $|f - k|$ for whichever integer k brings it into $[0, 1/2]$. This is **aliasing**, and it is irreversible.

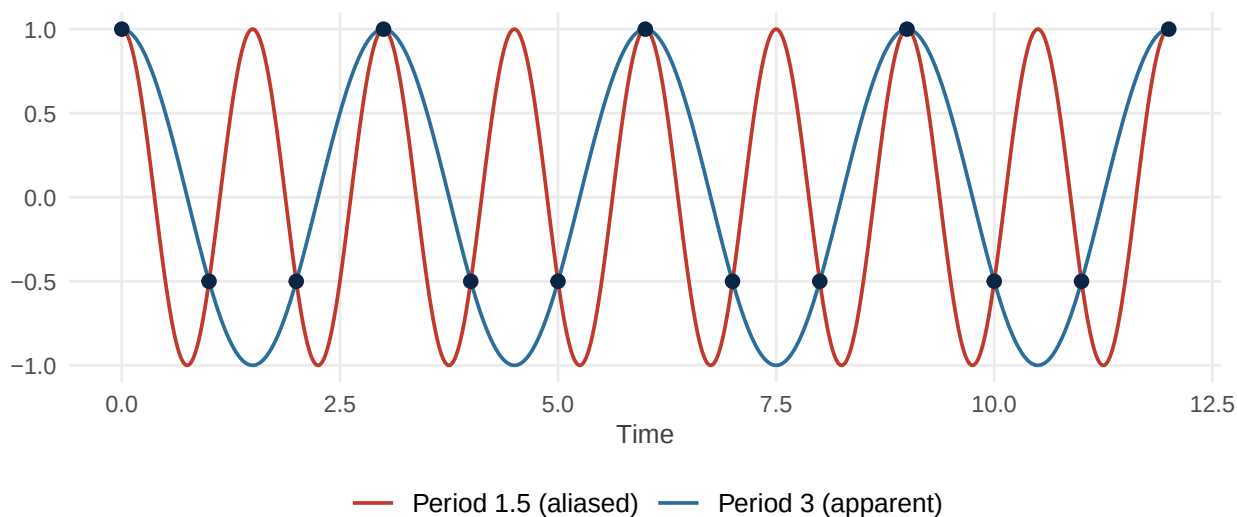


Figure 50: Two sinusoids with periods 1.5 and 3 observations. Sampled at the integers, marked by the points, they are indistinguishable.

No analysis recovers the original frequency, because the sampled values are numerically identical for both curves. The information is not obscured; it is absent.

The practical consequence is that the sampling interval must be chosen before the data are collected, with the fastest interesting oscillation in mind. Monthly data cannot answer a question about a weekly cycle.

8.4 A series as a sum of waves

The starting idea of spectral analysis is that any series can be written as a sum of sinusoids of different frequencies, and that the size of the coefficient at each frequency tells you how much of the series that oscillation accounts for.

A sinusoid with amplitude A , frequency f and phase φ can be written

$$A \cos(2\pi ft + \varphi) = a \cos(2\pi ft) + b \sin(2\pi ft),$$

with $A = \sqrt{a^2 + b^2}$. The right hand form is the useful one, because it is linear in a and b : fitting a sinusoid of known frequency is an ordinary regression with two columns.

Fit such a pair at every frequency of the form j/n , for $j = 1, \dots, \lfloor n/2 \rfloor$, and something convenient happens. These frequencies are the ones completing a whole number of cycles within the observed record, and at them the sine and cosine columns turn out to be mutually orthogonal. That means the fitted amplitudes do not interfere with one another, and the sum of squares of the series splits cleanly across frequencies with nothing left over.

The result is a decomposition of the variance of the series by frequency. It is an analysis of variance table whose rows are oscillations rather than treatments, and the periodogram of the next section is that table drawn as a picture. Nothing is estimated in the usual sense and nothing is lost: the transformation is a change of coordinates, writing the same information on a frequency axis instead of a time axis.

8.5 The periodogram

The **periodogram** $I(f)$ measures how much of the variation in a series is accounted for by an oscillation at frequency f . The total area under it is the sample variance of the series, so a tall ordinate at f means that oscillations of period $1/f$ explain a large share of the movement in the data.

That interpretation is the whole of what the periodogram means, and it is the sentence to remember. A peak at $f = 1/12$ in monthly data says that a twelve month oscillation accounts for a large part of the variance.

8.5.1 Computing one

Base R also supplies `spec.pgram()`, which is what most people use. Be aware that its defaults remove a linear trend, taper the ends of the series, pad it to a convenient length and rescale by `frequency(x)`. All four are usually desirable and all four change the numbers, so a hand computation will not match it unless those defaults are switched off.

8.5.2 A first example

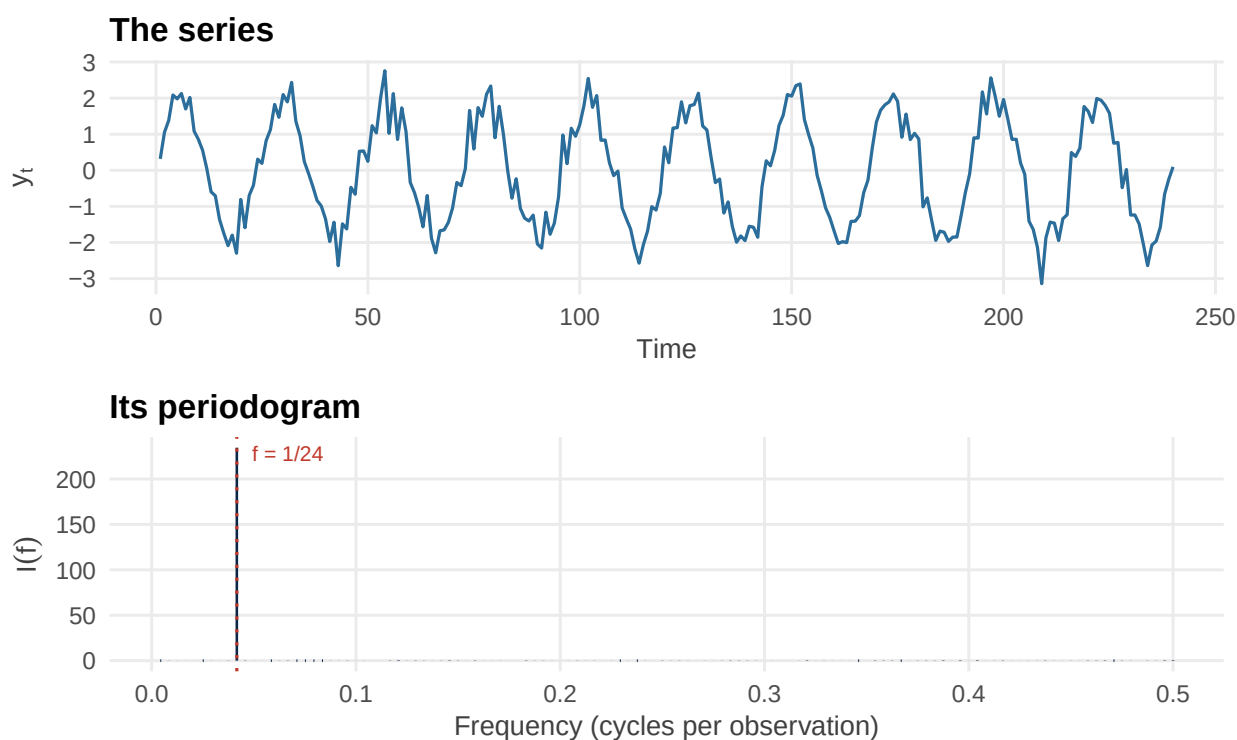


Figure 51: A cycle of period 24 buried in noise, and its periodogram. The single spike sits at $f = 1/24$.

The time plot shows a series that is evidently oscillating and gives no easy way to say how fast. The periodogram answers in one picture. This is the case for the frequency domain in its purest form, and it is worth noting how little the ACF of the same series would have added.

```
#> # A tibble: 3 x 3
```

```

#>      freq period      I
#>    <dbl> <dbl>   <dbl>
#> 1 0.0417   24    235.
#> 2 0.5       2     0.671
#> 3 0.121    8.28   0.615

```

8.6 Reading a periodogram

This section is the core of the chapter. Everything before it was preparation.

8.6.1 Step one: convert the peak to a period

Find the largest ordinate, read off its frequency, and invert it. Then attach the units of the data. A peak at $f = 0.0417$ in hourly data is a cycle of $1/0.0417 = 24$ hours, and that is how it should be reported.

8.6.2 Step two: check for harmonics

A deterministic seasonal pattern that is a pure sinusoid produces one peak. Almost no real seasonal pattern is a pure sinusoid.

Any repeating pattern of period m can be built from a sinusoid at the **fundamental** frequency $1/m$ together with **harmonics** at $2/m, 3/m$ and so on. The sharper the pattern, the more harmonics are needed, because building a narrow spike out of smooth curves takes many of them.

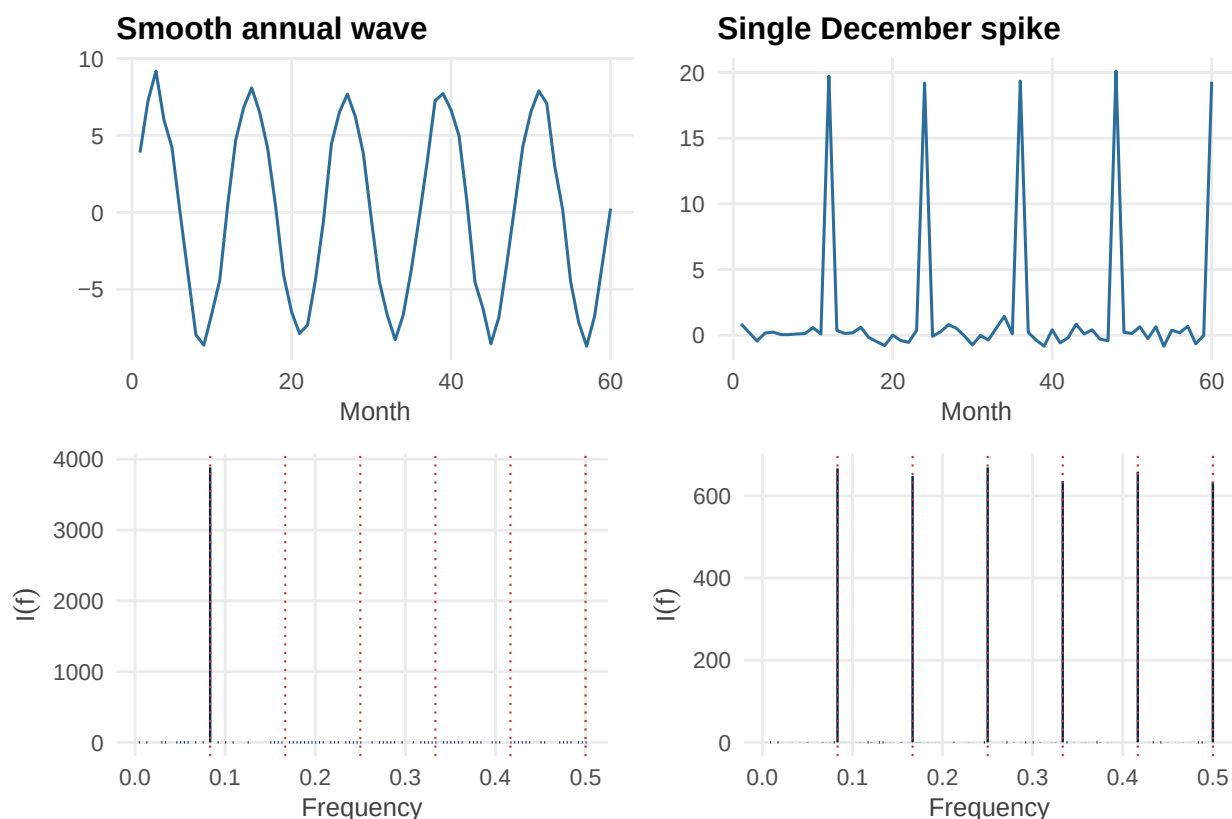


Figure 52: Two monthly seasonal patterns and their periodograms. The smooth wave needs only the fundamental at $1/12$; the December spike needs every harmonic.

The practical rule is that peaks at $1/12$, $2/12$ and $3/12$ in monthly data are usually one annual pattern that is not a pure sinusoid, not three separate cycles of twelve, six and four months. Report the fundamental and note the harmonics.

This also connects back to the ACF. Seasonal peaks appear in the ACF at lags $m, 2m, 3m$ and in the periodogram at frequencies $1/m, 2/m, 3/m$, and these are not the same phenomenon seen twice. The ACF peaks occur because the pattern repeats; the spectral peaks are the harmonics needed to construct the shape of one season. A perfectly sinusoidal seasonal gives ACF peaks at every multiple of m and only a single spectral line.

8.6.3 Step three: do not trust a single tall ordinate

Look again at the periodogram in Figure 51, away from the spike. The small ordinates are jagged, and they stay jagged. That is not a small sample problem.

The raw periodogram is **unbiased but not consistent**. Each ordinate $I(f)$ estimates $S(f)$ correctly on average, but its standard deviation is roughly equal to its own mean, and that does not shrink as n grows. A larger sample supplies more frequencies, each estimated just as badly as before.

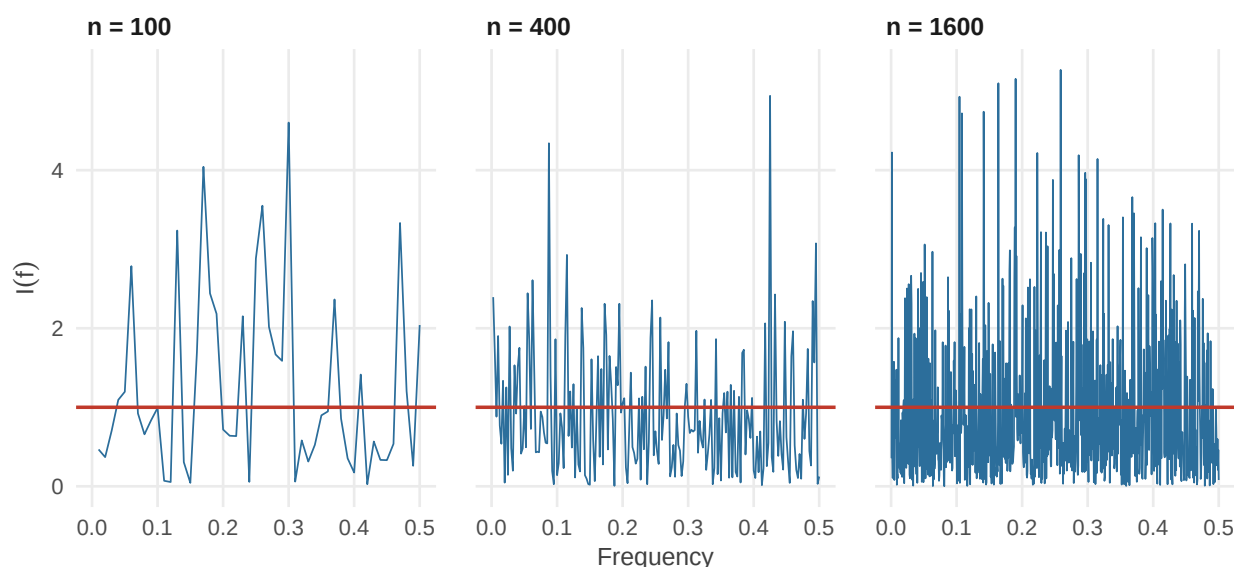


Figure 53: Raw periodograms of white noise at three sample sizes, against the true flat spectrum in red. The estimates do not settle down; there are simply more of them.

This single fact explains the rest of the chapter. A tall ordinate in a raw periodogram may be a real cycle or may be noise, and there are enough ordinates that some will always look impressive. Two remedies are used, and in practice both.

8.6.4 Step four: smooth, and see what survives

Since the underlying spectrum varies smoothly, neighbouring ordinates estimate nearly the same quantity, and averaging them reduces the noise. R's `span`s argument does this: `span = 9` averages roughly nine neighbouring ordinates, and a pair such as `span = c(9, 7)` applies two passes for a smoother result.

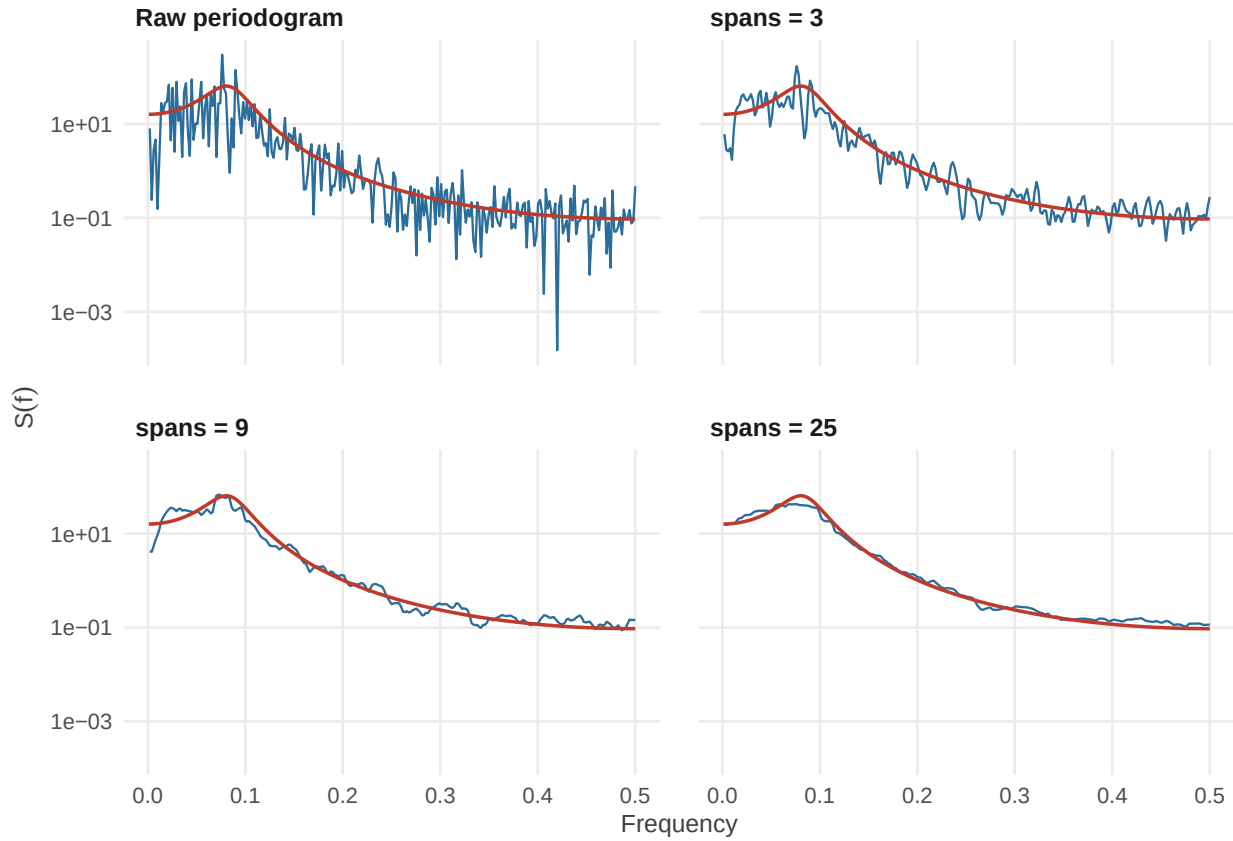


Figure 54: The same AR(2) series with no smoothing and with three increasing spans, against the true spectral density in red. Too little smoothing is noisy; too much flattens the peak.

There is a trade here and no formula that settles it. Too little smoothing leaves genuine features buried in noise; too much averages across them, flattening peaks and filling troughs. The working practice is to compute several and look at them together, keeping the least smoothing that produces a picture whose features survive when the span is increased a little. **A peak that vanishes when the span moves from nine to eleven was noise.**

Note also that spectra are conventionally plotted on a logarithmic vertical scale, and that R draws a single vertical bar in the corner rather than a band around the curve. The bar is a confidence interval, and on the log scale it has the same width at every frequency, so one bar serves for the whole plot.

8.6.5 Step five: test the largest peak

The eye finds patterns in noise, and Figure 53 shows how many tall ordinates white noise produces. A formal check is worth having. **Fisher's test** compares the largest ordinate with the total,

$$g = \frac{\max_j I(f_j)}{\sum_j I(f_j)},$$

and refers it to a null distribution that accounts for the fact that the maximum of many ordinates was selected after looking at them. Comparing the largest ordinate with a fixed threshold, without that correction, is the frequency domain version of testing the largest of many correlations at the five per cent level, and it manufactures cycles at a predictable rate.

```
#> [1] 0.2936269
#> [1] 2.495038e-140
```

One caution about interpretation. The null hypothesis is white noise, so the test asks whether a periodicity exists against *no* structure at all. On a series with ordinary serial correlation it will reject readily, because correlated noise has a non-flat spectrum whose largest ordinate is naturally large. If the question is whether a cycle exists over and above the autocorrelation, fit an ARMA model first and apply the test to the residuals.

8.7 Preparing a series

Three preparatory steps are routine, and all three are `spec.pgram()` defaults.

Remove the mean, and usually the trend. A non-zero mean puts an enormous ordinate at $f = 0$, and a trend puts enormous power at the lowest frequencies. That power does not stay put: it spreads across the whole band and can swamp everything of interest. Subtracting a fitted trend is the appropriate treatment here, and is gentler than differencing, which reshapes the whole spectrum rather than removing one feature.

Taper the ends. Observing a finite stretch of a series is like looking at it through a window with hard edges, and those edges cause power at one frequency to spill into neighbouring frequencies. This is called **leakage**, and it matters when the spectrum has a large dynamic range, since a strong low frequency component can leak enough power into the high frequency band to hide what is really there. Multiplying the first and last few per cent of the record by a smoothly rising and falling curve softens the edges and reduces the spill. R's default of ten per cent is modest; increase it when the spectrum is steep.

Padding is cosmetic. Appending zeros lengthens the series and so refines the frequency grid. It adds no information and improves no estimate. It can make a peak easier to locate by eye, and R's `fast = TRUE` pads for computational speed. Neither use is harmful provided nobody mistakes a finer grid for a better estimate.

8.8 What the spectrum looks like for familiar models

The population quantity being estimated is the **spectral density** $S(f)$, which distributes the variance of a stationary process across frequency. Its integral over all frequencies is the variance of the process, and the area over any band is the part of the variance contributed by oscillations in that band. That is the whole of its meaning.

Stationarity is doing real work here, exactly as it did in the definition of $\gamma(h)$. A process whose covariance structure drifts has no single $S(f)$ to estimate, and applying these methods to a non-stationary series produces a picture of something that does not exist.

For the models of the previous chapters the spectrum has a known shape, which makes the spectral view a useful check on time domain identification.

Process	Shape of $S(f)$
White noise	Flat at every frequency

Process	Shape of $S(f)$
MA(1)	Monotone, sloping up or down
AR(1), $\phi > 0$	Power concentrated at low frequencies
AR(1), $\phi < 0$	Power concentrated at high frequencies
AR(2), complex roots	A peak at an interior frequency
Seasonal	A narrow spike at $1/m$ and its harmonics

Table 33: Spectral signatures of familiar processes

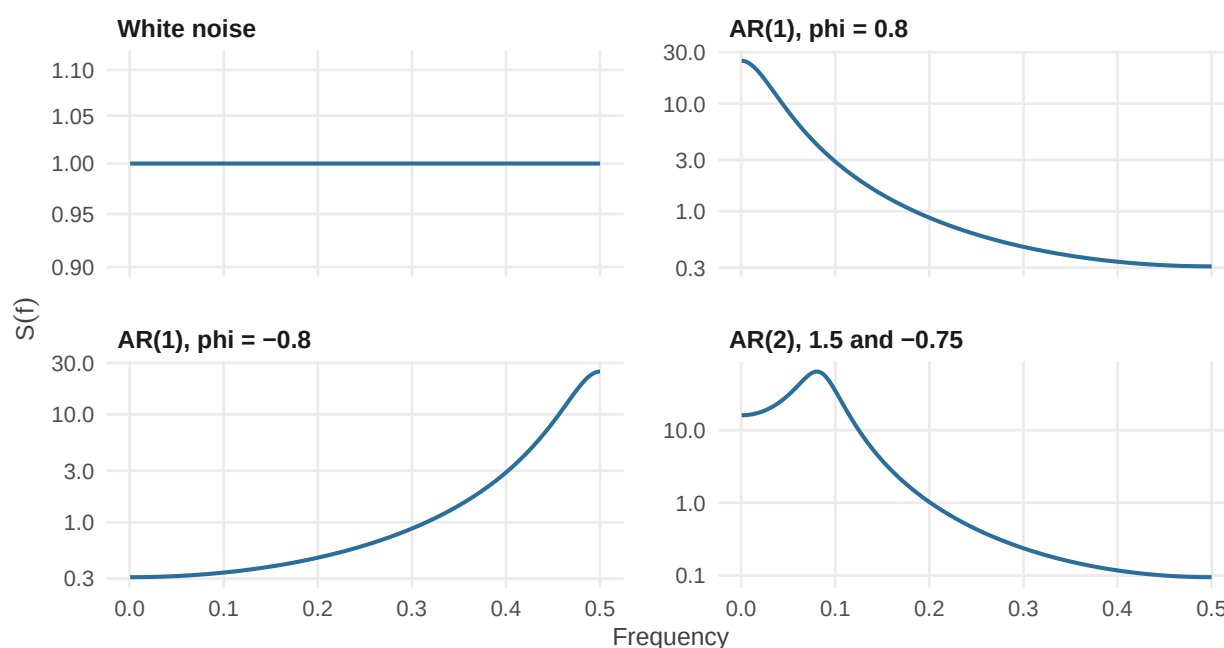


Figure 55: Theoretical spectral densities of four processes with unit innovation variance. The AR(2) peak sits near a period of twelve.

The white noise entry explains the name. A flat spectrum contains every frequency in equal measure, as white light contains every colour. By the same analogy an AR(1) with positive ϕ is often called red noise, since its power sits at the low frequency end, and one with negative ϕ blue noise.

The AR(2) panel is the interesting one, because it shows how a time domain model produces a cycle. When the roots of $\phi(z)$ are complex, the spectrum has a peak at an interior frequency, and the process oscillates with a length that is *approximately* constant. Note that the peak is broad rather than a spike. That width is the amount of variation in the cycle length, and it is the frequency domain version of the distinction the components chapter drew between a cycle and a seasonal pattern.

The AR(2) process $y_t = 1.5y_{t-1} - 0.75y_{t-2} + \varepsilon_t$ has complex roots, and its spectral peak sits at $f \approx 0.080$, a period of about 12.4 observations.

On monthly data this process would look seasonal to the eye and would not be. Its cycles average roughly a year in length but vary from one occurrence to the next, and nothing in the calendar ties them down. The broad peak in Figure 55 is the signature of that variation, and a truly seasonal

series would show a narrow spike instead.

This is a case where the spectrum settles a question the time plot cannot.

8.9 What differencing does to a spectrum

Filtering a series multiplies its spectrum, frequency by frequency, by a fixed curve determined by the filter. For the first difference $(1 - L)$ that curve is $4 \sin^2(\pi f)$, and it is worth looking at.

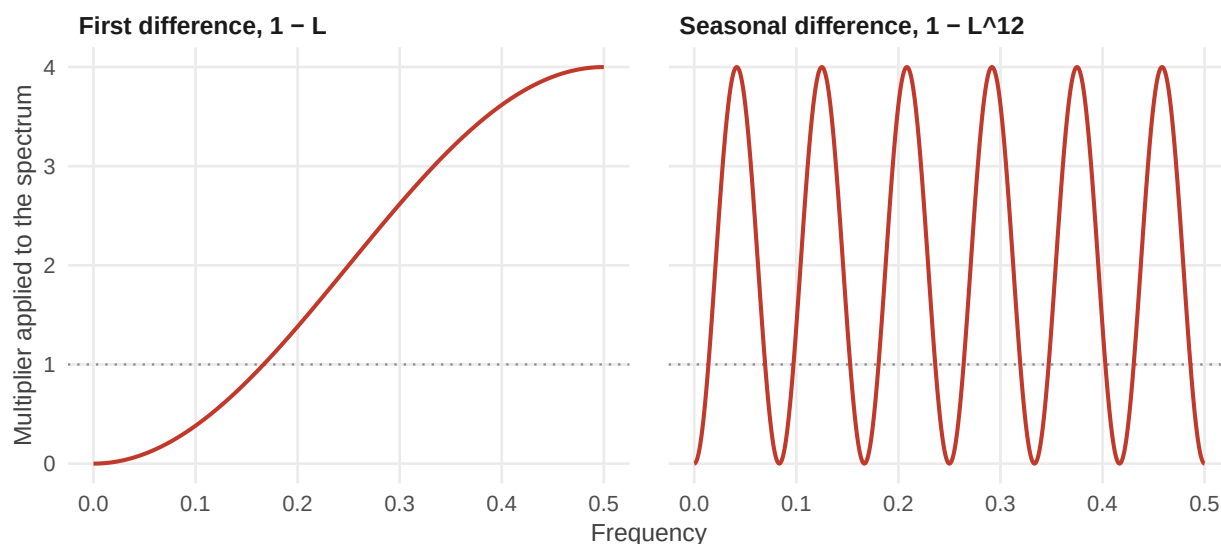


Figure 56: The effect of differencing on a spectrum. The first difference kills the zero frequency and quadruples the fastest one; the seasonal difference notches out the level and every seasonal harmonic.

Read the left panel. At $f = 0$ the multiplier is zero: differencing annihilates the zero frequency, which is the level and the trend, and that is what it was introduced to do. At $f = 1/2$ the multiplier is four: differencing *quadruples* the power at the fastest observable frequency.

So over-differencing is not a neutral act that merely fails to help. It removes a component that was not there and, in doing so, amplifies the fastest and least informative variation in the series. That is why the variance goes up rather than down, which was the diagnostic offered in the ARIMA chapter, and it is why the resulting series has $\rho(1)$ near -0.5 : a spectrum tilted steeply towards high frequencies is what an $MA(1)$ with $\theta = -1$ looks like from this side.

The right panel makes the same point about seasonal differencing. The operator $(1 - L^{12})$ removes the level and every seasonal harmonic exactly, and amplifies everything in between. Applied to a series whose seasonal pattern genuinely wanders, that is the right treatment. Applied to a fixed deterministic seasonal, whose power sits exactly on the notches, it removes something already exactly removable and leaves the amplified band behind, which is the over-differencing the SARIMA section warned about and the reason the fitted seasonal moving average parameter lands on -1 .

8.10 When to reach for a periodogram

Question	Time domain	Frequency domain
Is there dependence?	ACF and PACF	Spectrum departs from flat
How long is the cycle?	Awkward, from ACF peaks	Directly, from a spectral peak
Is the cycle regular?	Hard to judge	Peak width answers it
Are there several cycles at once?	Very hard	Immediate
What model should I fit?	Table 22	Not directly
Forecast the next value	Directly	Only via a fitted model
Handle non-stationarity	Differencing, transformation	Neither applies

Table 34: What each domain answers easily

The information content is identical. What differs is what has to be inferred and what can simply be read off, and the last two rows are why ARIMA rather than spectral analysis occupies most of a forecasting course. The frequency domain describes; it does not extrapolate, and it has nothing to say about a series whose statistical character is drifting.

The complementary case is equally clear. A series containing several superimposed periodicities of different lengths is close to unreadable in an ACF and immediately legible in a spectrum. When the question is *how long is the cycle*, reach for a periodogram.

8.11 Where this goes wrong

Applying it to a non-stationary series. There is no $S(f)$ to estimate. Remove the trend first, and remember that differencing is itself a filter that reshapes what you are about to look at.

Trusting the raw periodogram. Single ordinates are extremely noisy. Smooth, and check that features survive a change of span.

Believing a peak because it is tall. Use Fisher's test, and prewhiten first if the series is autocorrelated.

Reporting frequencies rather than periods. Convert, and give the units of the data.

Confusing R's frequency with f . The `ts()` argument is the period m . Passing $1/m$ produces a series R believes is annual, and every subsequent frequency axis will be wrong.

Ignoring the Nyquist limit. Nothing above $f = 1/2$ is observable, and anything faster in the underlying process has already been folded into the visible band before analysis begins.

Reading harmonics as separate cycles. Peaks at $1/12$, $2/12$ and $3/12$ in monthly data are usually one annual pattern that is not a pure sinusoid.

8.12 Summary

Spectral analysis rewrites a series as a sum of oscillations and asks how much of its variance each one accounts for. The periodogram displays that decomposition, so its peaks locate the dominant

cycles and the total area under it is the variance of the series.

Reading one is a five step routine: convert the peak to a period and attach units, check whether smaller peaks are harmonics of it rather than separate cycles, remember that raw ordinates are far too noisy to trust individually, smooth and keep only what survives, and test the largest peak formally before reporting it.

The value of the frequency domain is not that it contains information the ACF lacks, since it does not. It is that questions about cycle length, cycle regularity and several simultaneous periodicities are awkward in the time domain and immediate here, while questions about forecasting and non-stationarity run the other way. The two are complements, and serious work on a seasonal or cyclical series usually looks at both.